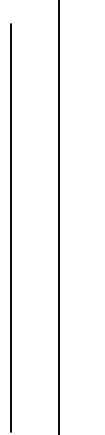


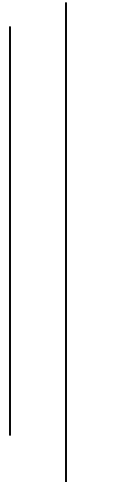
Friday, September 19, 2025

Complex Variable



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Complex Mapping

Complex Number:

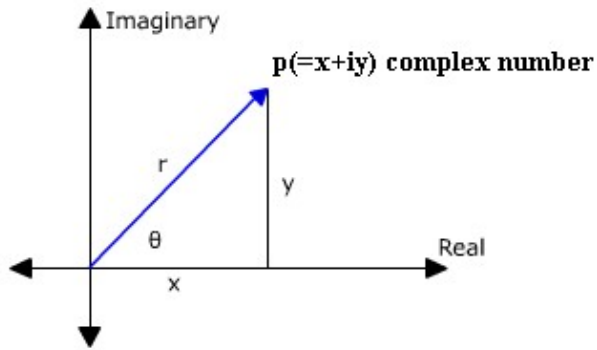


Figure 01

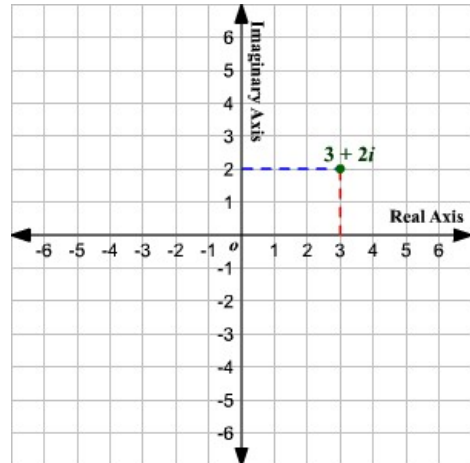


Figure 02

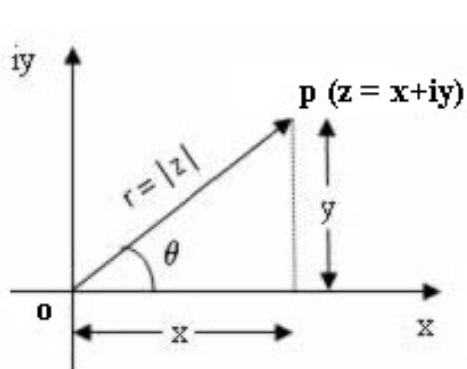


Figure 03

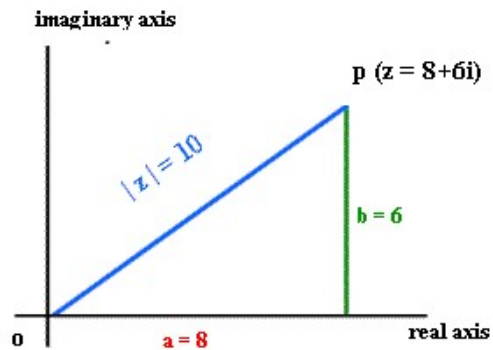


Figure 04

$$OP = |z| = \sqrt{x^2 + y^2}$$

$$OP = |z| = \sqrt{8^2 + 6^2} = \sqrt{100} = 10$$

Transformation

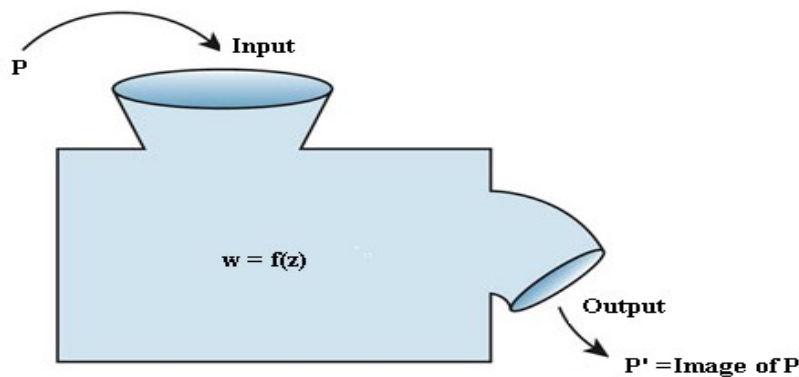


Figure 05

Where the point P in the z -plane and the point P' in the w -plane.

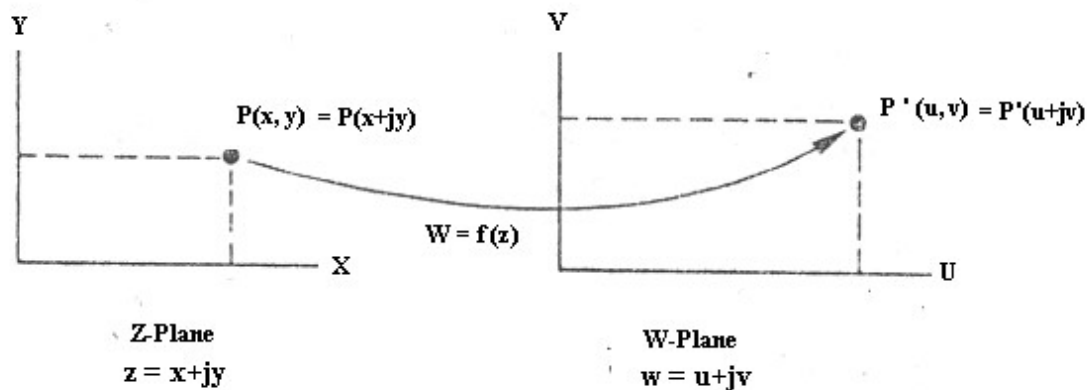


Figure 06

Definition: The transformation of P in the z -plane onto P' in the w -plane is said to be a mapping of P onto P' under the transformation $w = u + jv = f(z)$ and P' is sometimes referred to as the image of P .

Example 01

Determine the image of the point P , $z = 3 + j2$, on the w -plane under the transformation $w = 3z + 2 - j$

Answer:

We have,

$$w = 3z + 2 - j$$

$$w = f(z) = 3z + 2 - j$$

$$u + jv = f(z) = 3(x + jy) + 2 - j \quad [z = x + jy] \quad [w = u + jv]$$

$$u + jv = f(z) = 3x + 3jy + 2 - j$$

$$u + jv = f(z) = 3x + 2 + 3jy - j$$

$$u + jv = f(z) = 3x + 2 + j(3y - 1) \text{-----(i)}$$

Equating real and imaginary part, we get

$$u = 3x + 2 \text{-----(ii)}$$

$$v = 3y - 1 \text{-----(iii)}$$

Given, the point P, $z = 3 + j2$,

That is P (3, 2) -----(iv)

Here, $x = 3$, $y = 2$

Putting the values of x and y in (ii) & (iii), Then the point P ($z = 3 + j2$) transforms onto w-plane is

$$u = 3x + 2 \quad \text{and} \quad v = 3y - 1$$

$$u = 3.3 + 2 \quad v = 3.2 - 1$$

$$u = 11 \quad v = 5$$

$$\therefore w = u + jv = 11 + j5$$

The image of P is $P' (= u + jv = 11 + j5)$

That is $P'(11,5)$ -----(v)

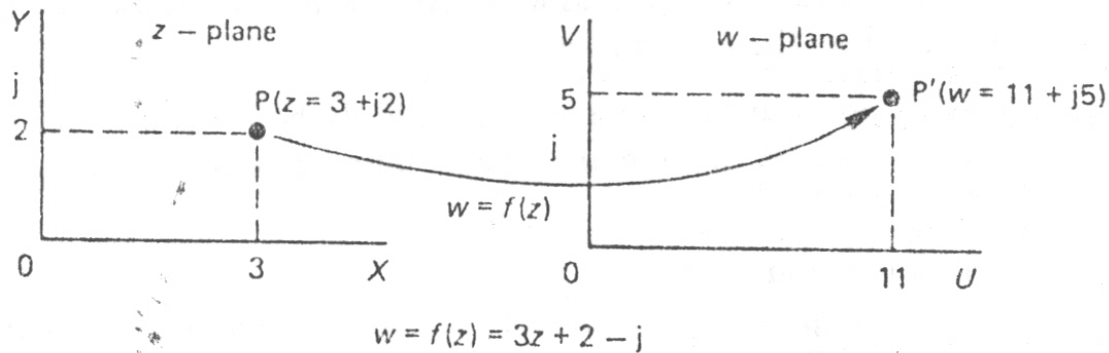


Figure 07

Example 02

Map the points $A(z = -2 + j)$ and $B(z = 3 + j4)$ on to the w-plane under the transformation $w = j2z + 3$ and illustrate the transformation on a diagram.

Solution:

We have,

$$w = f(z) = j2z + 3$$

$$u + jv = f(z) = j2z + 3 \quad [w = u + jv]$$

$$u + jv = j2(x + jy) + 3 \quad [z = x + jy]$$

$$u + jv = j2x + 2j^2y + 3$$

$$u + jv = j2x - 2y + 3 [\because j^2 = -1]$$

$$u + jv = (3 - 2y) + j2x \text{-----(i)}$$

Equating the coefficient of real and imaginary part, we get

$$\therefore u = 3 - 2y \text{-----(ii)}$$

$$\therefore v = 2x \text{ -----(iii)}$$

Given, $A(z = -2 + j.1)$

That is, $A(-2, 1)$ -----(iv)

Here,

$$x = -2, y = 1$$

Putting the value of x and y in (ii) and (iii),

$$u = 3 - 2y \qquad v = 2x$$

$$u = 3 - 2.1 \qquad v = 2.(-2)$$

$$u = 1 \qquad v = -4$$

$$\therefore w = u + jv = 1 - j.4$$

The image of A is $A'(w = 1 - j.4)$

That is $A'(1, -4)$ -----(v)

Again,

$$B(z = 3 + j4)$$

That is, $B(3, 4)$ -----(vi)

Here, $x = 3, y = 4$

Putting the value of x and y in (ii) and (iii),

$$u = 3 - 2y \qquad v = 2x$$

$$u = 3 - 2.4 \qquad v = 2.3$$

$$u = -5 \qquad v = 6$$

$$\therefore w = u + jv = -5 + j.6$$

The image of B is $B'(w = u + jv = -5 + j.6)$

That is $B'(-5, 6)$ -----(vii)

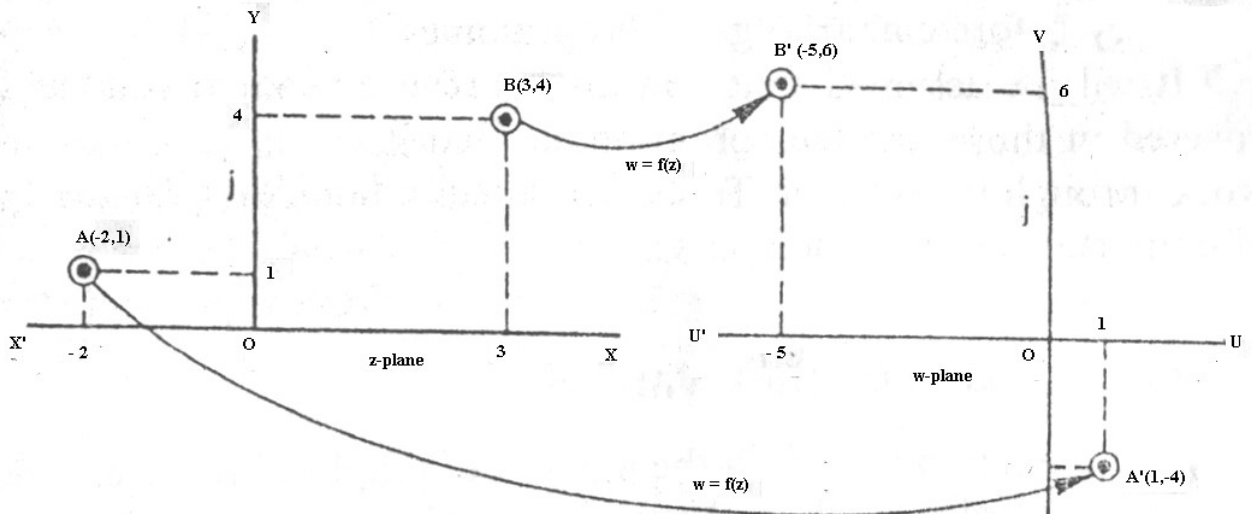


Figure 08

Example 03

Map the straight line joining $A(-2 + j)$ and $B(3 + j6)$ in the z -plane on to the w -plane when $w = u + jv = f(z) = 3 + j2z$

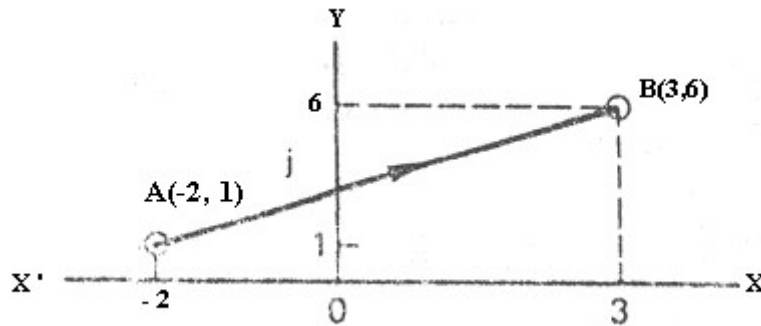


Figure 09

Answer:

We have,

$$w = f(z) = 3 + j2z$$

$$u + jv = f(z) = 3 + j2(x + jy) \quad [w = u + jv; z = x + jy]$$

$$u + jv = f(z) = 3 + j2x + j^2 2y$$

$$u + jv = f(z) = 3 + j2x - 2y [\because j^2 = -1]$$

$$u + jv = f(z) = 3 + j2x - 2y$$

$$u + jv = f(z) = 3 - 2y + j2x$$

$$u + jv = f(z) = (3 - 2y) + j2x \text{-----(i)}$$

Equating real and imaginary part, we get

$$u = 3 - 2y \text{-----(ii)}$$

$$v = 2x \text{-----(iii)}$$

Given, the point A, $z = -2 + j.1$

That is A (-2, 1) -----(iv)

Here, $x = -2, y = 1$

Putting the value of x and y in (ii) and (iii),

Then the point A, ($z = -2 + j$) transforms onto w -plane is

$$u = 3 - 2y \quad \text{and} \quad v = 2x$$

$$u = 3 - 2.1 \quad v = 2.(-2)$$

$$u = 1 \quad v = -4$$

$\therefore$ The image of A is $A' (w = u + jv = 1 - 4j)$

That is $A' (1, -4) \text{-----(v)}$

Again,

Given, the point B, $z = 3 + j6$

That is B (3, 6) -----(vi)

Here, $x = 3, y = 6$

Putting the value of x and y in (ii) and (iii),

Then the point B, $z = 3 + j6$ transforms onto w-plane is

$$\begin{aligned} u &= 3 - 2y & \text{and} & & v &= 2x \\ u &= 3 - 2.6 & & & v &= 2.3 \\ u &= -9 & & & v &= 6 \end{aligned}$$

$\therefore$ The image of B is B' ($w = u + jv = -9 + 6j$)

That is $B'(-9, 6)$ -----(vii)

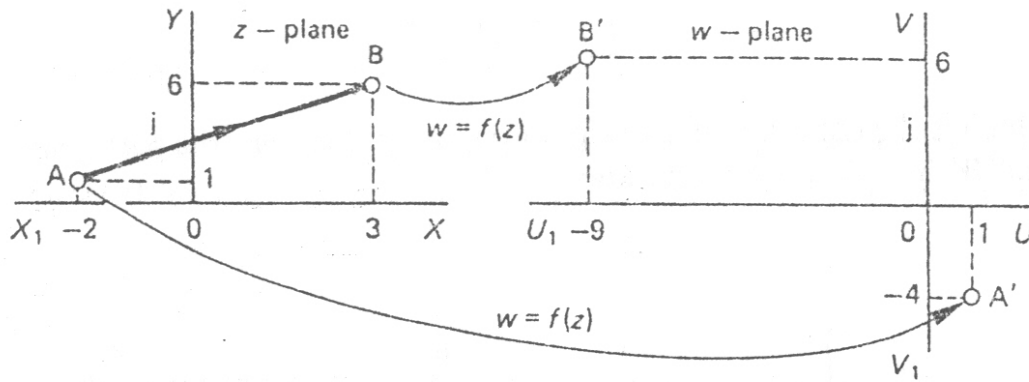


Figure 10

Again,

From Figure 09:

Given, the point A, $z = -2 + j.1$

That is A (-2, 1)

Here, $x_1 = -2$, $y_1 = 1$

Given, the point B, $z = 3 + j6$

That is, B (3, 6),

Here, $x_2 = 3$, $y_2 = 6$

The equation of the straight line AB is:

$$\begin{aligned} \frac{y - y_1}{y_1 - y_2} &= \frac{x - x_1}{x_1 - x_2} \\ \Rightarrow \frac{y - 1}{1 - 6} &= \frac{x - (-2)}{-2 - 3} \\ \Rightarrow \frac{y - 1}{-5} &= \frac{x + 2}{-5} \\ \Rightarrow y - 1 &= x + 2 \\ \Rightarrow y &= x + 3 \text{ -----(viii)} \end{aligned}$$

We have, from (ii) & (iii)

$$\begin{aligned} u &= 3 - 2y & v &= 2x \\ \Rightarrow 3 - 2y &= u & \Rightarrow 2x &= v \\ \Rightarrow -2y &= u - 3 & \Rightarrow x &= \frac{v}{2} \end{aligned}$$

$$\Rightarrow 2y = -u + 3$$

$$\Rightarrow 2y = 3 - u$$

$$\Rightarrow y = \frac{3-u}{2}$$

Putting the value of x and y in (viii),

$$\Rightarrow y = x + 3$$

$$\Rightarrow \frac{3-u}{2} = \frac{v}{2} + 3$$

$$\Rightarrow \frac{3-u}{2} = \frac{v+6}{2}$$

$$\Rightarrow 3-u = v+6$$

$$\Rightarrow v+6 = 3-u$$

$$\Rightarrow v = -6+3-u$$

$$\Rightarrow v = -3-u$$

$$\Rightarrow v = -u-3 \text{-----(ix)}$$

The equation (ix) is the equation of the straight line $A'B'$

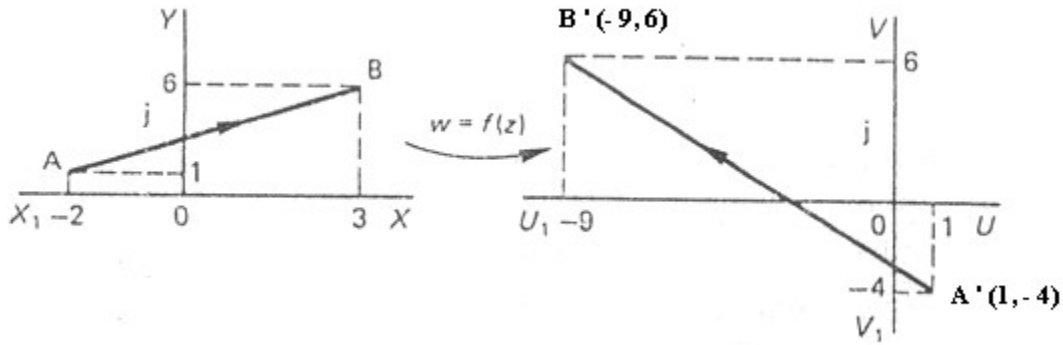


Figure 11

Justification:

We have

The image of A is $A' (w = u + jv = 1 - 4j)$

That is $A' (1, -4)$

Here, $u_1 = 1, v_1 = -4$

and

The image of B is $B' (w = u + jv = -9 + 6j)$

That is $B' (-9, 6)$

Here, $u_2 = -9, v_2 = 6$

The equation of the straight line $A'B'$ is:

$$\frac{v - v_1}{v_1 - v_2} = \frac{u - u_1}{u_1 - u_2} \quad \left[\text{As we know: } \frac{y - y_1}{y_1 - y_2} = \frac{x - x_1}{x_1 - x_2} \right]$$

$$\begin{aligned}
&\Rightarrow \frac{v - (-4)}{-4 - 6} = \frac{u - 1}{1 - (-9)} \\
&\Rightarrow \frac{v + 4}{-10} = \frac{u - 1}{1 + 9} \\
&\Rightarrow \frac{v + 4}{-10} = \frac{u - 1}{10} \\
&\Rightarrow \frac{v + 4}{-1} = \frac{u - 1}{1} \\
&\Rightarrow v + 4 = -1(u - 1) \\
&\Rightarrow v + 4 = -u + 1 \\
&\Rightarrow v = -u + 1 - 4 \\
&\Rightarrow v = -u - 3 \text{-----(x)}
\end{aligned}$$

Since the equation (ix) and (x) is same. **Hence proved**

Example 04: Graph of Parabola

$$y^2 = 4ax$$

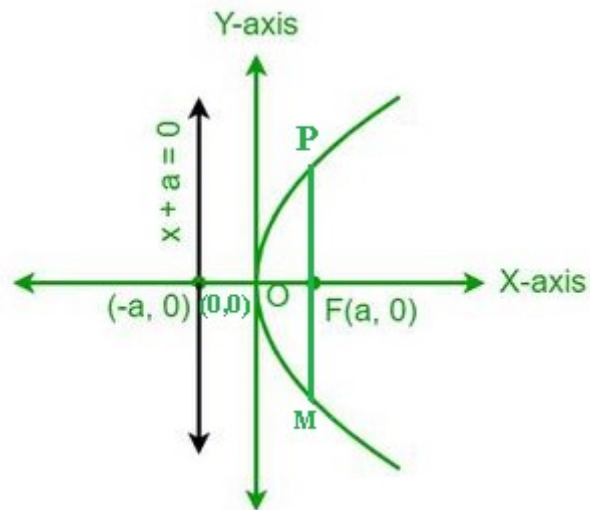


Figure 12

Focus F (a, 0)

PM= Latus Rectum = 4a

Example 05: Graph of Parabola

$$y^2 = -4ax$$

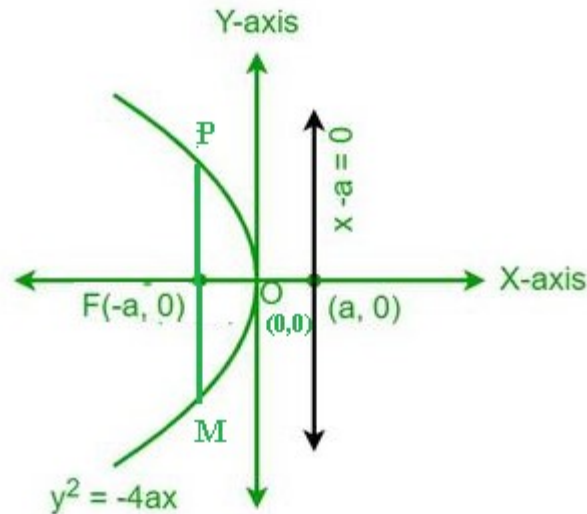


Figure 13

Focus $F(-a, 0)$

$PM = \text{Latus Rectum} = 4a$

Example 06: Graph of Parabola

$$x^2 = 4ay$$

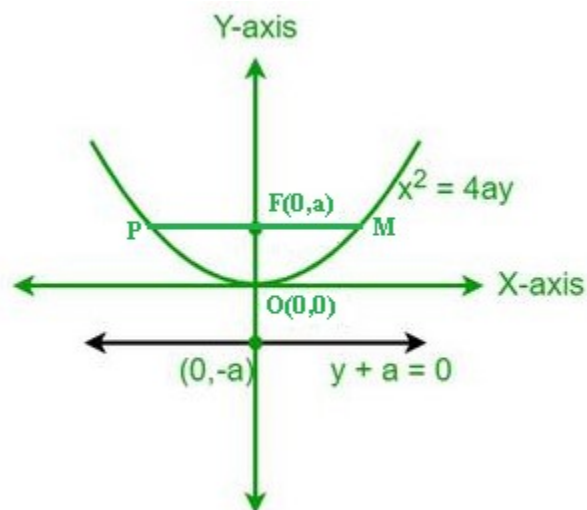


Figure 14

Focus $F(0, a)$

$PM = \text{Latus Rectum} = 4a$

Example 07: Graph of Parabola

$$x^2 = -4ay$$

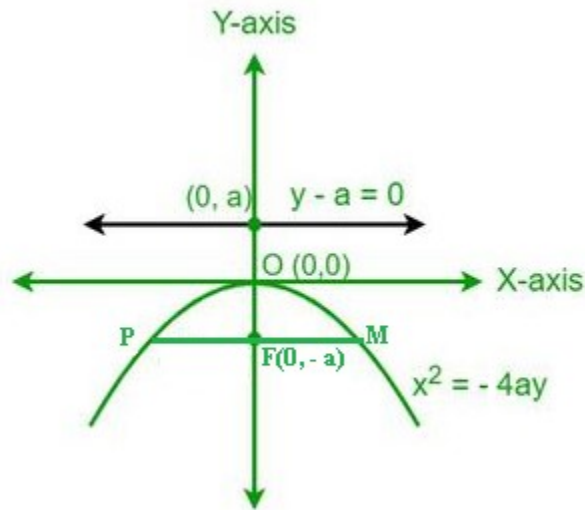


Figure 15

Focus F (0, -a)

PM= Latus Rectum = 4a

Example 08

A straight line joining A(2 + 0.j) and B(0 + 2j) in the z-plane is mapped onto the w-plane by the transformation equation $w = z^2$,

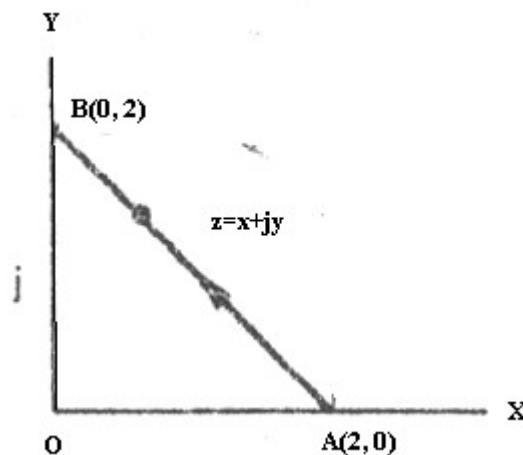


Figure 16

Solution:

We have, $w = f(z) = z^2$

$$u + jv = f(z) = (x + jy)^2 \quad [w = u + jv; z = x + jy]$$

$$u + jv = x^2 + j2xy + j^2y^2$$

$$u + jv = x^2 + j2xy - y^2 \quad [\because j^2 = -1]$$

$$u + jv = x^2 - y^2 + j2xy \text{ -----(i)}$$

Equating the coefficient of real and imaginary part, we get

$$\therefore u = x^2 - y^2 \quad \text{----- (ii)}$$

$$\therefore v = 2xy \quad \text{----- (iii)}$$

Given, $A(2 + j.0)$

That is, $A(2,0)$ -----(iv)

Here $x = 2, y = 0$

Putting the values of x and y in (ii) and (iii),

$$u = x^2 - y^2 \quad v = 2xy$$

$$u = 2^2 - 0^2 \quad v = 2.2.0$$

$$u = 4 - 0 \quad v = 0$$

$$u = 4$$

$$\therefore w = u + jv = 4 + j.0$$

The image of A is $A'(w = 4 + j.0)$

That is $A'(4,0)$ -----(v)

Again,

$$B(z = 0 + j2)$$

That is, $B(0,2)$ -----(vi)

Putting the values of x and y in (ii) and (iii),

$$u = x^2 - y^2 \quad v = 2xy$$

$$u = 0^2 - 2^2 \quad v = 2.0.2$$

$$u = 0 - 4 \quad v = 0$$

$$u = -4$$

$$\therefore w = u + jv = -4 + j.0$$

The image of B is $B'(w = u + jv = -4 + j.0)$

That is $B'(-4,0)$ -----(vii)

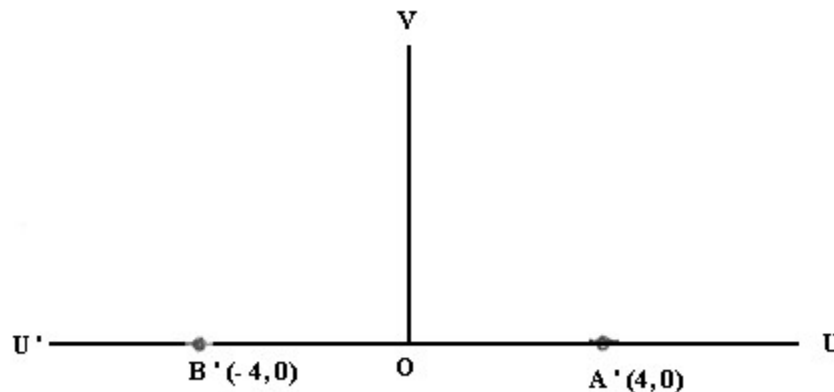


Figure 17

Given, From Figure 16:

We have: A (2, 0) and B (0, 2)

$$A : x_1 = 2, y_1 = 0$$

$$B : x_2 = 0, y_2 = 2$$

The equation of the line AB is,

$$\frac{y - y_1}{y_1 - y_2} = \frac{x - x_1}{x_1 - x_2}$$

$$\Rightarrow \frac{y - 0}{0 - 2} = \frac{x - 2}{2 - 0}$$

$$\Rightarrow \frac{y}{-2} = \frac{x - 2}{2}$$

$$\Rightarrow 2y = -2(x - 2)$$

$$\Rightarrow y = \frac{-2(x - 2)}{2}$$

$$\therefore y = -(x - 2)$$

$$\therefore y = -x + 2$$

$$\therefore y = 2 - x \quad \text{-----(viii)}$$

Putting the value of y in (ii),

$$\therefore u = x^2 - y^2$$

$$\Rightarrow u = x^2 - (2 - x)^2$$

$$\Rightarrow u = x^2 - (4 - 4x + x^2)$$

$$\Rightarrow u = x^2 - 4 + 4x - x^2$$

$$\Rightarrow u = 4x - 4$$

$$\Rightarrow u + 4 = 4x$$

$$\Rightarrow x = \frac{u + 4}{4} \quad \text{-----(ix)}$$

Putting the value of y in (iii),

$$v = 2xy$$

$$v = 2x(2 - x)$$

$$\therefore v = 4x - 2x^2 \quad \text{-----(x)}$$

Now putting the value of x in equation (x)

$$\therefore v = 4x - 2x^2$$

$$v = 4\left(\frac{u + 4}{4}\right) - 2\left(\frac{u + 4}{4}\right)^2$$

$$\Rightarrow v = u + 4 - 2\left(\frac{u^2 + 8u + 16}{16}\right)$$

$$\Rightarrow v = u + 4 - \frac{1}{8}(u^2 + 8u + 16)$$

$$\begin{aligned}
\Rightarrow v &= u + 4 - \frac{1}{8}u^2 - u - 2 \\
\Rightarrow v &= 2 - \frac{1}{8}u^2 \\
\Rightarrow v &= \frac{16 - u^2}{8} \\
\Rightarrow v &= -\frac{1}{8}(u^2 - 16) \\
\Rightarrow 8v &= -(u^2 - 16) \\
\Rightarrow 8v &= -u^2 + 16 \\
\Rightarrow 8v &= -u^2 + 16 \\
\Rightarrow -u^2 &= 8v - 16 \\
\Rightarrow u^2 &= -8(v - 2) \\
\Rightarrow u^2 &= -4 \cdot 2(v - 2) \text{(xi)}
\end{aligned}$$

The equation (xi) represents an equation of a parabola.

Let,

$$U = u \text{ and } V = v - 2 \text{(xii)}$$

From (xii),

When,

$$U = 0 \text{ then } u = 0$$

and

$$V = 0 \text{ then}$$

$$\Rightarrow 0 = v - 2$$

$$\Rightarrow v = 2$$

$$\therefore \text{Vertex} = (u, v) = (0, 2)$$

And latus rectum $A'B' =$

$$= 4a$$

$$= 4 \cdot 2$$

$$= 8$$

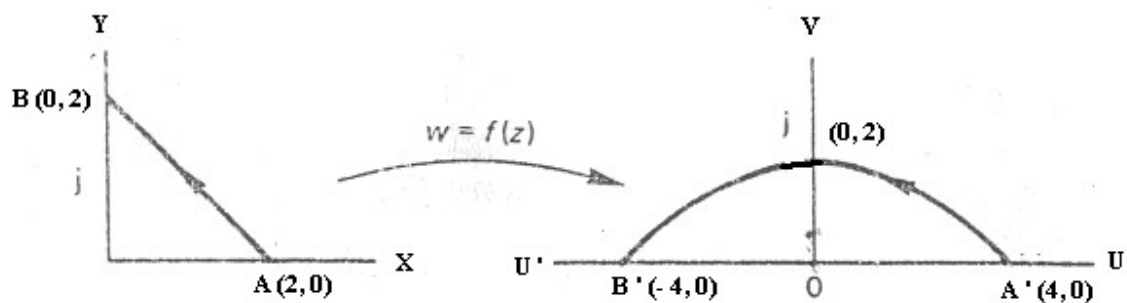


Figure 18

Example 09

The straight line AB in the z-plane as shown is mapped onto the w-plane by $w = z^2$

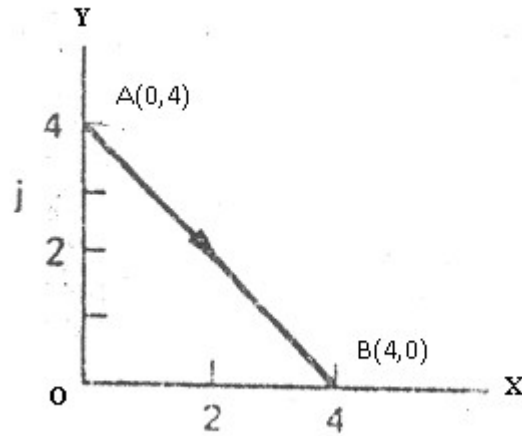


Figure 19

Solution:

We have,

$$w = f(z) = z^2$$

$$w = f(z) = (x + jy)^2 \quad [z = x + jy]$$

$$u + jv = f(z) = x^2 + j2xy + j^2y^2 \quad [w = u + jv]$$

$$u + jv = x^2 + j2xy - y^2 \quad [\because j^2 = -1]$$

$$u + jv = x^2 - y^2 + j2xy \quad \text{-----(i)}$$

Equating the coefficient of real and imaginary part, we get

$$\therefore u = x^2 - y^2 \quad \text{-----(ii)}$$

$$\therefore v = 2xy \quad \text{-----(iii)}$$

Given, $A(0 + j.4)$

That is, $A(0,4) \quad \text{-----(iv)}$

Here $x = 0, y = 4$

Putting the value of x and y in (ii) and (iii),

$$u = x^2 - y^2 \quad v = 2xy$$

$$u = 0^2 - 4^2 \quad v = 2.0.4$$

$$u = 0 - 16 \quad v = 0$$

$$u = -16$$

$$\therefore w = u + jv = -16 + j.0$$

The image of A is $A'(w = -16 + j.0)$

That is $A'(-16,0) \quad \text{-----(v)}$

Again, $B(z = 4 + j.0)$

That is, $B(4,0) \quad \text{-----(vi)}$

Here, $x = 4, y = 0$

Putting the value of x and y in (ii) and (iii),

$$u = x^2 - y^2 \quad v = 2xy$$

$$u = 4^2 - 0^2 \quad v = 2 \cdot 4 \cdot 0$$

$$u = 16 \quad v = 0$$

$$u = 16$$

$$\therefore w = u + jv = 16 + j \cdot 0$$

The image of B is $B' (w = u + jv = 16 + j \cdot 0)$

That is $B' (16, 0)$ -----(vii)

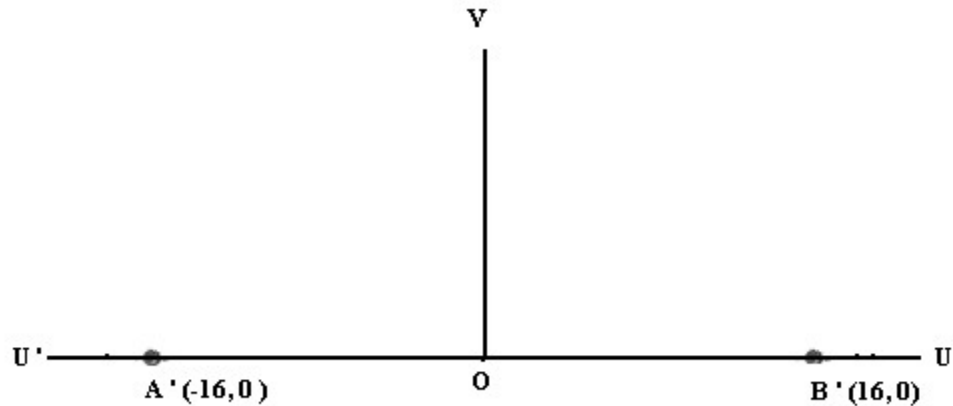


Figure 20

Given, From Figure 19:

For $A(0, 4)$ and $B(4, 0)$

$$A : x_1 = 0, y_1 = 4$$

$$B : x_2 = 4, y_2 = 0$$

The equation of the line AB is,

$$\frac{y - y_1}{y_1 - y_2} = \frac{x - x_1}{x_1 - x_2}$$

$$\Rightarrow \frac{y - 4}{4 - 0} = \frac{x - 0}{0 - 4}$$

$$\Rightarrow \frac{y - 4}{4} = \frac{x}{-4}$$

$$\Rightarrow -4(y - 4) = 4x$$

$$\Rightarrow -(y - 4) = x$$

$$\Rightarrow (y - 4) = -x$$

$$\Rightarrow y = 4 - x \quad \text{-----(viii)}$$

Putting the value of y in (ii),

$$\therefore u = x^2 - y^2$$

$$\begin{aligned}
\Rightarrow u &= x^2 - (4 - x)^2 \\
\Rightarrow u &= x^2 - (16 - 8x + x^2) \\
\Rightarrow u &= x^2 - 16 + 8x - x^2 \\
\Rightarrow u &= 8x - 16 \\
\Rightarrow u + 16 &= 8x \\
\Rightarrow x &= \frac{u + 16}{8} \text{-----} (ix)
\end{aligned}$$

Putting the value of y in (iii),

$$\begin{aligned}
v &= 2xy \\
v &= 2x(4 - x) \\
\therefore v &= 8x - 2x^2 \text{-----} (x)
\end{aligned}$$

Putting the value of x in (x), we get

$$\begin{aligned}
\Rightarrow v &= 8\left(\frac{u + 16}{8}\right) - 2\left(\frac{u + 16}{8}\right)^2 \\
\Rightarrow v &= (u + 16) - \frac{2}{64}(u + 16)^2 \\
\Rightarrow v &= (u + 16) - \frac{1}{32}(u + 16)^2 \\
\Rightarrow v &= u + 16 - \frac{1}{32}(u^2 + 32u + 256) \\
\Rightarrow v &= u + 16 - \frac{1}{32}u^2 - u - 8 \\
\Rightarrow v &= 8 - \frac{1}{32}u^2 \\
\Rightarrow v &= -\frac{1}{32}u^2 + 8 \\
\Rightarrow v &= -\left(\frac{u^2 - 256}{32}\right) \\
\Rightarrow 32v &= -(u^2 - 256) \\
\Rightarrow -u^2 + 256 &= 32v \\
\Rightarrow -u^2 &= 32v - 256 \\
\Rightarrow u^2 &= -32v + 256 \\
\Rightarrow u^2 &= -32(v - 8) \\
\Rightarrow u^2 &= -4 \cdot 8(v - 8) \text{-----} (xi)
\end{aligned}$$

The equation (xi) represents an equation of a parabola.

Let,

$$U = u \text{ and } V = v - 8 \text{-----} (xii)$$

From (xii),

When,

$$U = 0 \text{ then } u = 0$$

and

$$V = 0 \text{ then}$$

$$\Rightarrow 0 = v - 8$$

$$\Rightarrow v = 8$$

$$\therefore \text{Vertex} = (u, v) = (0, 8)$$

And latus rectum $A'B' =$

$$= 4a$$

$$= 4 \cdot 8$$

$$= 32$$

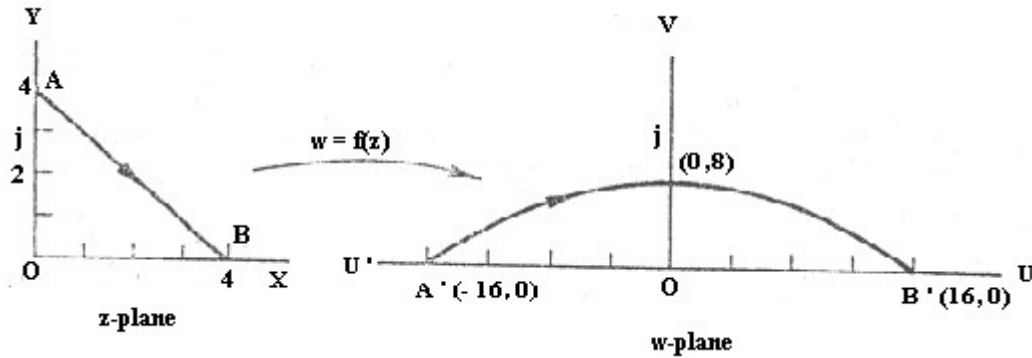


Figure 21

Example 10

A triangle consisting of AB, BC and CA in the z -plane is mapped to the w -plane by the transformation $w = z^2$

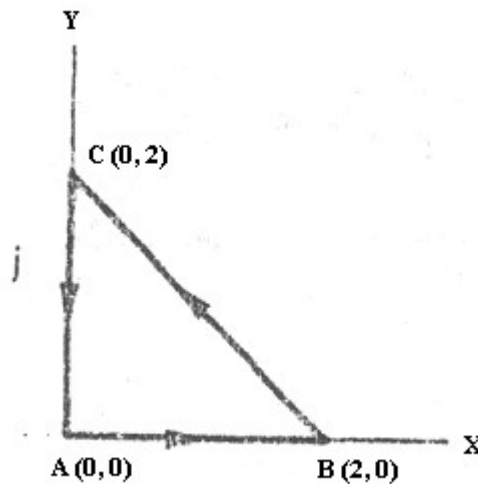


Figure 22

Solution:

We have,

$$w = f(z) = z^2$$

$$w = f(z) = (x + jy)^2 \quad [z = x + jy]$$

$$u + jv = f(z) = x^2 + j2xy + j^2y^2 \quad [w = u + jv]$$

$$u + jv = f(z) = x^2 + j2xy - y^2 [\because j^2 = -1]$$

$$u + jv = x^2 - y^2 + j2xy \text{ -----(i)}$$

Equating the coefficient of real and imaginary part, we get

$$\therefore u = x^2 - y^2 \text{ -----(ii)}$$

$$\therefore v = 2xy \text{ -----(iii)}$$

$$\text{Given, } A(0,0) \text{ -----(iv)}$$

That is, $x = 0, y = 0$

Putting the value of x and y in (ii) and (iii)

$$u = x^2 - y^2 \qquad v = 2xy$$

$$u = 0^2 - 0^2 \qquad v = 2.0.0$$

$$u = 0 - 0 \qquad v = 0$$

$$u = 0$$

$$\therefore w = u + jv = 0 + j.0$$

The image of A is $A'(w = 0 + j.0)$

$$\text{That is } A'(0,0) \text{ -----(v)}$$

$$\text{Again, } B(2,0) \text{ -----(vi)}$$

That is, $x = 2, y = 0$

Putting the value of x and y in (ii) and (iii),

$$u = x^2 - y^2 \qquad v = 2xy$$

$$u = 2^2 - 0^2 \qquad v = 2.2.0$$

$$u = 4 \qquad v = 0$$

$$u = 4$$

$$\therefore w = u + jv = 4 + j.0$$

The image of B is $B'(w = u + jv = 4 + j.0)$

$$\text{That is } B'(4,0) \text{ -----(vii)}$$

$$\text{Again, } C(0,2) \text{ -----(viii)}$$

That is, $x = 0, y = 2$

Putting the value of x and y in (ii) and (iii)

$$u = x^2 - y^2 \qquad v = 2xy$$

$$u = 0^2 - 2^2 \qquad v = 2.0.2$$

$$u = -4 \qquad v = 0$$

$$u = -4$$

$$\therefore w = u + jv = -4 + j.0$$

The image of C is

$$C'(w = u + jv = -4 + j.0)$$

That is

$$C'(-4,0) \text{ -----(ix)}$$

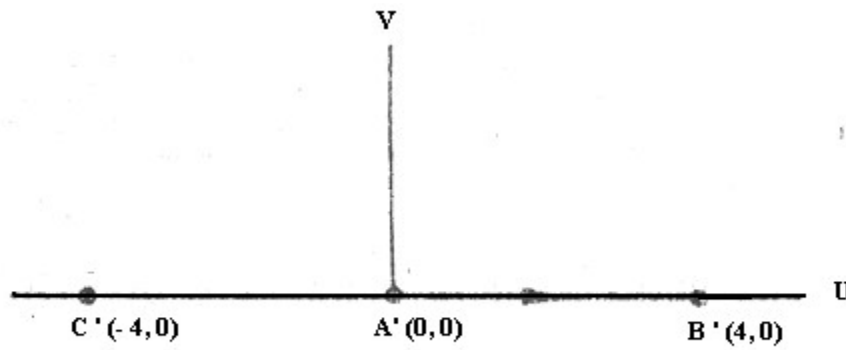


Figure 23

From Figure 22:

Given: $A(0,0)$

That is,

$$x_1 = 0, y_1 = 0$$

$B(2,0)$

That is, $x_2 = 2, y_2 = 0$

The equation of the line AB is,

$$\frac{y - y_1}{y_1 - y_2} = \frac{x - x_1}{x_1 - x_2}$$

$$\Rightarrow \frac{y - 0}{0 - 0} = \frac{x - 0}{0 - 2}$$

$$\Rightarrow \frac{y}{0} = \frac{x}{-2}$$

$$\Rightarrow -2y = 0$$

$$\Rightarrow y = 0$$

------(x)

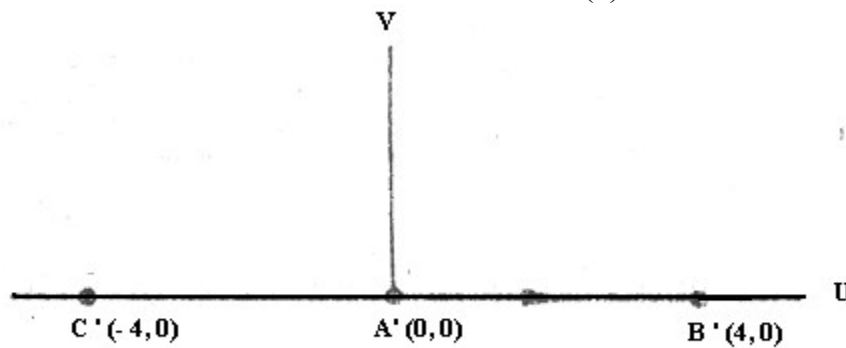


Figure 24

Now, Putting the value of y from (x) in (ii) & (iii),

$$u = x^2 - y^2$$

$$v = 2xy$$

$$\Rightarrow u = x^2 - 0$$

$$\therefore v = 0$$

$$\Rightarrow u = x^2$$

Given:

From Figure 22

B(2,0)

Then, $x_1 = 2, y_1 = 0$

Again

C(0,2)

Then, $x_2 = 0, y_2 = 2$

The equation of the line BC is,

$$\frac{y - y_1}{y_1 - y_2} = \frac{x - x_1}{x_1 - x_2}$$

$$\Rightarrow \frac{y - 0}{0 - 2} = \frac{x - 2}{2 - 0}$$

$$\Rightarrow \frac{y}{-2} = \frac{x - 2}{2}$$

$$\Rightarrow 2y = -2(x - 2)$$

$$\Rightarrow y = \frac{-2(x - 2)}{2}$$

$$\therefore y = -(x - 2)$$

$$\therefore y = 2 - x \text{ -----(xi)}$$

Putting the value of y in (ii),

$$\therefore u = x^2 - y^2$$

$$\Rightarrow u = x^2 - (2 - x)^2$$

$$\Rightarrow u = x^2 - (4 - 4x + x^2)$$

$$\Rightarrow u = x^2 - 4 + 4x - x^2$$

$$\Rightarrow u = 4x - 4$$

$$\Rightarrow u + 4 = 4x$$

$$\Rightarrow x = \frac{u + 4}{4} \text{ -----(xii)}$$

Putting the value of y in (iii),

$$v = 2xy$$

$$v = 2x(2 - x)$$

$$\therefore v = 4x - 2x^2 \text{ -----(xiii)}$$

Now putting the value of x in equation (xiii)

$$v = 4\left(\frac{u + 4}{4}\right) - 2\left(\frac{u + 4}{4}\right)^2$$

$$\Rightarrow v = u + 4 - 2\left(\frac{u^2 + 8u + 16}{16}\right)$$

$$\begin{aligned}
\Rightarrow v &= u + 4 - \frac{1}{8}(u^2 + 8u + 16) \\
\Rightarrow v &= u + 4 - \frac{1}{8}u^2 - u - 2 \\
\Rightarrow v &= 2 - \frac{1}{8}u^2 \\
\Rightarrow v &= \frac{16 - u^2}{8} \\
\Rightarrow v &= -\frac{1}{8}(u^2 - 16) \\
\Rightarrow 8v &= -(u^2 - 16) \\
\Rightarrow 8v &= -u^2 + 16 \\
\Rightarrow 8v &= -u^2 + 16 \\
\Rightarrow -u^2 &= 8v - 16 \\
\Rightarrow u^2 &= -8(v - 2) \\
\Rightarrow u^2 &= -4 \cdot 2(v - 2) \quad \text{----- (xiv)}
\end{aligned}$$

The equation (xiv) represents an equation of a parabola.

Let,

$$U = u \text{ and } V = v - 2 \quad \text{----- (xv)}$$

From (xv),

When,

$$U = 0 \text{ then } u = 0$$

and

$$V = 0 \text{ then}$$

$$\Rightarrow 0 = v - 2$$

$$\Rightarrow v = 2$$

$$\therefore \text{Vertex} = (u, v) = (0, 2)$$

And latus rectum $B'C' =$

$$= 4a$$

$$= 4 \cdot 2 = 8$$

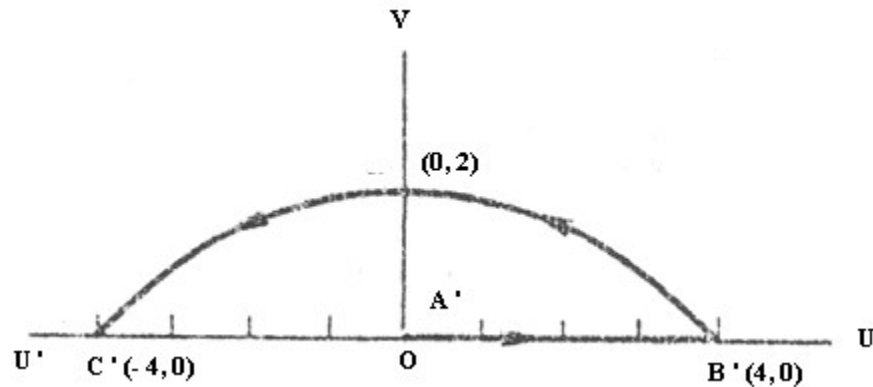


Figure 25

So, finally we get

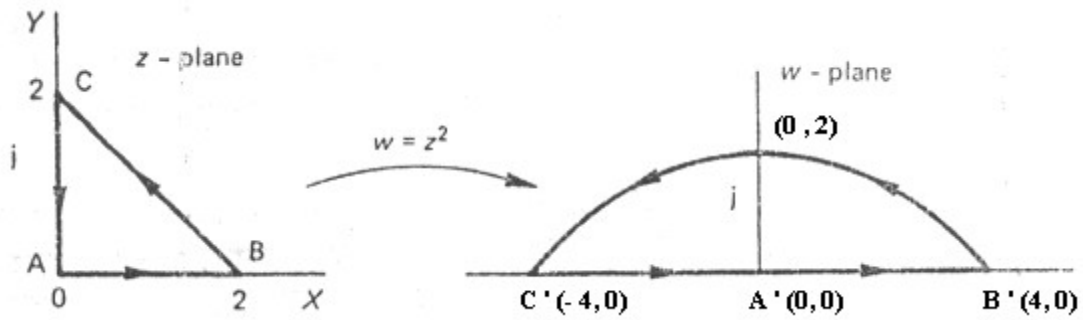


Figure 26

Example 11

A straight line joining $A(-j)$ and $B(2 + j)$ in the z -plane is mapped onto the w -plane by

the transformation equation $w = \frac{1}{z}$

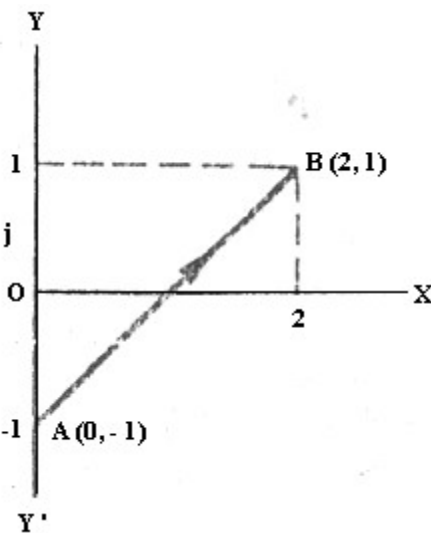


Figure 27

Solution:

Given,

$$w = \frac{1}{z}$$

$$w = \frac{1}{x + jy}$$

$$[z = x + jy]$$

$$w = \frac{x - jy}{(x + jy)(x - jy)}$$

[Multiplying by $x - jy$]

$$w = \frac{x - jy}{x^2 - jxy + jxy - j^2 y^2}$$

$$w = \frac{x - jy}{x^2 + y^2}$$

$$[\because j^2 = -1]$$

$$u + jv = \frac{x - jy}{x^2 + y^2} \quad [w = u + jv]$$

$$u + jv = \frac{x}{x^2 + y^2} - j \frac{y}{x^2 + y^2} \quad \text{-----(i)}$$

Equating the coefficient of real and imaginary part, we get,

$$u = \frac{x}{x^2 + y^2} \quad \text{----- (ii)}$$

$$v = \frac{-y}{x^2 + y^2} \quad \text{----- (iii)}$$

Given, $A(0 - j.1)$

That is, $A(0, -1)$ -----(iv)

Here,

$$x = 0, y = -1$$

Putting the value of x and y in (ii) and (iii)

$$u = \frac{x}{x^2 + y^2} \quad v = \frac{-y}{x^2 + y^2}$$

$$u = \frac{0}{0 + (-1)^2} \quad v = \frac{-(-1)}{0^2 + (-1)^2}$$

$$u = \frac{0}{0 + 1} \quad v = \frac{1}{1}$$

$$u = \frac{0}{1} \quad v = 1$$

$$u = 0 \quad v = 1$$

$$\therefore w = u + jv = 0 + j.1$$

The image of A is $A'(w = 0 + j.1)$

That is $A'(0, 1)$ -----(v)

Again,

$$B(z = 2 + j.1)$$

That is, $B(2, 1)$ -----(vi)

Here, $x = 2, y = 1$

Putting the value of x and y in (ii) and (iii),

$$u = \frac{x}{x^2 + y^2} \quad v = \frac{-y}{x^2 + y^2}$$

$$u = \frac{2}{2^2 + 1^2} \quad v = \frac{-1}{2^2 + 1^2}$$

$$u = \frac{2}{5} \quad v = \frac{-1}{5}$$

$$\therefore w = u + jv = \frac{2}{5} - j\frac{1}{5}$$

The image of B is $B'(w = \frac{2}{5} - j\frac{1}{5})$

That is $B'(\frac{2}{5}, -\frac{1}{5})$ -----(vii)

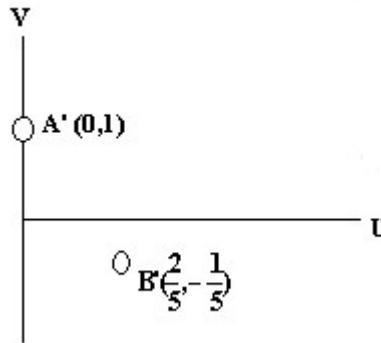


Figure 28

From Figure 27:

Given A(0,-1) and B(2,1)

The equation of the line AB is,

$$\begin{aligned} \frac{y - y_1}{y_1 - y_2} &= \frac{x - x_1}{x_1 - x_2} \\ \Rightarrow \frac{y - (-1)}{-1 - 1} &= \frac{x - 0}{0 - 2} \\ \Rightarrow \frac{y + 1}{-1 - 1} &= \frac{x - 0}{0 - 2} \\ \Rightarrow \frac{y + 1}{-2} &= \frac{x}{-2} \\ \Rightarrow y + 1 &= x \\ \therefore y &= x - 1 \end{aligned} \quad \text{-----(viii)}$$

Again, Given

$$w = \frac{1}{z}$$

$$\therefore z = \frac{1}{w}$$

$$z = \frac{1}{u + jv}$$

$$[w = u + jv]$$

$$z = \frac{u - jv}{(u + jv)(u - jv)}$$

$$z = \frac{u - jv}{u^2 - (jv)^2}$$

$$z = \frac{u - jv}{u^2 + v^2} \quad [\because j^2 = -1]$$

$$x + jy = \frac{u - jv}{u^2 + v^2} \quad [z = x + jy]$$

$$\text{i.e. } x + jy = \frac{u}{u^2 + v^2} - j \frac{v}{u^2 + v^2} \text{-----}(ix)$$

Equating the coefficient of real and imaginary part, we get,

$$x = \frac{u}{u^2 + v^2} ; \quad y = \frac{-v}{u^2 + v^2} \text{-----}(x)$$

Putting the value of x and y in (viii),

$$y = x - 1$$

$$\Rightarrow \frac{-v}{u^2 + v^2} = \frac{u}{u^2 + v^2} - 1$$

$$\Rightarrow \frac{-v}{u^2 + v^2} = \frac{u - u^2 - v^2}{u^2 + v^2}$$

$$\Rightarrow -v = u - u^2 - v^2$$

$$\Rightarrow u - u^2 - v^2 + v = 0$$

$$\Rightarrow -u + u^2 + v^2 - v = 0$$

$$\Rightarrow u^2 - u + v^2 - v = 0$$

$$\Rightarrow (u^2 - u) + (v^2 - v) = 0$$

$$\Rightarrow u^2 - 2 \cdot u \cdot \frac{1}{2} + \left(\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2 + v^2 - 2 \cdot v \cdot \frac{1}{2} + \left(\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2 = 0$$

$$\Rightarrow u^2 - 2 \cdot u \cdot \frac{1}{2} + \left(\frac{1}{2}\right)^2 + v^2 - 2 \cdot v \cdot \frac{1}{2} + \left(\frac{1}{2}\right)^2 - \frac{1}{4} - \frac{1}{4} = 0$$

$$\Rightarrow \left(u - \frac{1}{2}\right)^2 + \left(v - \frac{1}{2}\right)^2 - \frac{2}{4} = 0$$

$$\Rightarrow \left(u - \frac{1}{2}\right)^2 + \left(v - \frac{1}{2}\right)^2 - \frac{1}{2} = 0$$

$$\Rightarrow \left(u - \frac{1}{2}\right)^2 + \left(v - \frac{1}{2}\right)^2 = \frac{1}{2}$$

$$\Rightarrow \left(u - \frac{1}{2}\right)^2 + \left(v - \frac{1}{2}\right)^2 = \left(\frac{1}{\sqrt{2}}\right)^2 \text{-----}(xi)$$

The equation (xi) represents an equation of a circle whose centre $C\left(\frac{1}{2}, \frac{1}{2}\right)$ and

Radius = $\frac{1}{\sqrt{2}}$

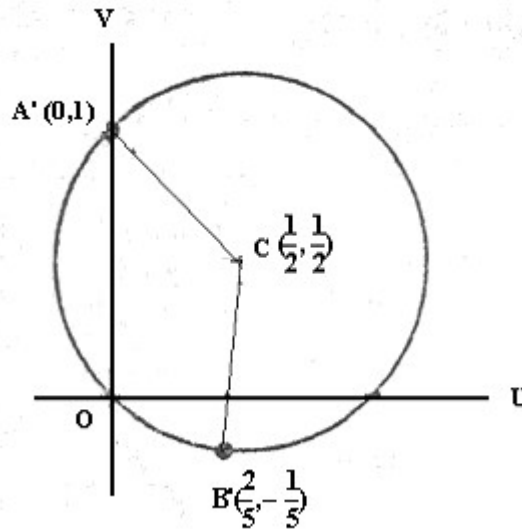


Figure 29

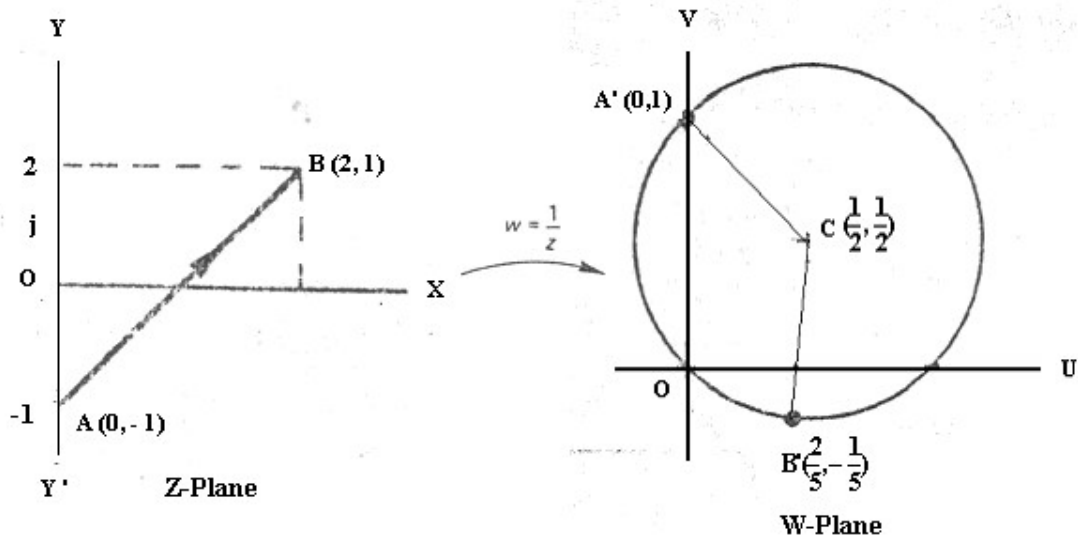


Figure 30

Justification of radius of the circle:

We have $A'(w = 0 + j.1)$ that is the coordinate of $A'(0,1)$ and the center $C\left(\frac{1}{2}, \frac{1}{2}\right)$

$$\therefore A'C = \sqrt{(u_1 - u_2)^2 + (v_1 - v_2)^2}$$

$$\therefore A'C = \sqrt{(0 - \frac{1}{2})^2 + (1 - \frac{1}{2})^2}$$

$$\therefore A'C = \sqrt{(\frac{1}{2})^2 + (\frac{1}{2})^2}$$

$$\therefore A'C = \sqrt{\frac{1}{4} + \frac{1}{4}}$$

$$\therefore A'C = \sqrt{\frac{2}{4}}$$

$$\therefore A'C = \sqrt{\frac{1}{2}}$$

$$\therefore A'C = \frac{1}{\sqrt{2}} \text{ (Proved)}$$

$$\therefore \text{Radius} = \frac{1}{\sqrt{2}}$$

We have $B'(\frac{2}{5}, -\frac{1}{5})$ and $C(\frac{1}{2}, \frac{1}{2})$

$$\therefore B'C = \sqrt{(u_1 - u_2)^2 + (v_1 - v_2)^2}$$

$$B'C = \sqrt{(\frac{2}{5} - \frac{1}{2})^2 + (-\frac{1}{5} - \frac{1}{2})^2}$$

$$B'C = \sqrt{(\frac{4-5}{10})^2 + (\frac{-2-5}{10})^2}$$

$$B'C = \sqrt{(\frac{-1}{10})^2 + (\frac{-7}{10})^2}$$

$$B'C = \sqrt{\frac{1}{100} + \frac{49}{100}}$$

$$B'C = \sqrt{\frac{50}{100}}$$

$$B'C = \sqrt{\frac{1}{2}}$$

$$\therefore B'C = \frac{1}{\sqrt{2}} \text{ (Proved)}$$

$$\therefore \text{Radius} = \frac{1}{\sqrt{2}}$$

Example 12

A circle in the z -plane has its centre at $z = 3$ and a radius of 2 units. Determine its image in the w -plane when transformation by $w = \frac{1}{z}$

Where c is the circle $|z - 3| = 2$

We have,

$$z = x + jy$$

$$z - 3 = x + jy - 3$$

$$z - 3 = x - 3 + jy$$

$$\therefore |z - 3| = \sqrt{(x - 3)^2 + y^2}$$

Given,

$$|z - 3| = 2$$

$$\therefore |z - 3| = \sqrt{(x - 3)^2 + y^2} = 2$$

$$\therefore \sqrt{(x - 3)^2 + y^2} = 2$$

$$\therefore (x - 3)^2 + y^2 = 2^2$$

$$\therefore (x - 3)^2 + (y - 0)^2 = 2^2$$

This is the equation of the circle whose centre (3,0) and radius 2

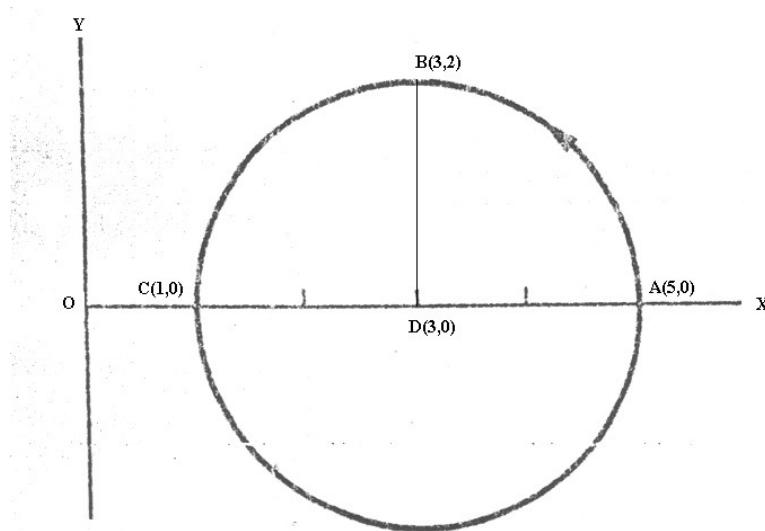


Figure 31

Solution:

Given,

$$z = 3$$

$$\Rightarrow x + jy = 3$$

$$\Rightarrow x + jy = 3 + 0.j \text{ -----(i)}$$

Equating the coefficient of real and imaginary part, we get,

$$x = 3, \quad y = 0$$

Hence, we can write,

$$(x, y) = (3, 0)$$

and given, radius = 2

So, Equation of the circle

$$(x - 3)^2 + (y - 0)^2 = 2^2 \quad \text{-----(ii)}$$

$$[\because (x - a)^2 + (y - b)^2 = r^2]$$

That is centre of the circle is (3,0) and radius 2

$$(x - 3)^2 + y^2 = 4$$

$$\Rightarrow x^2 - 6x + 9 + y^2 = 4$$

$$\Rightarrow x^2 + y^2 - 6x + 5 = 0 \quad \text{----- (iii)}$$

Again, Given,

$$w = \frac{1}{z}$$

$$w = \frac{1}{x + jy} \quad [z = x + jy]$$

$$w = \frac{x - jy}{(x + jy)(x - jy)} \quad [\text{Multiplying by } x - jy]$$

$$w = \frac{x - jy}{x^2 - jxy + jxy - j^2 y^2}$$

$$w = \frac{x - jy}{x^2 + y^2} \quad [\because j^2 = -1]$$

$$u + jv = \frac{x - jy}{x^2 + y^2} \quad [w = u + jv]$$

$$u + jv = \frac{x}{x^2 + y^2} - j \frac{y}{x^2 + y^2} \quad \text{-----(iv)}$$

Equating the coefficient of real and imaginary part, we get,

$$u = \frac{x}{x^2 + y^2} \quad \text{----- (v)}$$

$$v = \frac{-y}{x^2 + y^2} \quad \text{----- (vi)}$$

Again, Given

$$w = \frac{1}{z}$$

$$\therefore z = \frac{1}{w}$$

$$z = \frac{1}{u + jv} \quad [w = u + jv]$$

$$z = \frac{u - jv}{(u + jv)(u - jv)}$$

$$z = \frac{u - jv}{u^2 - (jv)^2}$$

$$z = \frac{u - jv}{u^2 + v^2} \quad [\because j^2 = -1]$$

$$x + jy = \frac{u - jv}{u^2 + v^2} \quad [z = x + jy]$$

$$\text{i.e. } x + jy = \frac{u}{u^2 + v^2} - j \frac{v}{u^2 + v^2} \text{----- (vii)}$$

Equating the coefficient of real and imaginary part, we get,

$$x = \frac{u}{u^2 + v^2} ; \quad y = \frac{-v}{u^2 + v^2} \text{----- (viii)}$$

Substituting the values of x and y in (iii),

$$x^2 + y^2 - 6x + 5 = 0$$

$$\Rightarrow \left(\frac{u}{u^2 + v^2} \right)^2 + \left(\frac{-v}{u^2 + v^2} \right)^2 - 6 \left(\frac{u}{u^2 + v^2} \right) + 5 = 0$$

$$\Rightarrow \frac{u^2}{(u^2 + v^2)^2} + \frac{v^2}{(u^2 + v^2)^2} - \frac{6u}{u^2 + v^2} + 5 = 0$$

$$\Rightarrow \frac{u^2 + v^2}{(u^2 + v^2)^2} - \frac{6u}{u^2 + v^2} + 5 = 0$$

$$\Rightarrow \frac{1}{u^2 + v^2} - \frac{6u}{u^2 + v^2} + 5 = 0$$

$$\Rightarrow \frac{1 - 6u + 5(u^2 + v^2)}{u^2 + v^2} = 0$$

$$\Rightarrow 5(u^2 + v^2) - 6u + 1 = 0$$

$$\Rightarrow u^2 + v^2 - \frac{6}{5}u + \frac{1}{5} = 0 \text{ [dividing by 5]}$$

$$\Rightarrow u^2 + v^2 - 2 \cdot \frac{3}{5} \cdot u + 2 \cdot 0 \cdot v + \frac{1}{5} = 0$$

$$\Rightarrow u^2 + v^2 + 2 \cdot \left(-\frac{3}{5}\right) \cdot u + 2 \cdot 0 \cdot v + \frac{1}{5} = 0 \text{----- (ix)}$$

We know the general equation of the circle is

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Whose centre is $(-g, -f)$ and radius is $\sqrt{g^2 + f^2 - c}$

Hence from (ix)

Here $g = -\frac{3}{5}$, $f = 0$ and $c = \frac{1}{5}$

The centre of the new circle of (ix) is $(-g, -f) = (-(-\frac{3}{5}), -0) = (\frac{3}{5}, 0)$

That is, centre of new circle in the w-plane is, $D(\frac{3}{5}, 0)$ [From figure 32]

$$\begin{aligned} \text{Radius is } \sqrt{g^2 + f^2 - c} &= \sqrt{(-\frac{3}{5})^2 + 0^2 - \frac{1}{5}} \\ &= \sqrt{(-\frac{3}{5})^2 + 0^2 - \frac{1}{5}} = \sqrt{\frac{9}{25} + 0 - \frac{1}{5}} \\ &= \sqrt{\frac{9}{25} - \frac{1}{5}} \\ &= \sqrt{\frac{9-5}{25}} \\ &= \sqrt{\frac{4}{25}} \\ &= \frac{2}{5} \end{aligned}$$

That is, radius of new circle in the w-plane is, $\frac{2}{5}$

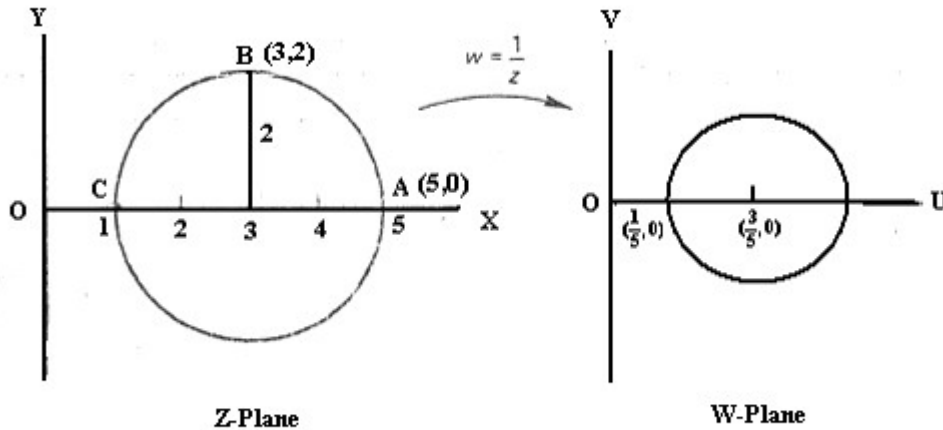


Figure 32

Taking three sample points A, B, C as shown, that is:

A(5,0), B(3,2), C(1,0)

Putting the values of **A(5,0), B(3,2), C(1,0)** in (v) and (vi)

We have,

$$u = \frac{x}{x^2 + y^2} \quad v = \frac{-y}{x^2 + y^2}$$

$$\text{For } A(5,0); u = \frac{x}{x^2 + y^2} = \frac{5}{5^2 + 0^2} = \frac{5}{25 + 0} = \frac{5}{25} = \frac{1}{5}$$

$$\text{For } A(5,0); v = \frac{-y}{x^2 + y^2} = \frac{-0}{5^2 + 0^2} = \frac{0}{25 + 0} = 0$$

$$\therefore \text{For } A(5,0); w = u + jv = \frac{1}{5} + j.0$$

$$\text{The image of A is } A'(w = u + jv = \frac{1}{5} + j.0) = \frac{1}{5} + j.0$$

$$\text{That is } A'(\frac{1}{5}, 0) \quad \text{-----}(x)$$

$$\text{For } B(3,2); u = \frac{x}{x^2 + y^2} = \frac{3}{3^2 + 2^2} = \frac{3}{9 + 4} = \frac{3}{13}$$

$$\text{For } B(3,2); v = \frac{-y}{x^2 + y^2} = \frac{-2}{3^2 + 2^2} = \frac{-2}{9 + 4} = \frac{-2}{13}$$

$$\therefore \text{For } B(3,2); w = u + jv = \frac{3}{13} + j.(\frac{-2}{13})$$

$$\text{The image of B is } B'(w = u + jv = \frac{3}{13} + j.(\frac{-2}{13})) = \frac{3}{13} - j. \frac{2}{13}$$

$$\text{That is } B'(\frac{3}{13}, -\frac{2}{13}) \quad \text{-----}(xi)$$

$$\text{For } C(1,0); u = \frac{x}{x^2 + y^2} = \frac{1}{1^2 + 0^2} = \frac{1}{1} = 1$$

$$\text{For } C(1,0); v = \frac{-y}{x^2 + y^2} = \frac{-0}{1^2 + 0^2} = \frac{-0}{1} = 0$$

$$\therefore \text{For } C(1,0); w = u + jv = 1 + j.(0)$$

$$\text{The image of C is } C'(w = u + jv = 1 + j.0 = 1 + j.0$$

$$\text{That is } C'(1,0) \quad \text{-----}(xii)$$

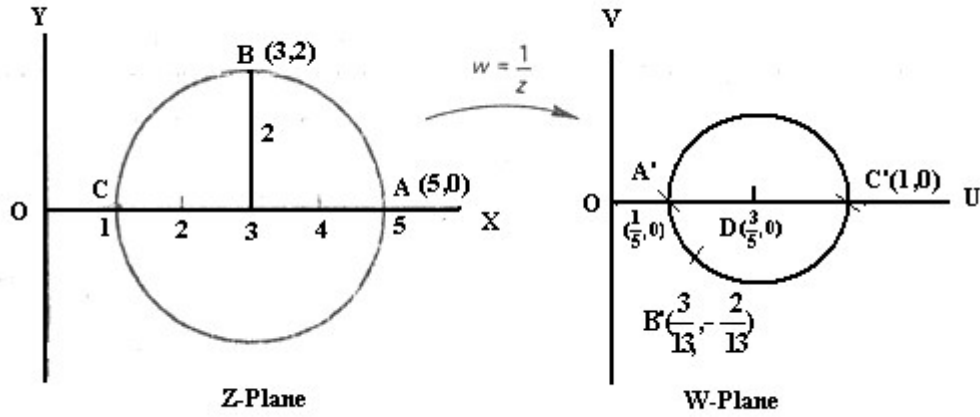


Figure 33

Justification:

We have $A'(\frac{1}{5}, 0), B'(\frac{3}{13}, -\frac{2}{13}), C'(1, 0), D(\frac{3}{5}, 0)$

Radius = $\frac{2}{5}$

We know the length of a line between two points (x_1, y_1) and (x_2, y_2) is:

$$\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

Here, in the w-plane

$$\therefore A'D = \sqrt{(u_1 - u_2)^2 + (v_1 - v_2)^2}$$

$$\therefore A'D = \sqrt{(\frac{1}{5} - \frac{3}{5})^2 + (0 - 0)^2}$$

$$\therefore A'D = \sqrt{(-\frac{2}{5})^2}$$

$$\therefore A'D = \sqrt{\frac{4}{25}}$$

$$\therefore A'D = \frac{2}{5} \text{ (Proved)}$$

Again,

$$\therefore B'D = \sqrt{(u_1 - u_2)^2 + (v_1 - v_2)^2}$$

$$B'D = \sqrt{(\frac{3}{13} - \frac{3}{5})^2 + (-\frac{2}{13} - 0)^2}$$

$$B'D = \sqrt{(\frac{15 - 39}{65})^2 + (\frac{4}{169})}$$

$$B'D = \sqrt{\left(\frac{-24}{65}\right)^2 + \frac{4}{169}}$$

$$B'D = \sqrt{\frac{576}{4225} + \frac{4}{169}}$$

$$B'D = \sqrt{\frac{114244}{714025}}$$

$$B'D = \frac{2}{5} \quad (\text{Proved})$$

$$\therefore C'D = \sqrt{(u_1 - u_2)^2 + (v_1 - v_2)^2}$$

$$C'D = \sqrt{\left(1 - \frac{3}{5}\right)^2 + (0 - 0)^2}$$

$$C'D = \sqrt{\left(\frac{5-3}{5}\right)^2}$$

$$C'D = \sqrt{\left(\frac{2}{5}\right)^2}$$

$$C'D = \frac{2}{5} \quad (\text{Proved})$$

Example 13

A circle $|z| = 1$ in the Z-plane is mapped onto the W-plane by $w = \frac{1}{z-2}$

Solution: from figure 34

$$OP = |z| = \sqrt{x^2 + y^2}$$

Given,

$$|z| = 1$$

$$\sqrt{x^2 + y^2} = 1$$

$$\therefore x^2 + y^2 = 1$$

$$(x-0)^2 + (y-0)^2 = 1^2 \quad \text{------(i)}$$

[We have, $(x-a)^2 + (y-b)^2 = r^2$]

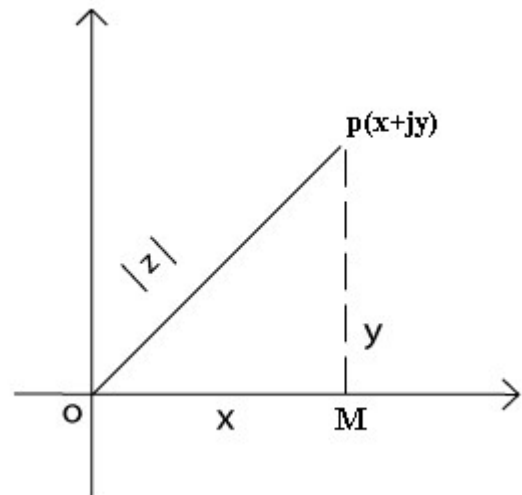
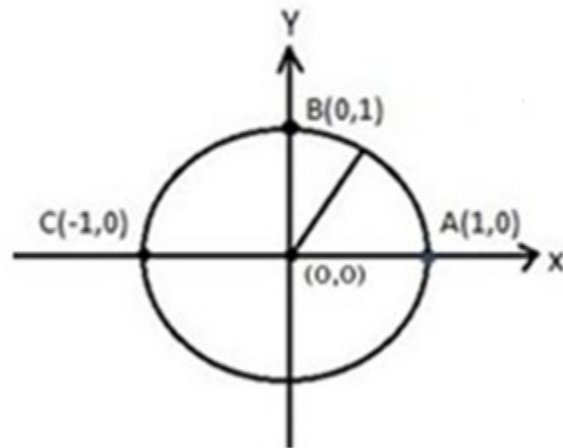


Figure 34

Which is the equation of a circle whose Center (0, 0), Radius=1



Z-Plane

Figure 35

From (i)

$$(x-0)^2 + (y-0)^2 = 1^2$$

$$x^2 + y^2 = 1$$

$$x^2 + y^2 - 1 = 0$$

------(ii)

Given,

$$w = \frac{1}{z-2}$$

$$z-2 = \frac{1}{w}$$

$$\therefore z = \frac{1}{w} + 2$$

i.e.

$$x + jy = \frac{1}{w} + 2$$

$$[z = x + jy]$$

$$x + jy = \frac{1}{u + jv} + 2$$

$$[w = u + jv]$$

$$x + jy = \frac{u - jv}{(u + jv)(u - jv)} + 2$$

[Multiplying by $u - jv$]

$$x + jy = \frac{u - jv}{u^2 - (jv)^2} + 2$$

$$x + jy = \frac{u - jv}{u^2 - j^2 v^2} + 2$$

$$x + jy = \frac{u - jv}{u^2 + v^2} + 2$$

$$[j^2 = -1]$$

$$\therefore x - 2 + jy = \frac{u - jv}{u^2 + v^2}$$

$$\therefore x - 2 + jy = \frac{u}{u^2 + v^2} - j \frac{v}{u^2 + v^2} \text{-----(iii)}$$

Equating the co-efficient of real and imaginary part on both sides, we get,

$$x - 2 = \frac{u}{u^2 + v^2} ; y = \frac{-v}{u^2 + v^2}$$

$$x = \frac{u}{u^2 + v^2} + 2 ; y = \frac{-v}{u^2 + v^2} \text{-----(iv)}$$

Substituting these values x and y in equation (ii);

$$x^2 + y^2 - 1 = 0$$

$$\left(\frac{u}{u^2 + v^2} + 2 \right)^2 + \left(\frac{-v}{u^2 + v^2} \right)^2 - 1 = 0$$

$$\left\{ \frac{u + 2(u^2 + v^2)}{u^2 + v^2} \right\}^2 + \frac{v^2}{(u^2 + v^2)^2} = 1$$

$$\frac{\{u + 2(u^2 + v^2)\}^2}{(u^2 + v^2)^2} + \frac{v^2}{(u^2 + v^2)^2} = 1$$

$$\frac{\{u + 2(u^2 + v^2)\}^2 + v^2}{(u^2 + v^2)^2} = 1$$

$$\{u + 2(u^2 + v^2)\}^2 + v^2 = (u^2 + v^2)^2$$

$$u^2 + 2 \times u \times 2(u^2 + v^2) + \{2(u^2 + v^2)\}^2 + v^2 = (u^2 + v^2)^2$$

$$u^2 + 4u(u^2 + v^2) + \{2(u^2 + v^2)\}^2 + v^2 = (u^2 + v^2)^2$$

$$u^2 + v^2 + 4u(u^2 + v^2) + \{2(u^2 + v^2)\}^2 = (u^2 + v^2)^2$$

$$(u^2 + v^2) + 4u(u^2 + v^2) + \{2(u^2 + v^2)\}^2 = (u^2 + v^2)^2$$

$$(u^2 + v^2) + 4u(u^2 + v^2) + 4(u^2 + v^2)^2 = (u^2 + v^2)^2$$

$$1 + 4u + 4(u^2 + v^2) = u^2 + v^2$$

[Dividing by $u^2 + v^2$]

$$1 + 4u + 4(u^2 + v^2) - (u^2 + v^2) = 0$$

$$1 + 4u + 3(u^2 + v^2) = 0$$

$$3(u^2 + v^2) + 4u + 1 = 0$$

$$(u^2 + v^2) + \frac{4}{3}u + \frac{1}{3} = 0$$

$$u^2 + \frac{4}{3}u + v^2 + \frac{1}{3} = 0$$

$$u^2 + \frac{4}{3}u + v^2 + \frac{1}{3} = 0$$

$$u^2 + v^2 + \frac{4}{3}u + \frac{1}{3} = 0$$

$$u^2 + v^2 + 2 \cdot \frac{2}{3} \cdot u + 2 \cdot 0 \cdot v + \frac{1}{3} = 0 \text{ -----(v)}$$

We know the general equation of the circle is

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Whose centre is $(-g, -f)$ and radius is $\sqrt{g^2 + f^2 - c}$

Hence from (v)

$$\text{Here } g = \frac{2}{3}, f = 0 \text{ and } c = \frac{1}{3}$$

The centre of new circle in w-plane is $(-g, -f) = ((-\frac{2}{3}), -0) = (-\frac{2}{3}, 0)$

$$\text{Radius is } \sqrt{g^2 + f^2 - c} = \sqrt{(\frac{2}{3})^2 + 0^2 - \frac{1}{3}}$$

$$= \sqrt{\frac{4}{9} - \frac{1}{3}}$$

$$= \sqrt{\frac{4-3}{9}}$$

$$= \sqrt{\frac{1}{9}}$$

$$= \frac{1}{3}$$

Now, we have to find out the image point $A', B' \text{ and } C'$

Taking three sample points from figure 35, $A(1,0)$, $B(0,1)$ & $C(-1,0)$

Given,

$$w = \frac{1}{z-2}$$

$$w = \frac{1}{x+jy-2}$$

$$[z = x + jy]$$

$$= \frac{1}{x-2+jy}$$

$$= \frac{x-2-jy}{(x-2+jy)(x-2-jy)}$$

[Multiplying by $x-2-jy$]

$$= \frac{x-2-jy}{(x-2)^2 - (jy)^2}$$

$$= \frac{x-2-jy}{(x-2)^2 + y^2}$$

$$[[j^2 = -1; a^2 - b^2 = (a+b)(a-b)]]$$

$$w = \frac{x-2-jy}{(x-2)^2 + y^2}$$

$$w = \frac{x-2}{(x-2)^2 + y^2} - j \frac{y}{(x-2)^2 + y^2}$$

$$u + jv = \frac{x-2}{(x-2)^2 + y^2} - j \frac{y}{(x-2)^2 + y^2} \quad [w = u + jv]$$

$$u + jv = \frac{x-2}{(x-2)^2 + y^2} - j \frac{y}{(x-2)^2 + y^2} \quad \text{-----(vi)}$$

Equating the co-efficient of real and imaginary part from (vi), we get,

$$u = \frac{x-2}{(x-2)^2 + y^2} \quad \text{-----(vii)}$$

$$v = \frac{-y}{(x-2)^2 + y^2} \quad \text{-----(viii)}$$

For $A(1,0)$: we get from vii & viii

$$u = \frac{x-2}{(x-2)^2 + y^2} = \frac{1-2}{(1-2)^2 + 0^2} = -\frac{1}{1} = -1$$

$$v = \frac{-y}{(x-2)^2 + y^2} = \frac{-0}{(1-2)^2 + 0^2} = 0$$

$\therefore A'(w = u + jv = -1 + j.0)$ is the image of A.

That is $A'(-1,0)$ -----(ix)

For $B(0,1)$: we get from vii & viii

$$u = \frac{x-2}{(x-2)^2 + y^2} = \frac{0-2}{(0-2)^2 + 1^2} = -\frac{2}{5}$$

$$v = \frac{-y}{(x-2)^2 + y^2} = \frac{-1}{(0-2)^2 + 1^2} = -\frac{1}{5}$$

$\therefore B'(w = u + jv = -\frac{2}{5} - \frac{1}{5}j)$ is the image of B.

That is $B'(-\frac{2}{5}, -\frac{1}{5})$ -----(x)

For $C(-1,0)$: we get from vii & viii

$$u = \frac{x-2}{(x-2)^2 + y^2} = \frac{-1-2}{(-1-2)^2 + 0^2} = -\frac{3}{9} = -\frac{1}{3}$$

$$v = \frac{-y}{(x-2)^2 + y^2} = \frac{-0}{(-1-2)^2 + 0^2} = 0$$

$\therefore C'(w = u + jv = -\frac{1}{3} + j.0)$ is the image of C.

That is $C'(-\frac{1}{3}, 0)$ -----(xi)

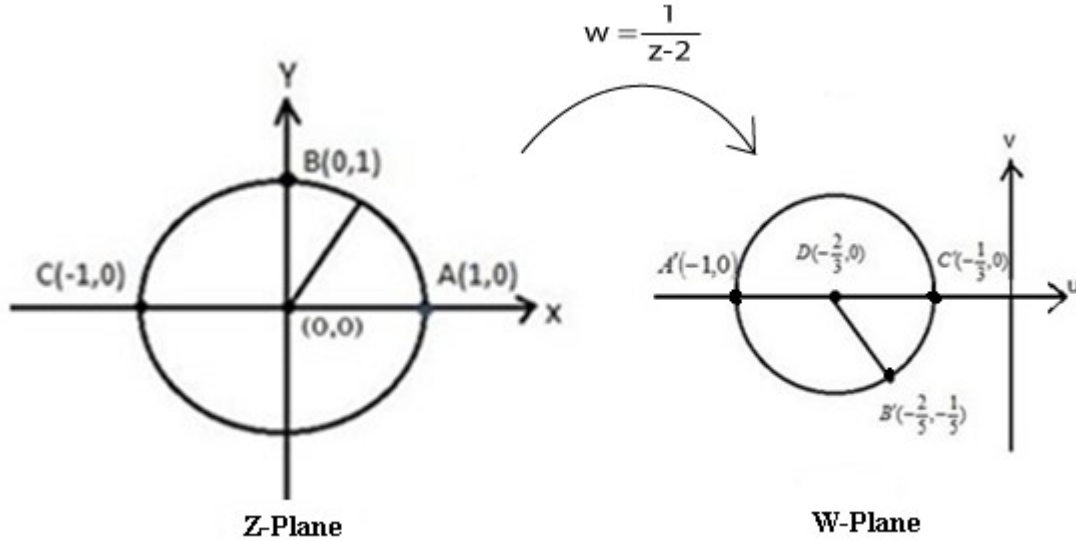


Figure 36

Justification:

We have, $A'(-1,0)$ $B'(-\frac{2}{5}, -\frac{1}{5})$ $C'(-\frac{1}{3}, 0)$ and the radius & centre of the new circle (v)

is Radius = $\frac{1}{3}$ and centre $D(-\frac{2}{3}, 0)$

We know the length of a line between two points (x_1, y_1) and (x_2, y_2) is:

$$\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

Here, in the w-plane

$$\begin{aligned} \therefore A'D &= \sqrt{(u_1 - u_2)^2 + (v_1 - v_2)^2} \\ &= \sqrt{\left(-1 - \left(-\frac{2}{3}\right)\right)^2 + (0 - 0)^2} \\ &= \sqrt{\left(-1 + \frac{2}{3}\right)^2 + 0} = \sqrt{\left(\frac{-3+2}{3}\right)^2} \\ &= \sqrt{\left(\frac{-1}{3}\right)^2} = \sqrt{\frac{1}{9}} = \frac{1}{3} \end{aligned}$$

$$\begin{aligned} \therefore B'D &= \sqrt{(u_1 - u_2)^2 + (v_1 - v_2)^2} \\ &= \sqrt{\left(-\frac{2}{5} - \left(-\frac{2}{3}\right)\right)^2 + \left(-\frac{1}{5} - 0\right)^2} \\ &= \sqrt{\left(-\frac{2}{5} + \frac{2}{3}\right)^2 + \left(-\frac{1}{5} + 0\right)^2} \end{aligned}$$

$$\begin{aligned}
&= \sqrt{\left(\frac{-6+10}{15}\right)^2 + \frac{1}{25}} \\
&= \sqrt{\frac{16}{225} + \frac{1}{25}} \\
&= \sqrt{\frac{16+9}{225}} = \sqrt{\frac{25}{225}} = \sqrt{\frac{1}{9}} = \frac{1}{3}
\end{aligned}$$

$$\begin{aligned}
\therefore C'D &= \sqrt{(u_1 - u_2)^2 + (v_1 - v_2)^2} \\
&= \sqrt{\left(-\frac{1}{3} - \left(-\frac{2}{3}\right)\right)^2 + (0-0)^2} \\
&= \sqrt{\left(-\frac{1}{3} + \frac{2}{3}\right)^2 + (0-0)^2} \\
&= \sqrt{\left(\frac{-1+2}{3}\right)^2} = \sqrt{\frac{1}{9}} = \frac{1}{3}
\end{aligned}$$

Example 14: Determine the image in w-plan of circle $|z| = 2$ in the z-plane under the transformation $w = \frac{1}{z-2}$ and show the region in w-plan onto which the region within the circle is mapped.

Answer: from figure 37

$$OP = |z| = \sqrt{x^2 + y^2}$$

Given,

$$|z| = 2$$

$$\sqrt{x^2 + y^2} = 2$$

$$x^2 + y^2 = 2^2 \quad [\text{Squaring}]$$

$$\therefore x^2 + y^2 = 4$$

$$(x-0)^2 + (y-0)^2 = 2^2 \text{-----(i)}$$

$$[\text{We have, } (x-a)^2 + (y-b)^2 = r^2]$$

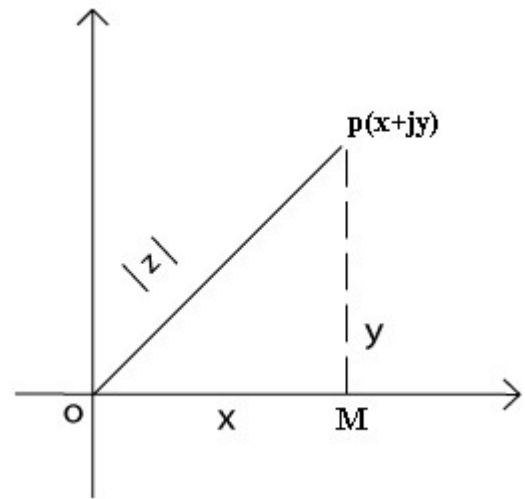
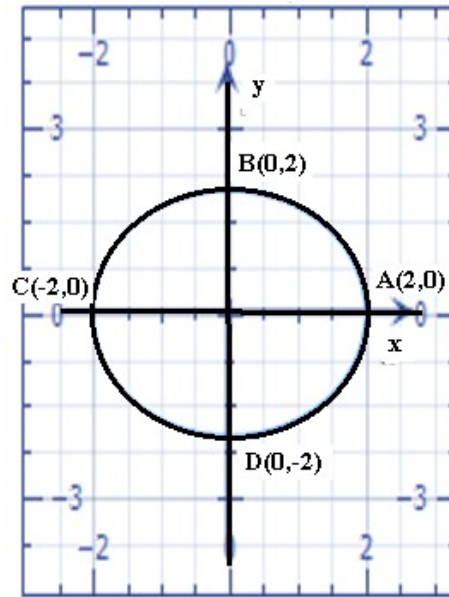


Figure no 37

The equation (i) represents a circle whose Center (0, 0) & radius = 2



Z-Plane

Figure no 38

Again, from (i)

$$(x-0)^2 + (y-0)^2 = 2^2$$

$$x^2 + y^2 = 4$$

$$x^2 + y^2 - 4 = 0 \quad \text{-----(ii)}$$

Given,

$$w = \frac{1}{z-2}$$

$$\Rightarrow z-2 = \frac{1}{w}$$

$$\Rightarrow z = \frac{1}{w} + 2$$

$$\Rightarrow x + jy = \frac{1}{u + jv} + 2$$

$$[z = x + jy] \text{ \& } [w = u + jv]$$

$$\Rightarrow x + jy = \frac{u - jv}{(u + jv)(u - jv)} + 2$$

$$\Rightarrow x + jy = \frac{u - jv}{u^2 - (jv)^2} + 2$$

$$\Rightarrow x + jy = \frac{u - jv}{u^2 - j^2 v^2} + 2$$

$$\Rightarrow x + jy = \frac{u - jv}{u^2 + v^2} + 2$$

$$[\because j^2 = -1]$$

$$\Rightarrow x + jy - 2 = \frac{u - jv}{u^2 + v^2}$$

$$\Rightarrow x + jy - 2 = \frac{u}{u^2 + v^2} - j \frac{v}{u^2 + v^2}$$

$$\therefore x - 2 + jy = \frac{u}{u^2 + v^2} - j \frac{v}{u^2 + v^2} \quad \text{-----(iii)}$$

Now, equating real and imaginary part,

$$x - 2 = \frac{u}{u^2 + v^2}$$

$$\therefore x = \frac{u}{u^2 + v^2} + 2 \quad \text{-----(iv)}$$

&

$$y = -\frac{v}{u^2 + v^2} \quad \text{-----(v)}$$

Substituting these values x and y in equation (ii)

$$x^2 + y^2 - 4 = 0$$

$$\Rightarrow \left(\frac{u}{u^2 + v^2} + 2\right)^2 + \left(-\frac{v}{u^2 + v^2}\right)^2 - 4 = 0$$

$$\Rightarrow \left[\frac{u + 2(u^2 + v^2)}{u^2 + v^2}\right]^2 + \frac{v^2}{(u^2 + v^2)^2} - 4 = 0$$

$$\Rightarrow \frac{\{u + 2(u^2 + v^2)\}^2}{(u^2 + v^2)^2} + \frac{v^2}{(u^2 + v^2)^2} = 4$$

$$\Rightarrow \frac{\{u + 2(u^2 + v^2)\}^2 + v^2}{(u^2 + v^2)^2} = 4$$

$$\Rightarrow \{u + 2(u^2 + v^2)\}^2 + v^2 = 4(u^2 + v^2)^2$$

$$\Rightarrow u^2 + 2 \times u \times 2(u^2 + v^2) + \{2(u^2 + v^2)\}^2 + v^2 = 4(u^2 + v^2)^2$$

$$\Rightarrow u^2 + 4u(u^2 + v^2) + 4(u^2 + v^2)^2 + v^2 = 4(u^2 + v^2)^2$$

$$\Rightarrow (u^2 + v^2) + 4u(u^2 + v^2) + 4(u^2 + v^2)^2 = 4(u^2 + v^2)^2$$

$$\Rightarrow 1 + 4u + 4(u^2 + v^2) = 4(u^2 + v^2)$$

$$\Rightarrow 1 + 4u + 4(u^2 + v^2) - 4(u^2 + v^2) = 0$$

$$\Rightarrow 1 + 4u = 0$$

$$\Rightarrow 4u = -1$$

$$\therefore u = -\frac{1}{4} \quad \text{-----(vi)}$$

Which is the equation of a straight line parallel to the y-axis in the w-plane

Now, we have to find out the image point A', B', C' & D' for A (2, 0), B (0, 2), C (-2, 0) & D (0, -2) from figure no 38

We have, given

$$w = \frac{1}{z-2}$$

$$w = \frac{1}{x+jy-2} \quad [z = x + jy]$$

$$= \frac{1}{x-2+jy}$$

$$= \frac{x-2-jy}{(x-2+jy)(x-2-jy)} \quad [\text{Multiplying by } x-2-jy \text{ on numerator and denominator}]$$

$$= \frac{x-2-jy}{(x-2)^2 - (jy)^2}$$

$$w = \frac{x-2-jy}{(x-2)^2 - (jy)^2}$$

$$w = \frac{x-2-jy}{(x-2)^2 + y^2} \quad [\because j^2 = -1]$$

$$\Rightarrow u + jv = \frac{x-2-jy}{(x-2)^2 + y^2} \quad [w = u + jv]$$

$$\Rightarrow u + jv = \frac{x-2}{(x-2)^2 + y^2} - j \frac{y}{(x-2)^2 + y^2} \text{-----(vii)}$$

Now equating real and imaginary part

$$u = \frac{x-2}{(x-2)^2 + y^2} \text{-----(viii)}$$

$$v = \frac{-y}{(x-2)^2 + y^2} \text{-----(ix)}$$

For A (2, 0); we get from viii & ix

$$\Rightarrow u = \frac{2-2}{(2-2)^2 + 0^2} = \frac{0}{0} \quad ; \text{undefined}$$

$$\Rightarrow v = -\frac{0}{(2-2)^2 + 0^2} = \frac{0}{0} \quad ; \text{undefined}$$

So the image of A is A' is undefined.

For B (0, 2); we get from viii & ix

$$\Rightarrow u = \frac{0-2}{(0-2)^2 + 2^2} = \frac{-2}{4+4} = -\frac{2}{8} = -\frac{1}{4}$$

$$\Rightarrow v = -\frac{2}{(0-2)^2 + 2^2} = -\frac{2}{4+4} = -\frac{2}{8} = -\frac{1}{4}$$

So the image of B is B' ($w = u + jv = -\frac{1}{4} - j\frac{1}{4}$)

$$\text{That is } B'(-\frac{1}{4}, -\frac{1}{4}) \text{-----(x)}$$

For C (-2, 0); we get from viii & ix

$$\Rightarrow u = \frac{-2-2}{(-2-2)^2 + 0^2} = -\frac{4}{16} = -\frac{1}{4}$$

$$\Rightarrow v = -\frac{0}{(-2-2)^2 + 0^2} = -\frac{0}{16} = 0$$

So the image of C is $C'(w = u + jv = -\frac{1}{4} + j.0)$

That is $C'(-\frac{1}{4}, 0)$ -----(xi)

For D (0,-2) ; we get from viii & ix

$$\Rightarrow u = \frac{0-2}{(0-2)^2 + (-2)^2} = \frac{-2}{4+4} = \frac{-2}{8} = -\frac{1}{4}$$

$$\Rightarrow v = -\frac{-2}{(0-2)^2 + (-2)^2} = \frac{2}{4+4} = \frac{2}{8} = \frac{1}{4}$$

So the image of D is $D'(w = u + jv = -\frac{1}{4} + j\frac{1}{4})$

That is $D'(-\frac{1}{4}, \frac{1}{4})$ -----(xii)

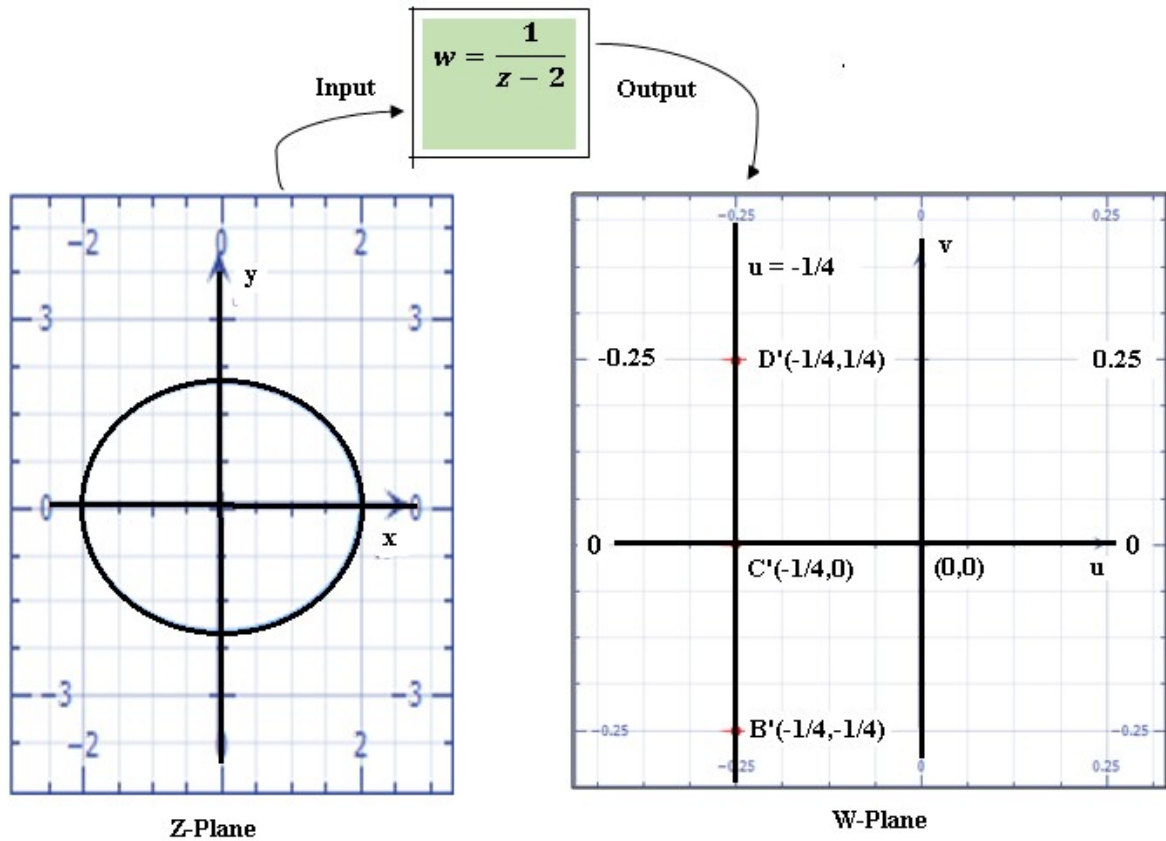
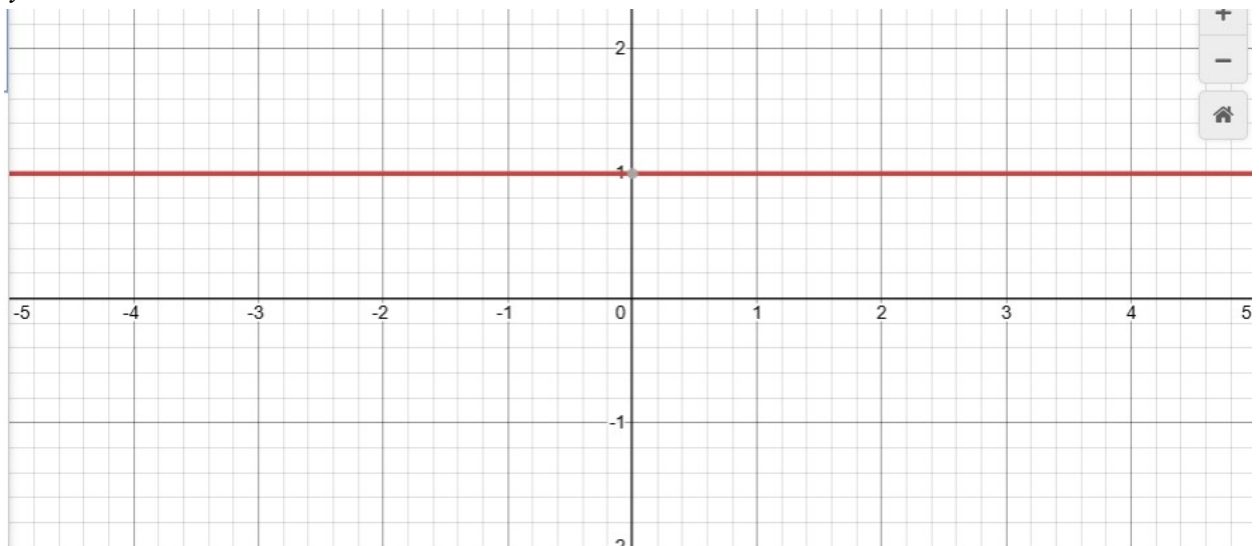


Figure 39

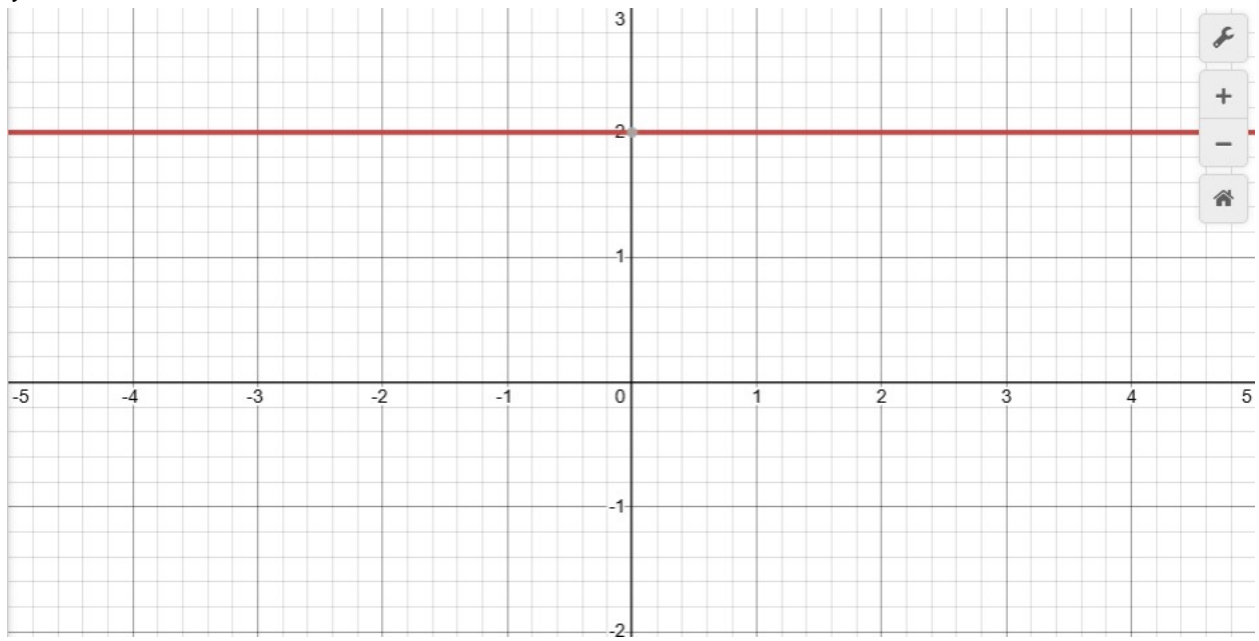
$$y = 0.x + 1$$

x	-2	-3	-1	0	1	2	3	-4	4
y	1	1	1	1	1	1	1	1	1

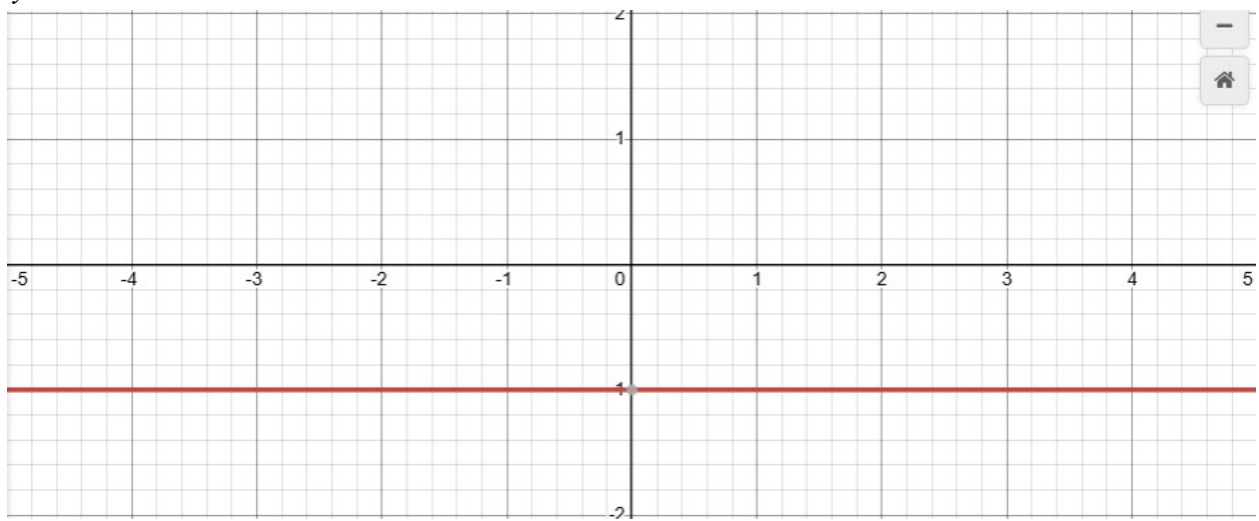
$$y = 1$$



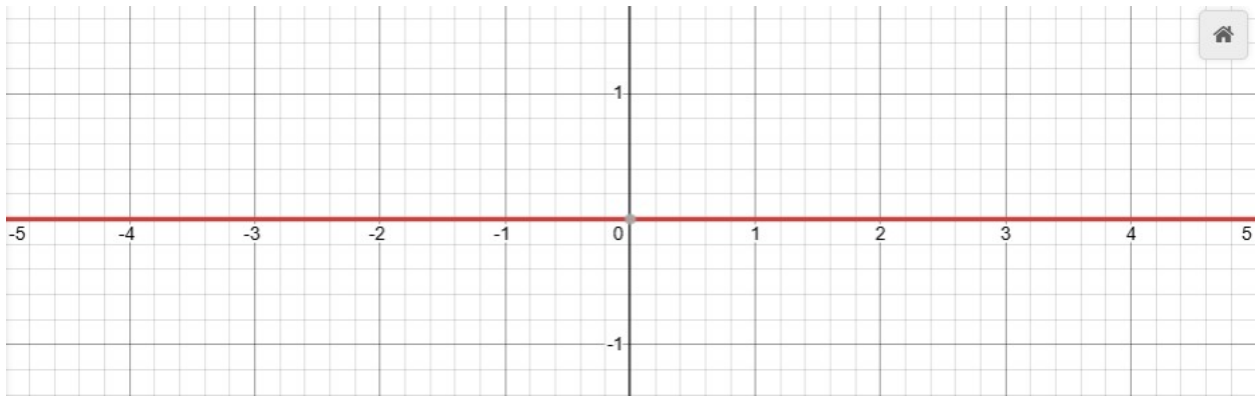
$$y = 2$$



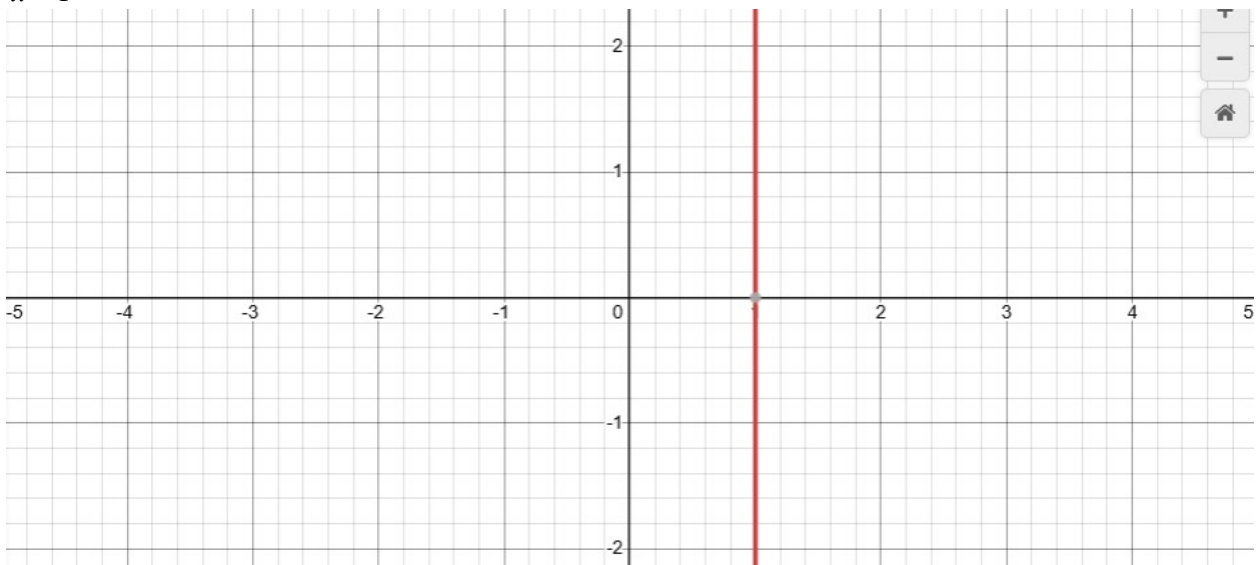
$$y = -1$$



$$y = 0$$



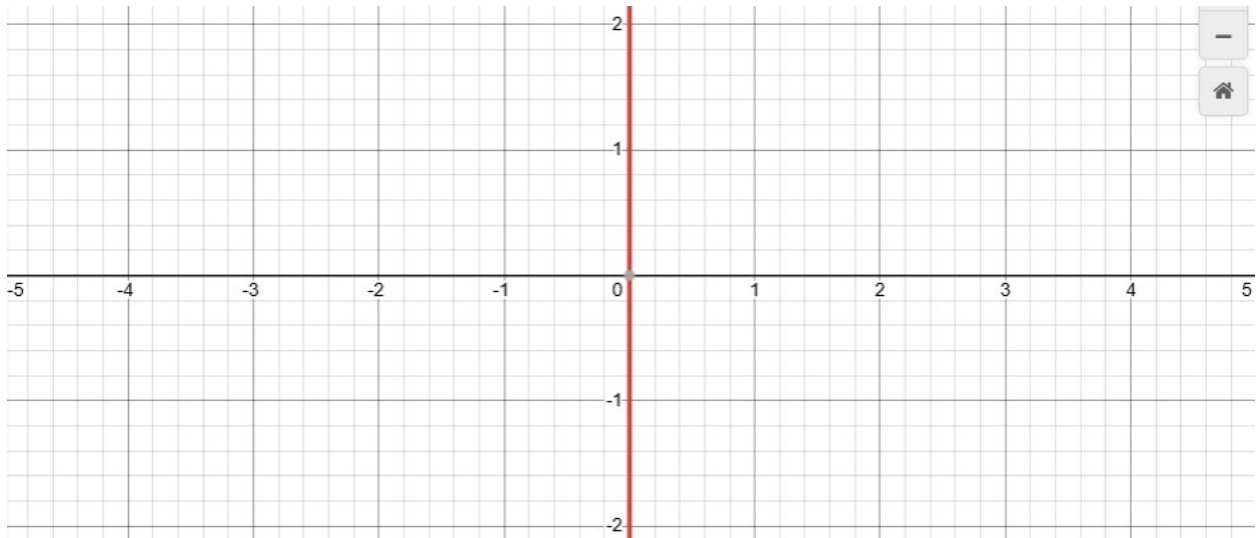
$$x = 1$$



$$x = -1$$



$$x = 0$$



Home Task:

01. Determine the image in the w-plane of the circle $|z| = 2$ in the z-plane under the transformation $w = \frac{z+j}{z-j}$ and show the region in the w-plane onto which the region within the circle is mapped.

02. Draw a graph of $y = \frac{1}{x}$

Answer:

x	-2	-1.5	-1	-0.5	-0.4	-0.2	0	0.2	0.4	0.5	1	1.5	2
$y = \frac{1}{x}$	-0.5	-0.6	-1	-2	-2.5	-5	inf	5	2.5	2	1	0.6	0.5

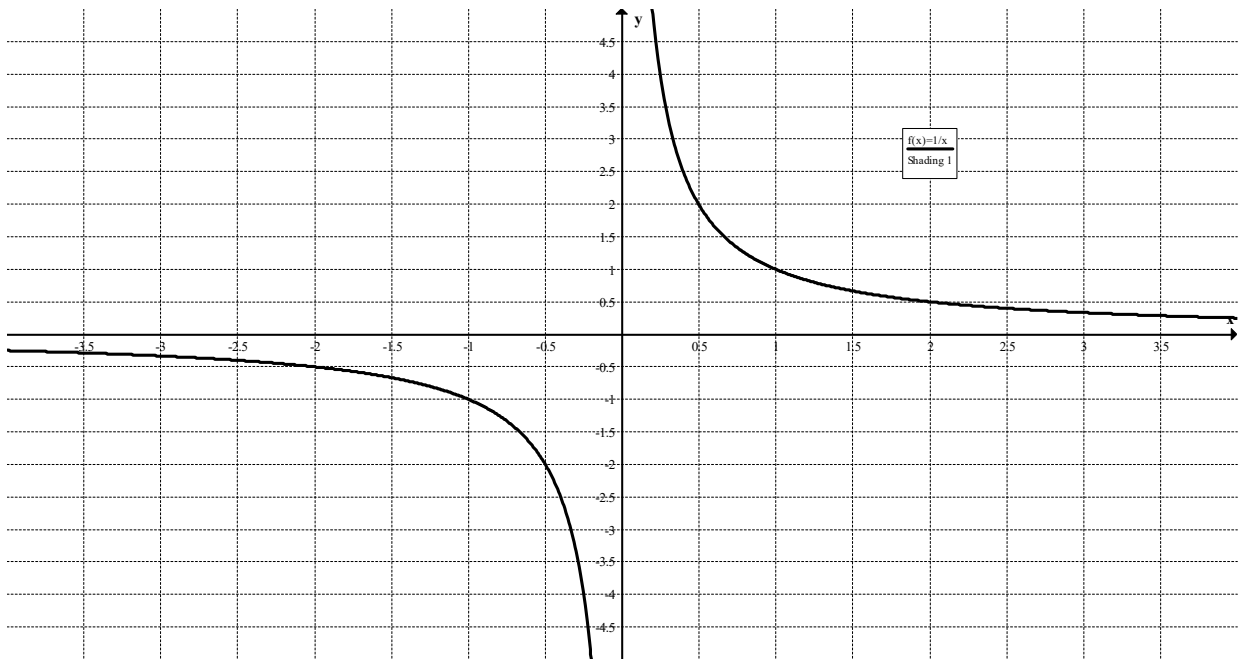


Figure 40: Graph of $y = \frac{1}{x}$

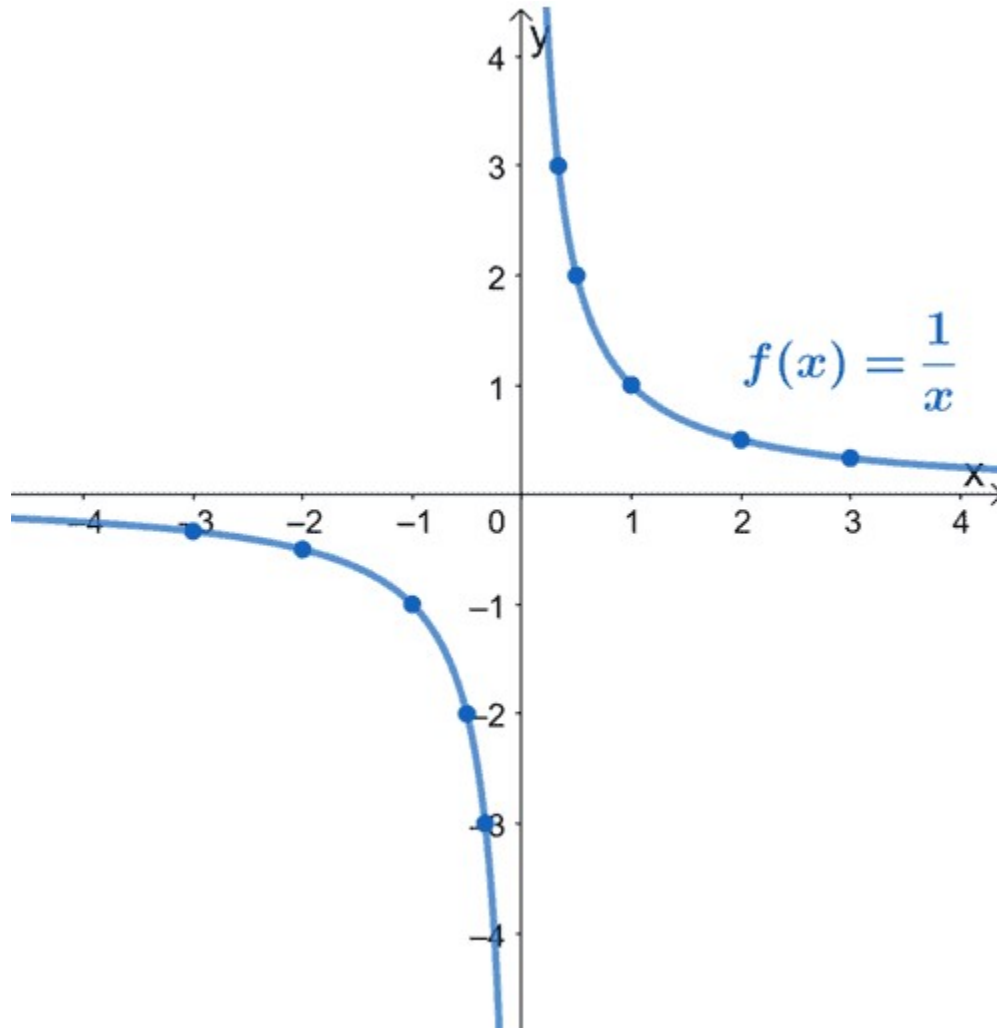


Figure 41

Regular Function/ Analytic Function:

A function $w = f(z)$ is said to be regular or analytic at a point $z = z_0$, if it is defined and single-valued and has a derivative (rate of change) at every point at and around $z = z_0$.

Singular Point:

A point at which **a function $f(z)$ is not analytic** is known as a singular point or singularity of the function.

For example, the function $\frac{1}{z-2}$ has a singular point at $z = 2$.

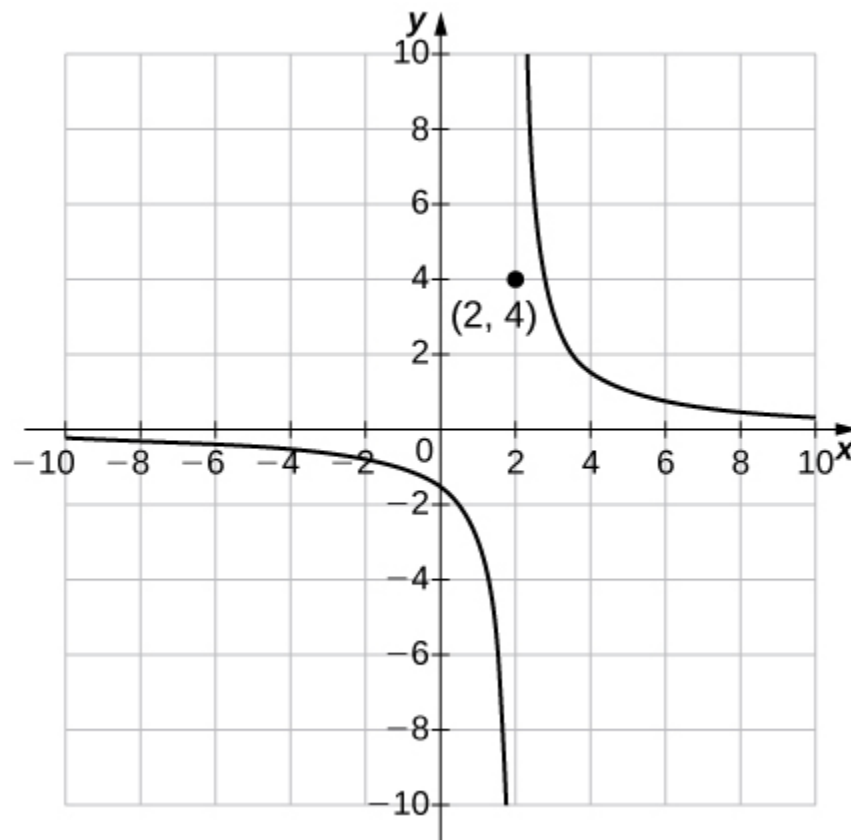
$$[\because \frac{1}{z-2} = \frac{1}{2-2} = \frac{1}{0} = \infty]$$

Points in a region where $f(z)$ ceases (বিরত থাকা) to be regular (disjoint/discontinuous/disconnected/irregular) are called singular points or singularities.

The point at which the function is not differentiable is called a singular point of the function.

A point or region of infinite mass density at which space and time are infinitely distorted by gravitational forces and which is held to be the final state of matter falling into a black hole and point at which the derivative of a given function of a complex variable does not exist but every neighborhood of which contains points for which the derivative does exist.

Mathematical singularity, a point at which a given mathematical object is not defined or not "well-behaved" for example infinite or not differentiable. In real analysis singularities are either discontinuities or discontinuities of the derivative (sometimes also discontinuities of higher order derivatives). A singularity, as most commonly used, is a point at which expected rules break down. The term comes from mathematics, where a point on a curve that has a sudden break in slope is considered to have a slope of undefined or infinite value; such a point is known as a singularity and A point at which an analytic function $f(z)$ is not analytic, i.e. at which $f'(z)$ fails to exist, is called a singular point or singularity of the function



Necessary Condition for $f(z)$ to be Analytic: The necessary and sufficient condition for a function $f(z) = u + jv$ to be analytic at all the points in a region R are:

Cauchy-Riemann Test

We have Cauchy-Riemann Equations are

$$\frac{\delta u}{\delta x} = \frac{\delta v}{\delta y} \quad \text{and} \quad \frac{\delta v}{\delta x} = -\frac{\delta u}{\delta y}$$

We said earlier that where a function fails to be regular, a singular point or singularity occurs. i.e. where $w = f(z)$ is not continuous or where the **Cauchy-Riemann Test fails**.

Determine where each of the following functions fails to be regular, i.e. where singularities occur.

As for example

$$w = f(z) = \frac{1}{(z-2)(z-3)}$$

Singularities at $z = 2$ and $z = 3$ *Answer*

Example 15

Determine the function $w = f(z) = z^2 - 4$ is analytic or not.

Answer:

Given,

$$w = f(z) = z^2 - 4$$

$$\Rightarrow u + iv = (x + iy)^2 - 4 \quad [\because w = u + iv \text{ \& } z = x + iy]$$

$$\Rightarrow u + iv = x^2 + 2xy + i^2 y^2 - 4$$

$$\Rightarrow u + iv = x^2 - y^2 - 4 + 2xyi \quad [i^2 = -1] \text{------(i)}$$

Equating real and imaginary part,

$$u = x^2 - y^2 - 4 \text{------(ii)}$$

$$v = 2xy \text{------(iii)}$$

From (ii)

$$u = x^2 - y^2 - 4$$

$$\frac{\delta u}{\delta x} = \frac{\delta}{\delta x} (x^2 - y^2 - 4)$$

$$\frac{\delta u}{\delta x} = 2x - 0 - 0$$

$$\frac{\delta u}{\delta x} = 2x \text{------(iv)}$$

Again

$$u = x^2 - y^2 - 4$$

$$\frac{\delta u}{\delta y} = \frac{\delta}{\delta y} (x^2 - y^2 - 4)$$

$$\frac{\delta u}{\delta y} = 0 - 2y - 0$$

$$\frac{\delta u}{\delta y} = -2y \text{-----(v)}$$

From (iii)

$$v = 2xy$$

$$\frac{\delta v}{\delta x} = \frac{\delta}{\delta x}(2xy)$$

$$\frac{\delta v}{\delta x} = 2y \text{-----(vi)}$$

Again

$$v = 2xy$$

$$\frac{\delta v}{\delta y} = \frac{\delta}{\delta y}(2xy)$$

$$\frac{\delta v}{\delta y} = 2x \text{-----(vii)}$$

We have Cauchy-Riemann Equations are

$$\frac{\delta u}{\delta x} = \frac{\delta v}{\delta y} \quad \& \quad \frac{\delta u}{\delta y} = -\frac{\delta v}{\delta x} \text{-----(viii)}$$

Putting the values of $\frac{\delta u}{\delta x}, \frac{\delta v}{\delta y}, \frac{\delta u}{\delta y}, \frac{\delta v}{\delta x}$ in (viii)

$$\begin{array}{ll} \frac{\delta u}{\delta x} = \frac{\delta v}{\delta y} & \& \frac{\delta u}{\delta y} = -\frac{\delta v}{\delta x} \\ \Rightarrow 2x = 2x & \Rightarrow -2y = -2y \\ \text{L.H.S} = \text{R.H.S} & \text{L.H.S} = \text{R.H.S} \end{array}$$

Since the Cauchy-Riemann Equations are satisfied by the function $w = f(z) = z^2 - 4$.
Hence the function $w = f(z) = z^2 - 4$ is analytic.

Example 16

Determine the function $w = f(z) = z \bar{z}$ is analytic or not.

Answer:

Given,

$$w = f(z) = z \bar{z}$$

$$u + iv = (x + iy)(x - iy) \quad [\because w = u + iv \ \& \ z = x + iy \ \& \ \bar{z} = x - iy]$$

$$u + iv = x^2 - xiy + xiy - i^2 y^2$$

$$u + iv = x^2 + y^2 \quad [i^2 = -1] \text{-----(i)}$$

Equating real and imaginary part,

$$u = x^2 + y^2 \text{-----(ii)}$$

$$v = 0 \text{-----(iii)}$$

From (ii)

$$u = x^2 + y^2$$

$$\frac{\delta u}{\delta x} = \frac{\delta}{\delta x}(x^2 + y^2)$$

$$\frac{\delta u}{\delta x} = 2x \quad \text{-----(iv)}$$

From (iii)

$$v = 0$$

$$\frac{\delta v}{\delta x} = 0 \quad \text{-----(v)}$$

Again,

$$u = x^2 + y^2$$

$$\frac{\delta u}{\delta y} = \frac{\delta}{\delta y}(x^2 + y^2)$$

$$\frac{\delta u}{\delta y} = 2y \quad \text{-----(vi)}$$

and

$$v = 0$$

$$\frac{\delta v}{\delta y} = 0 \quad \text{-----(vii)}$$

We have Cauchy-Riemann Equations are

$$\frac{\delta u}{\delta x} = \frac{\delta v}{\delta y} \quad \& \quad \frac{\delta u}{\delta y} = -\frac{\delta v}{\delta x} \quad \text{-----(viii)}$$

Putting the values of $\frac{\delta u}{\delta x}, \frac{\delta v}{\delta y}, \frac{\delta u}{\delta y}, \frac{\delta v}{\delta x}$ in (viii)

$$\frac{\delta u}{\delta x} = \frac{\delta v}{\delta y} \quad \& \quad \frac{\delta u}{\delta y} = -\frac{\delta v}{\delta x}$$

$$\Rightarrow 2x = 0 \quad \Rightarrow 2y = -0$$

$$\text{L.H.S} \neq \text{R.H.S}$$

$$\text{L.H.S} \neq \text{R.H.S}$$

Since the Cauchy-Riemann Equations are not satisfied by the function $w = f(z) = z \bar{z}$.

Hence the function $w = f(z) = z \bar{z}$ is not analytic.

Example 17

Determine the function $w = f(z) = e^z = e^x(\cos y + i \sin y)$ is analytic or not. Also find its derivative that is $f'(z) = ?$

Answer:

We have,

$$e^x = 1 + \frac{x^1}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \frac{x^6}{6!} + \frac{x^7}{7!} + \dots$$

Put $x = ix$,

$$e^{ix} = 1 + \frac{ix}{1!} + \frac{(ix)^2}{2!} + \frac{(ix)^3}{3!} + \frac{(ix)^4}{4!} + \frac{(ix)^5}{5!} + \frac{(ix)^6}{6!} + \frac{(ix)^7}{7!} + \dots$$

$$[i^2 = -1; i^3 = i^2 \cdot i = -i; i^4 = i^2 \cdot i^2 = (-1) \cdot (-1) = +1; i^5 = i^4 \cdot i = i]$$

$$e^{ix} = 1 + \frac{ix}{1!} + \frac{-x^2}{2!} + \frac{-ix^3}{3!} + \frac{x^4}{4!} + \frac{ix^5}{5!} + \frac{-x^6}{6!} + \frac{-ix^7}{7!} + \dots$$

$$e^{ix} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots + \left(\frac{ix^1}{1!} - \frac{ix^3}{3!} + \frac{ix^5}{5!} - \frac{ix^7}{7!} + \dots \right)$$

$$e^{ix} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots + i \left(\frac{x^1}{1!} - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots \right)$$

$$[\because \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots; \sin x = \frac{x^1}{1!} - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots]$$

$$\therefore e^{ix} = \cos x + i \sin x$$

$$\therefore e^{iy} = \cos y + i \sin y \text{ -----(i)}$$

We have,

$$w = f(z) = e^z = e^{x+iy} \quad [z = x + iy]$$

$$w = f(z) = e^z = e^x \cdot e^{iy}$$

$$w = f(z) = e^x (\cos y + i \sin y) \quad [\text{From (i): } e^{iy} = \cos y + i \sin y]$$

$$w = u + iv = e^x \cos y + i e^x \sin y \quad [\because w = u + iv] \text{ -----(ii)}$$

Equating real and imaginary part,

$$u = e^x \cos y \text{ -----(iii)}$$

$$v = e^x \sin y \text{ -----(iv)}$$

From (iii)

$$u = e^x \cos y$$

Differentiating (iii) partially with respect to x

$$\frac{\partial u}{\partial x} = \frac{\partial}{\partial x} (e^x \cos y)$$

$$\frac{\partial u}{\partial x} = \cos y \frac{\partial}{\partial x} (e^x)$$

$$\frac{\partial u}{\partial x} = \cos y e^x$$

$$\frac{\partial u}{\partial x} = e^x \cos y \text{ -----(v)}$$

Again, Differentiating (iii) partially with respect to y

$$u = e^x \cos y$$

$$\frac{\partial u}{\partial y} = \frac{\partial}{\partial y} (e^x \cos y)$$

$$\frac{\partial u}{\partial y} = e^x \frac{\partial}{\partial y} (\cos y)$$

$$\frac{\partial u}{\partial y} = -e^x \sin y \quad \text{-----(vi)}$$

From (iv),

$$v = e^x \sin y$$

Differentiating (iv) partially with respect to x

$$\frac{\partial v}{\partial x} = \frac{\partial}{\partial x} (e^x \sin y)$$

$$\frac{\partial v}{\partial x} = \sin y \frac{\partial}{\partial x} (e^x)$$

$$\frac{\partial v}{\partial x} = \sin y e^x$$

$$\frac{\partial v}{\partial x} = e^x \sin y \quad \text{-----(vii)}$$

Again, Differentiating (iv) partially with respect to y

$$\frac{\partial v}{\partial y} = \frac{\partial}{\partial y} (e^x \sin y)$$

$$\frac{\partial v}{\partial y} = e^x \frac{\partial}{\partial y} (\sin y)$$

$$\frac{\partial v}{\partial y} = e^x \cos y \quad \text{-----(viii)}$$

We have Cauchy-Riemann Equations are

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \& \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \quad \text{-----(ix)}$$

Putting the values of $\frac{\partial u}{\partial x}, \frac{\partial v}{\partial y}, \frac{\partial u}{\partial y}, \frac{\partial v}{\partial x}$ in (ix)

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$$

$$\& \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$$\Rightarrow e^x \cos y = e^x \cos y$$

$$\text{L.H.S} = \text{R.H.S}$$

$$\Rightarrow -e^x \sin y = -e^x \sin y$$

$$\text{L.H.S} = \text{R.H.S}$$

Since the Cauchy-Riemann Equations are satisfied by the function $w = f(z) = e^x(\cos y + i \sin y)$. Hence the function $w = f(z) = e^x(\cos y + i \sin y)$ is analytic.

2nd Part: We have,

From (ii)

$$w = f(z) = u + iv = e^x \cos y + ie^x \sin y$$

$$w = f(z) = u + iv \quad \text{-----(x)}$$

Differentiating (x) with respect to x

$$f'(z) = \frac{\delta u}{\delta x} + i \frac{\delta v}{\delta x}$$

$$f'(z) = e^x \cos y + ie^x \sin y \quad \left[\text{From (v) \& (vii); } \frac{\delta u}{\delta x} = e^x \cos y, \frac{\delta v}{\delta x} = e^x \sin y \right]$$

$$f'(z) = e^x(\cos y + i \sin y)$$

$$f'(z) = e^x e^{iy} \quad \left[\text{From (i): } e^{iy} = \cos y + i \sin y \right]$$

$$f'(z) = e^{x+iy}$$

$$f'(z) = e^z \quad \text{Answer} \quad [z = x + iy]$$

Example 18

Determine the function $w = f(z) = z^3$ is analytic or not.

Answer:

Given,

$$w = f(z) = z^3$$

$$\Rightarrow u + iv = (x + iy)^3 \quad [\because w = u + iv \text{ \& } z = x + iy]$$

$$\Rightarrow u + iv = x^3 + 3x^2(iy) + 3x(iy)^2 + (iy)^3 \quad [(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3]$$

$$\Rightarrow u + iv = x^3 + i3x^2y - 3xy^2 - iy^3 \quad [i^2 = -1]$$

$$\Rightarrow u + iv = x^3 - 3xy^2 + i(3x^2y - y^3) \quad \text{-----(i)}$$

Equating real and imaginary part

$$u = x^3 - 3xy^2 \quad \text{-----(ii)}$$

$$v = 3x^2y - y^3 \quad \text{-----(iii)}$$

Differentiating (ii) partially with respect to x

$$u = x^3 - 3xy^2$$

$$\frac{\delta u}{\delta x} = \frac{\delta}{\delta x}(x^3 - 3xy^2)$$

$$\frac{\delta u}{\delta x} = 3x^2 - 3y^2 \quad \text{-----(iv)}$$

Again, Differentiating (ii) partially with respect to y

$$u = x^3 - 3xy^2$$

$$\frac{\delta u}{\delta y} = \frac{\delta}{\delta y}(x^3 - 3xy^2)$$

$$\frac{\delta u}{\delta y} = \frac{\delta}{\delta y}(x^3) - 3x \frac{\delta}{\delta y}(y^2)$$

$$\frac{\delta u}{\delta y} = 0 - 6xy$$

$$\frac{\delta u}{\delta y} = -6xy \quad \text{-----(v)}$$

Differentiating (iii) partially with respect to x

$$v = 3x^2y - y^3$$

$$\frac{\delta v}{\delta x} = \frac{\delta}{\delta x}(3x^2y - y^3)$$

$$\frac{\delta v}{\delta x} = \frac{\delta}{\delta x}(3x^2y) - \frac{\delta}{\delta x}(y^3)$$

$$\frac{\delta v}{\delta x} = y \frac{\delta}{\delta x}(3x^2) - \frac{\delta}{\delta x}(y^3)$$

$$\frac{\delta v}{\delta x} = 6xy - 0$$

$$\frac{\delta v}{\delta x} = 6xy \quad \text{-----(vi)}$$

Again, Differentiating (iii) partially with respect to y

$$v = 3x^2y - y^3$$

$$\frac{\delta v}{\delta y} = \frac{\delta}{\delta y}(3x^2y - y^3)$$

$$\frac{\delta v}{\delta y} = \frac{\delta}{\delta y}(3x^2y) - \frac{\delta}{\delta y}(y^3)$$

$$\frac{\delta v}{\delta y} = 3x^2 \frac{\delta}{\delta y}(y) - \frac{\delta}{\delta y}(y^3)$$

$$\frac{\delta v}{\delta y} = 3x^2 \cdot 1 - 3y^2$$

$$\frac{\delta v}{\delta y} = 3x^2 - 3y^2 \quad \text{-----(vii)}$$

We have Cauchy-Riemann Equations are

$$\frac{\delta u}{\delta x} = \frac{\delta v}{\delta y} \quad \& \quad \frac{\delta u}{\delta y} = -\frac{\delta v}{\delta x} \quad \text{-----(viii)}$$

Putting the values of $\frac{\delta u}{\delta x}, \frac{\delta v}{\delta y}, \frac{\delta u}{\delta y}, \frac{\delta v}{\delta x}$ in (viii)

$$\frac{\delta u}{\delta x} = \frac{\delta v}{\delta y} \quad \& \quad \frac{\delta u}{\delta y} = -\frac{\delta v}{\delta x}$$

$$\Rightarrow 3x^2 - 3y^2 = 3x^2 - 3y^2$$

L.H.S = R.H.S

$$\Rightarrow -6xy = -6xy$$

L.H.S = R.H.S

Since the Cauchy-Riemann Equations are satisfied by the function $w = f(z) = z^3$. Hence the function $w = f(z) = z^3$ is analytic.

Home Task:

$f(z) = f(x + iy) = (x^3 - 3xy^2 - 2x) + i(3x^2y - y^3 - 2y)$ is analytic or not.

Example 19

Derive Laplace's Equation from Cauchy-Riemann Equations

Answer:

We have Cauchy-Riemann Equations are

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}$$

That is

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{-----(i)}$$

$$\frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y} \quad \text{-----(ii)}$$

Differentiating (i) with respect to x we get,

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$$

$$\frac{\partial}{\partial x} \left(\frac{\partial u}{\partial x} \right) = \frac{\partial}{\partial x} \left(\frac{\partial v}{\partial y} \right)$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 v}{\partial x \partial y} \quad \text{-----(iii)}$$

Differentiating (ii) with respect to y we get,

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$$

$$\frac{\partial}{\partial y} \left(\frac{\partial u}{\partial x} \right) = \frac{\partial}{\partial y} \left(\frac{\partial v}{\partial y} \right)$$

$$\frac{\partial^2 u}{\partial y \partial x} = \frac{\partial^2 v}{\partial y^2}$$

$$\frac{\partial^2 v}{\partial y^2} = \frac{\partial^2 u}{\partial y \partial x} \quad \text{-----(iv)}$$

Again

Differentiating (ii) with respect to x we get,

$$\frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}$$

$$\frac{\delta}{\delta x} \left(\frac{\delta v}{\delta x} \right) = - \frac{\delta}{\delta x} \left(\frac{\delta u}{\delta y} \right)$$

$$\frac{\delta^2 v}{\delta x^2} = - \frac{\delta^2 u}{\delta x \delta y} \text{-----(v)}$$

Differentiating (ii) with respect to y we get,

$$\frac{\delta v}{\delta x} = - \frac{\delta u}{\delta y}$$

$$\frac{\delta}{\delta y} \left(\frac{\delta v}{\delta x} \right) = - \frac{\delta}{\delta y} \left(\frac{\delta u}{\delta y} \right)$$

$$\frac{\delta^2 v}{\delta y \delta x} = - \frac{\delta^2 u}{\delta y^2}$$

$$\frac{\delta^2 u}{\delta y^2} = - \frac{\delta^2 v}{\delta y \delta x} \text{-----(vi)}$$

Adding (iii) & (vi)

$$\frac{\delta^2 u}{\delta x^2} + \frac{\delta^2 u}{\delta y^2} = \frac{\delta^2 \cancel{v}}{\delta y \delta x} - \frac{\delta^2 \cancel{v}}{\delta x \delta y}$$

$$\frac{\delta^2 u}{\delta x^2} + \frac{\delta^2 u}{\delta y^2} = 0 \text{-----(vii)}$$

Adding (iv) & (v)

$$\frac{\delta^2 v}{\delta x^2} + \frac{\delta^2 v}{\delta y^2} = \frac{\delta^2 \cancel{u}}{\delta x \delta y} - \frac{\delta^2 \cancel{u}}{\delta x \delta y}$$

$$\frac{\delta^2 v}{\delta x^2} + \frac{\delta^2 v}{\delta y^2} = 0 \text{-----(viii)}$$

The Equation no (vii) & (viii) are called Laplace's Equation

What is Harmonic Function?

Any function which satisfies the Laplace's Equation is known as a harmonic function. If $f(z) = u + jv$ is an analytic function, then u and v are both harmonic function..

A function $f(x, y, z)$ is called a harmonic function if its second-order partial derivatives

exist and if it satisfies Laplace's equation: $\frac{\delta^2 f}{\delta x^2} + \frac{\delta^2 f}{\delta y^2} + \frac{\delta^2 f}{\delta z^2} = 0$

Example 20:

Prove that $u = x^2 - y^2$ and $v = \frac{y}{x^2 + y^2}$ are harmonic functions of (x,y)

Answer: Given

$$u = x^2 - y^2 \quad \text{-----(i)}$$

Differentiating (i) with respect to x

$$\frac{\delta u}{\delta x} = \frac{\delta}{\delta x}(x^2 - y^2)$$

$$\frac{\delta u}{\delta x} = 2x + 0$$

$$\frac{\delta u}{\delta x} = 2x \quad \text{-----(ii)}$$

Again differentiating (ii) with respect to x

$$\frac{\delta u}{\delta x} = 2x$$

$$\frac{\delta}{\delta x} \left(\frac{\delta u}{\delta x} \right) = \frac{\delta}{\delta x}(2x)$$

$$\frac{\delta^2 u}{\delta x^2} = 2 \quad \text{-----(iii)}$$

Now differentiating (i) with respect to y

$$u = x^2 - y^2$$

$$\frac{\delta u}{\delta y} = \frac{\delta}{\delta y}(x^2 - y^2)$$

$$\frac{\delta u}{\delta y} = 0 - 2y$$

$$\frac{\delta u}{\delta y} = 0 - 2y \quad \text{-----(iv)}$$

Again differentiating (iv) with respect to y

$$\frac{\delta u}{\delta y} = -2y$$

$$\frac{\delta}{\delta y} \left(\frac{\delta u}{\delta y} \right) = -\frac{\delta}{\delta y}(2y)$$

$$\frac{\delta^2 u}{\delta y^2} = -2 \quad \text{-----(v)}$$

Adding (iii) and (v)

$$\frac{\delta^2 u}{\delta x^2} + \frac{\delta^2 u}{\delta y^2} = 2 - 2 = 0$$

$$\frac{\delta^2 u}{\delta x^2} + \frac{\delta^2 u}{\delta y^2} = 0$$

Since $u(x, y) = x^2 - y^2$ satisfies Laplace's equation. Hence $u(x, y) = x^2 - y^2$ is a harmonic function.

Now, given

$$v = \frac{y}{x^2 + y^2} \text{-----(vi)}$$

Differentiating (vi) with respect to x

$$\begin{aligned} \frac{\delta v}{\delta x} &= \frac{\delta}{\delta x} \left(\frac{y}{x^2 + y^2} \right) \\ \frac{\delta v}{\delta x} &= \frac{(x^2 + y^2) \frac{\delta}{\delta x} (y) - y \frac{\delta}{\delta x} (x^2 + y^2)}{(x^2 + y^2)^2} \quad \left[\because \frac{d}{dx} \left(\frac{u}{v} \right) = \frac{v \frac{d}{dx} (u) - u \frac{d}{dx} (v)}{v^2} \right] \\ &= \frac{(x^2 + y^2).0 - y(2x + 0)}{(x^2 + y^2)^2} \\ &= \frac{0 - 2xy}{(x^2 + y^2)^2} \end{aligned}$$

$$\frac{\delta v}{\delta x} = \frac{-2xy}{(x^2 + y^2)^2} \text{-----(vii)}$$

Again differentiating (vii) with respect to x

$$\begin{aligned} \frac{\delta}{\delta x} \left(\frac{\delta v}{\delta x} \right) &= - \frac{\delta}{\delta x} \left\{ \frac{2xy}{(x^2 + y^2)^2} \right\} \\ \frac{\delta^2 v}{\delta x^2} &= - \left[\frac{(x^2 + y^2)^2 \frac{\delta}{\delta x} (2xy) - 2xy \frac{\delta}{\delta x} (x^2 + y^2)^2}{\{(x^2 + y^2)^2\}^2} \right] \\ \frac{\delta^2 v}{\delta x^2} &= \frac{-(x^2 + y^2)^2 2y \frac{\delta}{\delta x} (x) + 2xy \times 2(x^2 + y^2)^{2-1} \frac{\delta}{\delta x} (x^2 + y^2)}{(x^2 + y^2)^4} \\ \frac{\delta^2 v}{\delta x^2} &= \frac{-(x^2 + y^2)^2 2y.1 + 4xy(x^2 + y^2)(2x + 0)}{(x^2 + y^2)^4} \\ \frac{\delta^2 v}{\delta x^2} &= \frac{-(x^2 + y^2)^2 2y + 8x^2 y(x^2 + y^2)}{(x^2 + y^2)^4} \end{aligned}$$

$$\begin{aligned} \frac{\delta^2 v}{\delta x^2} &= \frac{(x^2 + y^2)[-(x^2 + y^2)2y + 8x^2 y]}{(x^2 + y^2)^3(x^2 + y^2)} \\ \frac{\delta^2 v}{\delta x^2} &= \frac{[-2x^2 y - 2y^3 + 8x^2 y]}{(x^2 + y^2)^3} \\ \frac{\delta^2 v}{\delta x^2} &= \frac{6x^2 y - 2y^3}{(x^2 + y^2)^3} \text{-----(viii)} \end{aligned}$$

Again,

$$v = \frac{y}{x^2 + y^2}$$

Differentiating with respect to y

$$\begin{aligned}\frac{\delta v}{\delta y} &= \frac{\delta}{\delta y} \left(\frac{y}{x^2 + y^2} \right) \\ \frac{\delta v}{\delta y} &= \frac{(x^2 + y^2) \frac{\delta}{\delta y}(y) - y \frac{\delta}{\delta y}(x^2 + y^2)}{(x^2 + y^2)^2} \\ \frac{\delta v}{\delta y} &= \frac{(x^2 + y^2) \cdot 1 - y(0 + 2y)}{(x^2 + y^2)^2} \\ \frac{\delta v}{\delta y} &= \frac{(x^2 + y^2) - 2y^2}{(x^2 + y^2)^2} \\ \frac{\delta v}{\delta y} &= \frac{(x^2 - y^2)}{(x^2 + y^2)^2} \text{-----(ix)}\end{aligned}$$

Again, differentiating (ix) with respect to y

$$\begin{aligned}\frac{\delta}{\delta y} \left(\frac{\delta v}{\delta y} \right) &= \frac{\delta}{\delta y} \left\{ \frac{(x^2 - y^2)}{(x^2 + y^2)^2} \right\} \\ \frac{\delta^2 v}{\delta y^2} &= \frac{(x^2 + y^2)^2 \frac{\delta}{\delta y}(x^2 - y^2) - (x^2 - y^2) \frac{\delta}{\delta y}(x^2 + y^2)^2}{\{(x^2 + y^2)^2\}^2} \\ \frac{\delta^2 v}{\delta y^2} &= \frac{(x^2 + y^2)^2(0 - 2y) - (x^2 - y^2) \times 2(x^2 + y^2)^{2-1} \frac{\delta}{\delta y}(x^2 + y^2)}{(x^2 + y^2)^4} \\ \frac{\delta^2 v}{\delta y^2} &= \frac{(x^2 + y^2)^2(0 - 2y) - (x^2 - y^2) \times 2(x^2 + y^2)(0 + 2y)}{(x^2 + y^2)^4} \\ \frac{\delta^2 v}{\delta y^2} &= \frac{(x^2 + y^2)^2(-2y) - 2(x^2 - y^2)(x^2 + y^2)(2y)}{(x^2 + y^2)^4} \\ \frac{\delta^2 v}{\delta y^2} &= \frac{(x^2 + y^2)^2(-2y) - 4y(x^2 - y^2)(x^2 + y^2)}{(x^2 + y^2)^4} \\ \frac{\delta^2 v}{\delta y^2} &= \frac{(x^2 + y^2)[(x^2 + y^2)(-2y) - 4y(x^2 - y^2)]}{(x^2 + y^2)(x^2 + y^2)^3} \\ \frac{\delta^2 v}{\delta y^2} &= \frac{(x^2 + y^2)(-2y) - 4y(x^2 - y^2)}{(x^2 + y^2)^3}\end{aligned}$$

$$\frac{\delta^2 v}{\delta y^2} = \frac{-2x^2 y - 2y^3 - 4x^2 y + 4y^3}{(x^2 + y^2)^3}$$

$$\frac{\delta^2 v}{\delta y^2} = \frac{-6x^2 y + 2y^3}{(x^2 + y^2)^3} \text{-----(x)}$$

Adding (viii) and(x),

$$\frac{\delta^2 v}{\delta x^2} + \frac{\delta^2 v}{\delta y^2} = \frac{6x^2 y - 2y^3}{(x^2 + y^2)^3} + \frac{-6x^2 y + 2y^3}{(x^2 + y^2)^3}$$

$$\frac{\delta^2 v}{\delta x^2} + \frac{\delta^2 v}{\delta y^2} = \frac{6x^2 y - 2y^3 - 6x^2 y + 2y^3}{(x^2 + y^2)^3}$$

$$\frac{\delta^2 v}{\delta x^2} + \frac{\delta^2 v}{\delta y^2} = 0$$

Since $v(x, y) = \frac{y}{x^2 + y^2}$ satisfies Laplace's equation. Hence $v(x, y) = \frac{y}{x^2 + y^2}$ is a harmonic function. **(Proved)**

Example 21:

Prove that, $u = e^{-x}(x \sin y - y \cos y)$ is a harmonic function.

Answer:

Given, $u = e^{-x}(x \sin y - y \cos y)$(i)

Differentiating (i) with respect to x ,

$$\frac{\delta u}{\delta x} = \frac{\delta}{\delta x} [e^{-x}(x \sin y - y \cos y)]$$

$$\frac{\delta u}{\delta x} = e^{-x} \frac{\delta}{\delta x} (x \sin y - y \cos y) + (x \sin y - y \cos y) \frac{\delta}{\delta x} (e^{-x}) \left[\frac{d}{dx} (uv) = u \frac{d}{dx} v + v \frac{d}{dx} u \right]$$

$$\frac{\delta u}{\delta x} = e^{-x} \left[\frac{\delta}{\delta x} (x \sin y) - \frac{\delta}{\delta x} (y \cos y) \right] - (x \sin y - y \cos y) e^{-x} \left[\because \frac{d}{dx} (e^{-x}) = -e^{-x} \right]$$

$$\frac{\delta u}{\delta x} = e^{-x} \left[\left\{ x \frac{\delta}{\delta x} \sin y + \sin y \frac{\delta}{\delta x} x \right\} - \left\{ y \frac{\delta}{\delta x} \cos y + \cos y \frac{\delta}{\delta x} y \right\} \right] - e^{-x} (x \sin y - y \cos y)$$

$$\frac{\delta u}{\delta x} = e^{-x} [\{x(0) + \sin y.1\} - \{y(0) + \cos y.(0)\}] - e^{-x} (x \sin y - y \cos y)$$

$$\frac{\delta u}{\delta x} = e^{-x} [0 + \sin y - 0 - 0] - e^{-x} (x \sin y - y \cos y)$$

$$\frac{\delta u}{\delta x} = e^{-x} \sin y - e^{-x} (x \sin y - y \cos y)$$

$$\frac{\delta u}{\delta x} = e^{-x} \sin y - x e^{-x} \sin y + y e^{-x} \cos y$$

$$\frac{\delta u}{\delta x} = e^{-x} (\sin y - x \sin y + y \cos y) \text{.....(ii)}$$

Again Differentiating (ii) with respect to x ,

$$\frac{\partial u}{\partial x} = e^{-x}(\sin y - x \sin y + y \cos y)$$

$$\frac{\partial}{\partial x} \left(\frac{\partial u}{\partial x} \right) = e^{-x} \frac{\partial}{\partial x} (\sin y - x \sin y + y \cos y) + (\sin y - x \sin y + y \cos y) \frac{\partial}{\partial x} e^{-x}$$

$$\frac{\partial^2 u}{\partial x^2} = e^{-x} \frac{\partial}{\partial x} (\sin y - x \sin y + y \cos y) + (-)e^{-x} (\sin y - x \sin y + y \cos y)$$

$$\frac{\partial^2 u}{\partial x^2} = e^{-x} \left[\frac{\partial}{\partial x} (\sin y) - \frac{\partial}{\partial x} (x \sin y) + \frac{\partial}{\partial x} (y \cos y) \right] - e^{-x} (\sin y - x \sin y + y \cos y)$$

$$\frac{\partial^2 u}{\partial x^2} = e^{-x} \left[0 - \left\{ x \frac{\partial}{\partial x} (\sin y) + \sin y \frac{\partial x}{\partial x} \right\} + y \frac{\partial}{\partial x} (\cos y) + \cos y \frac{\partial}{\partial x} y \right] - (\sin y - x \sin y + y \cos y) e^{-x}$$

$$\frac{\partial^2 u}{\partial x^2} = e^{-x} [0 - x.0 - \sin y.1 + y.0 + \cos y.0] - e^{-x} (\sin y - x \sin y + y \cos y)$$

$$\frac{\partial^2 u}{\partial x^2} = e^{-x} [0 - 0 - \sin y + 0 + 0] - e^{-x} (\sin y - x \sin y + y \cos y)$$

$$\frac{\partial^2 u}{\partial x^2} = e^{-x} (-\sin y - \sin y + x \sin y - y \cos y)$$

$$\frac{\partial^2 u}{\partial x^2} = e^{-x} (-2 \sin y + x \sin y - y \cos y) \dots \dots \dots (iii)$$

Now differentiating (i) with respect to y ,

$$u = e^{-x}(x \sin y - y \cos y)$$

$$\frac{\partial u}{\partial y} = e^{-x} \frac{\partial}{\partial y} (x \sin y - y \cos y) + (x \sin y - y \cos y) \frac{\partial}{\partial y} (e^{-x})$$

$$\frac{\partial u}{\partial y} = e^{-x} \left\{ x \frac{\partial}{\partial y} (\sin y) + \sin y \frac{\partial x}{\partial y} - (y \frac{\partial}{\partial y} \cos y + \cos y \frac{\partial}{\partial y} (y)) \right\} + (x \sin y - y \cos y).0$$

$$\frac{\partial u}{\partial y} = e^{-x} \{ x \cos y + \sin y.0 - (-y \sin y + \cos y.1) \} + 0$$

$$\frac{\partial u}{\partial y} = e^{-x} (x \cos y + y \sin y - \cos y) \dots \dots \dots (iv)$$

Again differentiating (iv) with respect to y

$$\frac{\partial u}{\partial y} = e^{-x} (x \cos y + y \sin y - \cos y)$$

$$\frac{\partial}{\partial y} \left(\frac{\partial u}{\partial y} \right) = \frac{\partial}{\partial y} [e^{-x} (x \cos y + y \sin y - \cos y)]$$

$$\frac{\partial^2 u}{\partial y^2} = e^{-x} \frac{\partial}{\partial y} (x \cos y + y \sin y - \cos y) + (x \cos y + y \sin y - \cos y) \frac{\partial}{\partial y} (e^{-x})$$

$$\frac{\delta^2 u}{\delta y^2} = e^{-x} \left[\frac{\delta}{\delta y} (x \cos y) + \frac{\delta}{\delta y} (y \sin y) - \frac{\delta}{\delta y} (\cos y) \right] + (x \cos y + y \sin y - \cos y) \frac{\delta}{\delta y} (e^{-x})$$

$$\frac{\delta^2 u}{\delta y^2} = e^{-x} \left[x \frac{\delta}{\delta y} (\cos y) + \cos y \frac{\delta}{\delta y} x + y \frac{\delta}{\delta y} (\sin y) + \sin y \frac{\delta}{\delta y} y - \frac{\delta}{\delta y} (\cos y) \right] + (x \cos y + y \sin y - \cos y) \cdot 0$$

$$\frac{\delta^2 u}{\delta y^2} = e^{-x} [-x \sin y + \cos y \cdot 0 + y \cos y + \sin y + \sin y] + 0$$

$$\frac{\delta^2 u}{\delta y^2} = e^{-x} [-x \sin y + y \cos y + 2 \sin y] \dots \dots \dots (v)$$

Adding (iii) and (v),

$$\frac{\delta^2 u}{\delta x^2} + \frac{\delta^2 u}{\delta y^2} = e^{-x} [-2 \sin y + x \sin y - y \cos y] + e^{-x} [2 \sin y - x \sin y + y \cos y]$$

$$\frac{\delta^2 u}{\delta x^2} + \frac{\delta^2 u}{\delta y^2} = e^{-x} [-2 \sin y + x \sin y - y \cos y + 2 \sin y - x \sin y + y \cos y]$$

$$\frac{\delta^2 u}{\delta x^2} + \frac{\delta^2 u}{\delta y^2} = 0$$

Since $u(x, y) = e^{-x}(x \sin y - y \cos y)$ satisfies Laplace's equation,

hence $u(x, y) = e^{-x}(x \sin y - y \cos y)$ is a Harmonic function.

Home Task

- i. Prove that $f(z) = |z|^2$ is not harmonic functions but $f(z) = \ln(|z|^2)$ is harmonic.
- ii. Is $f(x, y, z) = x^2 + y^2 - 2z^2$ harmonic? What about $f(x, y, z) = x^2 - y^2 + z^2$?
- iii. Show that the function $u(x, y) = 3x^3 - 9xy^2$ is harmonic
- iv. Verify $u(x, y) = x^3 - 3xy^2 - 5y$ is harmonic
- v. Test the following functions harmonic or not
 - a) $u = x^3 - 3xy^2$
 - b) $u = e^{-y} \sin x$
 - c) $u = e^x \cos y$
 - d) $u(x, y) = 3x^2y + 2x^2 - y^3 - 2y^2$
 - e) $u(x, y) = e^x(x \cos y - y \sin y)$
 - f) $u = y^3 - 3x^2y$
 - g) $u = 2x(1 - y)$
 - h) $u(x, y) = x^3 - 3xy^2 + 3x^2 - 3y^2 + 1$
 - i) $u(x, y) = x^2 - y^2 - 2xy - 2x + 3y$

Complex Integration

Complex integration is a simple extension of the ideas we develop in calculus to the complex world. In real calculus, differentiation and integration are, roughly speaking, inverse operations (save for the additional interpretation of derivative as the slope of a function and integral as the area under the curve). We will see that the same inverse relationship between differentiation and integration exists in the complex domain;

Contour Integrals

Contour Integration contour integration is a method of evaluating certain integrals along paths in the complex plane.

Contour integrals also have important applications in physics, particularly in the study of waves and oscillations.

Contour integrals is basically the integration of the form, $\int_{z_0}^{z_1} f(z) dz$

Where z_0 and z_1 are complex numbers.

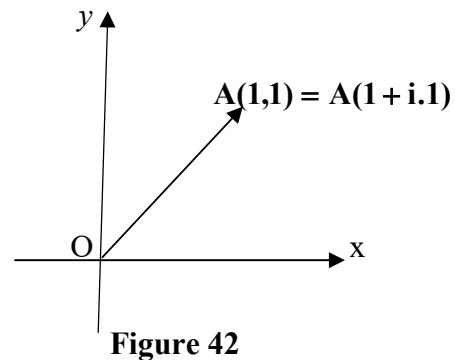
where f is a complex function of a complex variable and Γ is a given contour, that is $\int_{\Gamma} f(z) dz$

Example 22:

Find the value of integral $\int_0^{1+i} (x - y + ix^2) dz$

- (a) Along straight line from $z = 0$ to $z = 1 + i$
- (b) Along real axis from $z = 0$ to $z = 1$ and then along a line parallel to the imaginary axis from $z = 1$ to $z = 1 + i$.

Answer:



- a) Along OA line:

Given,

$$z = 0$$

$$x + iy = 0 \quad [z = x + iy]$$

$$x + iy = 0 + i.0$$

Equating real and imaginary part,

$$x = 0 \text{ \& } y = 0$$

That is coordinate of O (0, 0)

Again

Given,

$$z = 1 + i$$

$$x + iy = 1 + i.1 \quad [z = x + iy]$$

Equating real and imaginary part,

$$x = 1 \text{ \& } y = 1$$

That is coordinate of A (1,1)

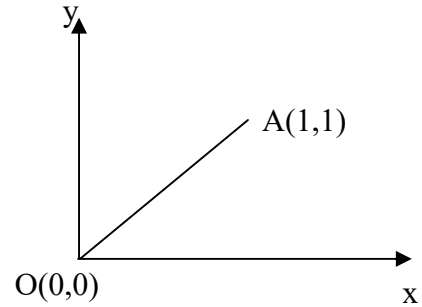


Figure 43

Now, The Equation of a straight line **OA** passing through **O(0,0)** and **A(1,1)** is

$$\frac{y - y_1}{y_1 - y_2} = \frac{x - x_1}{x_1 - x_2}$$

$$\Rightarrow \frac{y - 0}{0 - 1} = \frac{x - 0}{0 - 1}$$

$$\Rightarrow \frac{y}{-1} = \frac{x}{-1}$$

$$\Rightarrow y = x \quad \dots\dots\dots(i)$$

We have

$$z = x + iy$$

$$z = x + ix \quad [\because y = x \quad \text{from (i)}]$$

$$\frac{dz}{dx} = \frac{d}{dx}(x + ix)$$

$$\frac{dz}{dx} = 1 + i.1$$

$$dz = (1 + i)dx \quad \dots\dots\dots(ii)$$

Now,

$$\int_{OA} (x - y + ix^2) dz$$

$$= \int_0^1 (x - x + ix^2)(1 + i) dx \quad [\text{from (i) and (ii) ; } y = x \quad dz = (1 + i)dx]$$

$$= \int_0^1 (ix^2)(1 + i) dx$$

$$= (1 + i) \int_0^1 ix^2 dx$$

$$= (1 + i)i \int_0^1 x^2 dx$$

$$\begin{aligned}
&= (1+i) \cdot i \cdot \left[\frac{x^3}{3} \right]_0^1 \\
&= i(1+i) \left[\frac{1}{3} - \frac{0}{3} \right] \\
&= i(1+i) \left[\frac{1}{3} \right] \\
&= \frac{1}{3}(i-1) \quad [i^2 = -1]
\end{aligned}$$

b) Along OB and then along BA; Along OB from $z = 0$ to $z = 1$ and then along BA, from $z = 1$ to $z = i + 1$

Solution:

Along line OB

Given,

$$z = 0$$

$$x + iy = 0 \quad [z = x + iy]$$

$$x + iy = 0 + i \cdot 0$$

Equating real and imaginary part,

$$x = 0 \text{ \& } y = 0$$

That is coordinate of O (0, 0)

Again,

Given,

$$z = 1$$

$$x + iy = 1 \quad [z = x + iy]$$

$$x + iy = 1 + 0$$

$$x + iy = 1 + i \cdot 0$$

Equating real and imaginary part,

$$x = 1 \text{ \& } y = 0$$

That is coordinate of B (1,0)

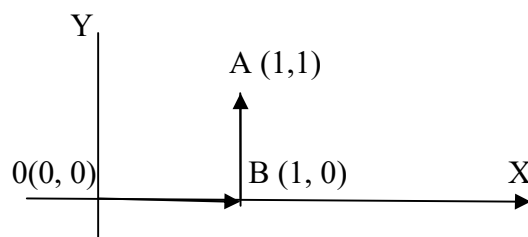


Figure 44

Required Integral,

$$\int_{OB} (x-y+ix^2)dz + \int_{BA} (x-y+ix^2)dz \dots\dots\dots(iii)$$

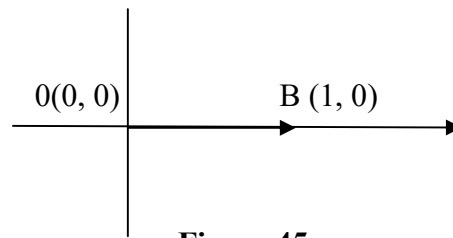


Figure 45

The Equation of OB is:

$$\frac{y-y_1}{y_1-y_2} = \frac{x-x_1}{x_1-x_2}$$

$$\frac{y-0}{0-0} = \frac{x-0}{0-1}$$

$$-y = 0$$

$$\therefore y = 0 \dots\dots\dots(iv)$$

We have,

$$z = x + iy$$

$$z = x + i.0 \quad [\because y = 0]$$

$$z = x$$

$$\frac{dz}{dx} = \frac{d}{dx}(x)$$

$$\frac{dz}{dx} = 1$$

$$dz = dx \dots\dots\dots(v)$$

Now first part of (iii),

Hence,

$$\int_{OB} (x-y+ix^2)dz$$

OB

$$= \int_0^1 (x-0+ix^2)dx \quad [\text{From (iv) and (v); } y = 0; dz = dx]$$

$$= \int_0^1 (x+ix^2)dx$$

$$= \left[\frac{x^2}{2} + i \frac{x^3}{3} \right]_0^1$$

$$= \frac{1}{2} + i \left(\frac{1}{3} \right) - \frac{0}{2} - i \frac{0}{3}$$

$$= \frac{1}{2} + i \cdot \frac{1}{3}$$

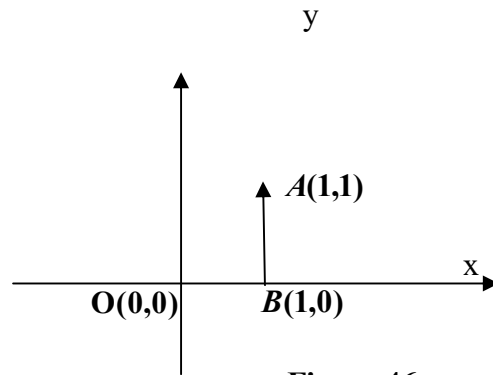


Figure 46

Along line BA:

The Equation of BA is:

$$\frac{y - y_1}{y_1 - y_2} = \frac{x - x_1}{x_1 - x_2}$$

$$\frac{y - 0}{0 - 1} = \frac{x - 1}{1 - 1}$$

$$\frac{y}{-1} = \frac{x - 1}{0}$$

$$y \cdot 0 = (x - 1)(-1)$$

$$0 = -x + 1$$

$$-x = -1$$

$$x = 1 \text{ -----(vi)}$$

$$\therefore dx = 0 \text{ -----(vii)}$$

[Differentiating vi]

We have,

$$z = x + iy$$

$$z = 1 + iy \quad [\text{from (vi); } x = 1]$$

$$z = 1 + iy$$

$$\frac{dz}{dy} = \frac{d}{dy}(1 + iy)$$

$$\frac{dz}{dy} = 0 + i \frac{dy}{dy}$$

$$\frac{dz}{dy} = 0 + i \cdot 1$$

$$dz = i dy \text{(viii)}$$

Second part of (iii)

Now,

$$\int_{BA} (x - y + ix^2) dz$$

BA

$$= \int_0^1 (1 - y + i \cdot 1^2) \, idy \quad [\text{from (vi),(viii); } x = 1; dz = idy]$$

$$= \int_0^1 (1 - y + i) \, idy$$

$$= i \int_0^1 (1 - y + i) \, dy$$

$$= i \left[y - \frac{y^2}{2} + iy \right]_0^1$$

$$= i \left[y - \frac{y^2}{2} + iy \right]_0^1$$

$$= i \left[1 - \frac{1}{2} + i - \left(0 - \frac{0}{2} + 0 \right) \right]$$

$$= i \left[\frac{1}{2} + i \right]$$

$$= \frac{i}{2} - 1 \dots \dots \dots (ix)$$

putting result in (iii),

$$\int_{OB} (x - y + ix^2) \, dz + \int_{BA} (x - y + ix^2) \, dz$$

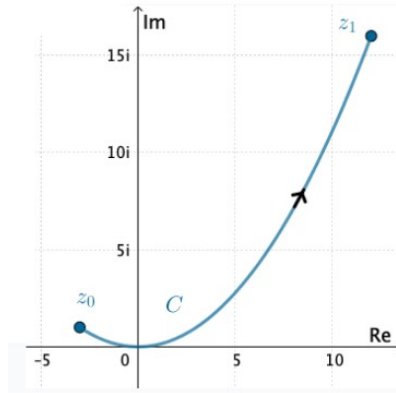
$$= \left(\frac{1}{2} + i \frac{1}{3} \right) + \left(\frac{i}{2} - 1 \right)$$

$$= \frac{1}{2} + \frac{i}{3} + \frac{i}{2} - 1$$

$$= -\frac{1}{2} + \frac{5}{6}i \quad \text{Answer}$$

Example 23:

Evaluate $\int_C \bar{z} dz$ where C is given by $x = 3t, y = t^2$ with $-1 \leq t \leq 4$



We have,

$$z = x + iy$$

$$z = 3t + it^2$$

$$\bar{z} = 3t - it^2$$

Again

$$z = 3t + it^2$$

$$\therefore \frac{dz}{dt} = \frac{d}{dt}(3t + it^2)$$

$$\therefore \frac{dz}{dt} = 3 + 2it$$

$$\therefore dz = (3 + 2it)dt$$

$$\text{So, } \int_C \bar{z} dz = \int_{-1}^4 (3t - it^2)(3 + 2it)dt$$

$$\int_C \bar{z} dz = \int_{-1}^4 (9t + 6it^2 - 3it^2 - 2i^2t^3)dt$$

$$\int_C \bar{z} dz = \int_{-1}^4 (9t + 3it^2 + 2t^3)dt$$

$$\int_C \bar{z} dz = \int_{-1}^4 (9t + 2t^3)dt + 3i \int_{-1}^4 t^2 dt$$

$$\int_C \bar{z} dz = \int_{-1}^4 (9t + 2t^3)dt + 3i \int_{-1}^4 t^2 dt$$

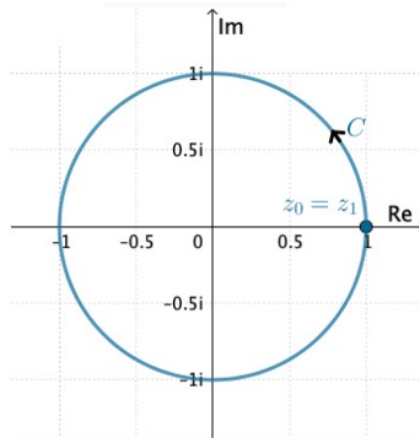
$$\int_C \bar{z} dz = \left[9 \frac{t^2}{2} + 2 \frac{t^4}{4} \right]_{-1}^4 + 3i \left[\frac{t^3}{3} \right]_{-1}^4$$

$$\int_C \bar{z} dz = \left[9 \frac{t^2}{2} + \frac{t^4}{2} \right]_{-1}^4 + i[t^3]_{-1}^4$$

$$\int_C \bar{z} dz = 195 + 65i$$

Example 24:

Evaluate $\int_C \frac{1}{z} dz$ where **C** is the circle $z = \cos t, y = \sin t$ with $0 \leq t \leq 2\pi$



We have,

$$z = x + iy$$

$$z = \cos t + i \sin t$$

$$z = e^{it}$$

$$\bar{z} = e^{-it}$$

Again

$$z = e^{it}$$

$$\therefore \frac{dz}{dt} = \frac{d}{dt}(e^{it})$$

$$\therefore \frac{dz}{dt} = ie^{it}$$

$$\therefore dz = ie^{it} dt$$

$$\text{So, } \int_C \frac{1}{z} dz = \int_C \frac{1}{e^{it}} (ie^{it}) dt$$

$$\text{So, } \int_C \frac{1}{z} dz = \int_C e^{-it} (ie^{it}) dt$$

$$\text{So, } \int_C \frac{1}{z} dz = \int_C ie^{-it+it} dt$$

$$\text{So, } \int_C \frac{1}{z} dz = \int_C ie^0 dt$$

$$\text{So, } \int_C \frac{1}{z} dz = \int_C i \cdot 1 dt$$

$$\text{So, } \int_C \frac{1}{z} dz = \int_0^{2\pi} i dt$$

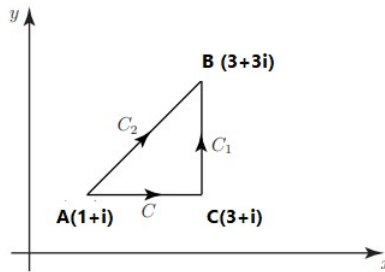
$$\text{So, } \int_C \frac{1}{z} dz = \int_0^{2\pi} i dt$$

$$\text{So, } \int_C \frac{1}{z} dz = [it]_0^{2\pi}$$

$$\text{So, } \int_C \frac{1}{z} dz = 2\pi i$$

Example 25: Obtain the complex integral $\int_C z dz$

Where **C** is the straight line path from $z = 1+i$ to $z = 3+3i$



Given,

$$z = 1 + i$$

$$x + iy = 1 + i.1 \quad [z = x + iy]$$

Equating real and imaginary part,

$$x = 1 \text{ \& } y = 1$$

That is coordinate of A (1,1)

Given,

$$z = 3 + 3i$$

$$x + iy = 3 + i.3 \quad [z = x + iy]$$

Equating real and imaginary part,

$$x = 3 \text{ \& } y = 3$$

That is coordinate of B (3,3)

Now, The Equation of a straight line **AB** passing through **A(1,1)** and **B(3,3)** is

$$\frac{y - y_1}{y_1 - y_2} = \frac{x - x_1}{x_1 - x_2}$$

$$\begin{aligned}
\Rightarrow \frac{y-1}{1-3} &= \frac{x-1}{1-3} \\
\Rightarrow \frac{y-1}{-2} &= \frac{x-1}{-2} \\
\Rightarrow y-1 &= x-1 \\
\Rightarrow y &= x-1+1 \\
\Rightarrow y &= x \text{ -----(i)}
\end{aligned}$$

We have

$$z = x + iy$$

$$z = x + ix \quad [\because y = x \quad \text{from(i)}]$$

$$\frac{dz}{dx} = \frac{d}{dx}(x + ix)$$

$$\frac{dz}{dx} = 1 + i.1$$

$$dz = (1 + i)dx \quad \text{.....(ii)}$$

$$\text{So, } \int_C z \, dz$$

$$\Rightarrow \int_{1+i}^{3+3i} (x + iy) \, dz$$

$$\Rightarrow \int_{1+i}^{3+3i} (x + i.x)(1 + i) \, dx$$

$$\Rightarrow (1 + i) \int_1^3 (x + i.x) \, dx$$

$$\Rightarrow (1 + i) \int_1^3 (x + i.x) \, dx$$

$$\Rightarrow (1 + i) \left[\frac{x^2}{2} + i \frac{x^2}{2} \right]_1^3$$

$$\Rightarrow (1 + i) \left[\frac{3^2}{2} + i \frac{3^2}{2} - \frac{1^2}{2} - i \frac{1^2}{2} \right]$$

$$\Rightarrow (1 + i) \left[\frac{9}{2} + i \frac{9}{2} - \frac{1}{2} - i \frac{1}{2} \right]$$

$$\Rightarrow (1 + i) \left[\frac{8}{2} + i \frac{8}{2} \right]$$

$$\Rightarrow (1 + i)[4 + 4i]$$

Example 26: Obtain the complex integral $\int_C z \, dz$

where c is the path from $z = 1+i$ to $z = 3+i$

Answer: We have,

$$w = f(z) = z$$

$$u + iv = f(z) = x + iy$$

We have

$$z = x + iy$$

$$z = x + ix \quad [\because y = x \quad \text{from (i)}]$$

$$\frac{dz}{dx} = \frac{d}{dx}(x + ix)$$

$$\frac{dz}{dx} = 1 + i.1$$

$$dz = (1 + i)dx \quad \dots\dots\dots\text{(i)}$$

Given,

$$z = 1 + i$$

$$x + iy = 1 + i.1 \quad [z = x + iy]$$

Equating real and imaginary part,

$$x = 1 \text{ \& } y = 1$$

That is coordinate of A (1,1)

Given,

$$z = 3 + i$$

$$x + iy = 3 + i.1 \quad [z = x + iy]$$

Equating real and imaginary part,

$$x = 3 \text{ \& } y = 1$$

That is coordinate of C (3,1)

Now, The Equation of a straight line AC passing through A(1,1) and C(3,1) is

$$\frac{y - y_1}{y_1 - y_2} = \frac{x - x_1}{x_1 - x_2}$$

$$\Rightarrow \frac{y - 1}{1 - 1} = \frac{x - 1}{1 - 3}$$

$$\Rightarrow \frac{y - 1}{0} = \frac{x - 1}{-2}$$

$$\Rightarrow y - 1 = 0$$

$$\Rightarrow y = 1$$

$$\Rightarrow y = 1 \quad \text{----- (ii)}$$

We have

$$z = x + iy$$

$$z = x + i.1 \quad [\because y = 1 \quad \text{from (ii)}]$$

$$\frac{dz}{dx} = \frac{d}{dx}(x + i.1)$$

$$\frac{dz}{dx} = 1 + 0$$

$$dz = dx \quad \dots\dots\dots(\text{iii})$$

$$\begin{aligned} \int_C z \, dz &= \int_{AC} (x + iy) \, dx \\ &= \int_{AC} (x + i \cdot 1) \, dx \\ &= \int_1^3 (x + i \cdot 1) \, dx \\ &= \left[\frac{x^2}{2} + ix \right]_1^3 \\ &= \left[\frac{3^2}{2} + 3i - \frac{1}{2} - i \right] \\ &= \left[\frac{9}{2} + 3i - \frac{1}{2} - i \right] \\ &= \left[\frac{8}{2} + 2i \right] \\ &= [4 + 2i] \end{aligned}$$

What is pole: pole is a certain type of singularity of a function can be found by substituting the denominator of the function equal to zero. Roots of denominator indicates poles. That is, Poles represents the points where a complex function cease to be analytic.

Cauchy's Theorem

The theorem states that if $f(z)$ is analytic everywhere within a simply-connected region then $\oint_C f(z) \, dz = 0$ for every simple closed path C lying in the region.

Cauchy's Integral Formula:

If $f(z)$ is analytic inside and on the boundary C of a simply-connected region then for

any point 'a' within the curve 'C': $\oint_C \frac{f(z)}{z-a} \, dz = 2\pi i \times f(a)$

Example 27:

Evaluate $\int_C \frac{z^2 - z + 1}{z-1} \, dz$

Where c is the circle i) $|z| = 1$ ii) $|z| = \frac{1}{2}$

We have,

$$z = x + jy$$

$$\therefore |z| = \sqrt{x^2 + y^2}$$

Given,

$$i) |z| = 1$$

$$\sqrt{x^2 + y^2} = 1$$

$$\therefore x^2 + y^2 = 1$$

$$(x-0)^2 + (y-0)^2 = 1^2 \quad \text{-----(i)}$$

[We have, $(x-a)^2 + (y-b)^2 = r^2$]

Which is the equation of a circle whose Center (0, 0), Radius = 1

Let $f(z) = z^2 - z + 1$ and singularity point is $a = 1$

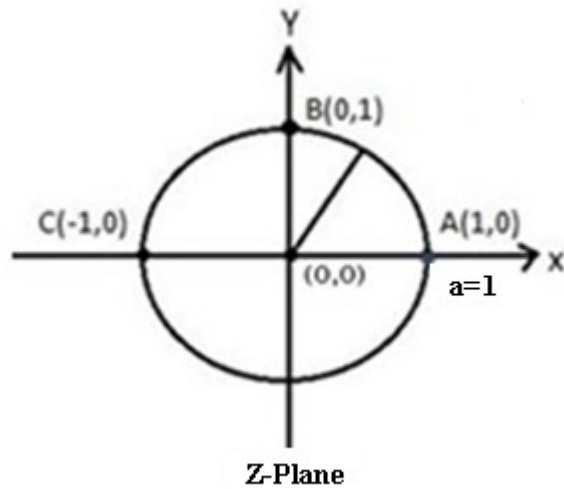


Figure 47

Pole $z-1=0$

$$\Rightarrow z = 1$$

$$\Rightarrow x + jy = 1 + 0.j$$

Equating the coefficient of real and imaginary part, we get,

$$x = 1, \quad y = 0$$

Hence, we can write,

$$(x,y) = (1,0)$$

$\therefore (x,y) = (1,0)$ is on the circle c, then by Cauchy's Integral formula

$$\int_c \frac{f(z)}{z-a} dz = 2\pi i \times f(a)$$

$$\Rightarrow \int_c \frac{z^2 - z + 1}{z - 1} dz = 2\pi i \times f(1) [a = 1]$$

$$\text{Here, } f(z) = z^2 - z + 1$$

$$\therefore f(1) = 1^2 - 1 + 1 = 1$$

$$\Rightarrow \int_c \frac{z^2 - z + 1}{z - 1} dz = 2\pi i \times f(1)$$

$$\Rightarrow \int_c \frac{z^2 - z + 1}{z - 1} dz = 2\pi i \times 1$$

$$\Rightarrow \int_c \frac{z^2 - z + 1}{z - 1} dz = 2\pi i \text{ Answer}$$

Given,

$$\text{i) } |z| = \frac{1}{2}$$

$$\sqrt{x^2 + y^2} = \frac{1}{2}$$

$$\therefore x^2 + y^2 = \frac{1}{4}$$

$$(x - 0)^2 + (y - 0)^2 = \left(\frac{1}{2}\right)^2 \text{ -----(i)}$$

$$[\text{We have, } (x - a)^2 + (y - b)^2 = r^2]$$

Which is the equation of a circle whose Center (0, 0), Radius = $\frac{1}{2}$

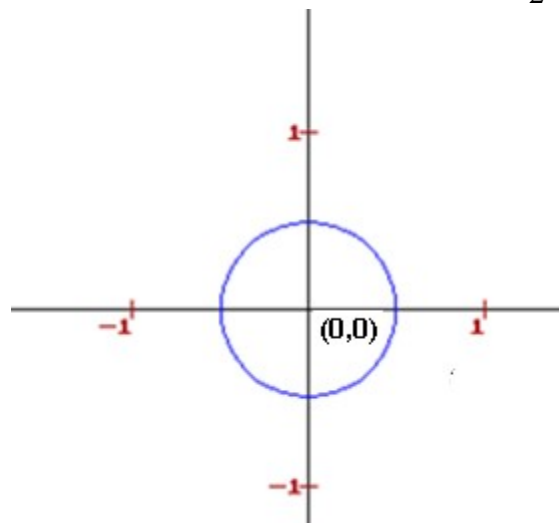


Figure 48

Pole $z - 1 = 0$

$$\Rightarrow z = 1$$

$$\Rightarrow x + jy = 1 + 0.j$$

Equating the coefficient of real and imaginary part, we get,

$$x = 1, \quad y = 0$$

Hence, we can write,

$$(x, y) = (1, 0)$$

$\therefore (x, y) = (1, 0)$ is outside the circle c , then by Cauchy's theorem

$$\oint_c f(z) dz = 0$$

$$\therefore \oint_c \frac{z^2 - z + 1}{z - 1} dz = 0$$

Example 28:

Evaluate $\int_c \frac{z}{z^2 - 3z + 2} dz$

Where c is the circle $|z - 2| = \frac{1}{2}$

We have,

$$z = x + jy$$

$$z - 2 = x + jy - 2$$

$$z - 2 = x - 2 + jy$$

$$\therefore |z - 2| = \sqrt{(x - 2)^2 + y^2}$$

Given,

$$|z - 2| = \frac{1}{2}$$

$$\therefore |z - 2| = \sqrt{(x - 2)^2 + y^2} = \frac{1}{2}$$

$$\therefore (x - 2)^2 + y^2 = \frac{1}{4}$$

$$\therefore (x - 2)^2 + (y - 0)^2 = \left(\frac{1}{2}\right)^2$$

Which is the equation of a circle whose Center $(2, 0)$, Radius $= \frac{1}{2}$

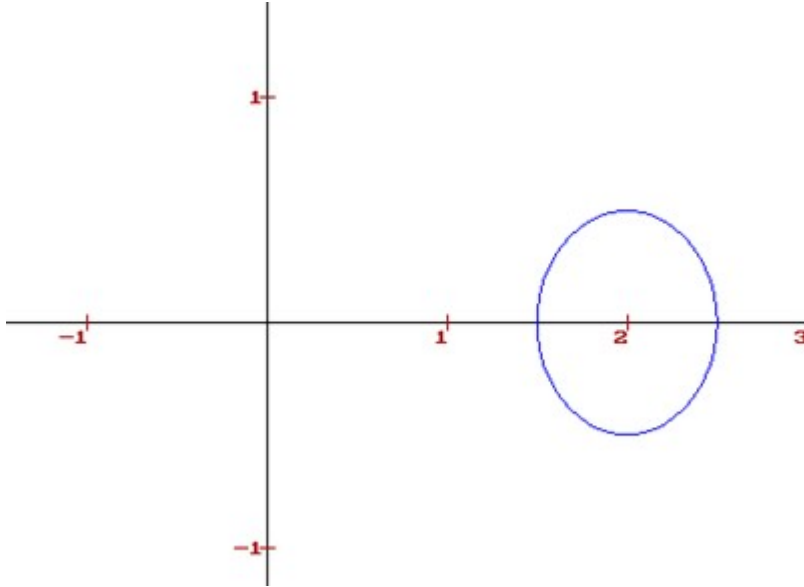


Figure 49

Poles: $z^2 - 3z + 2 = 0$

$$z^2 - 2z - z + 2 = 0$$

$$z(z-2) - 1(z-2) = 0$$

$$(z-2)(z-1) = 0$$

That is $z = 1, 2$

There is only one pole at $z = 2$ inside the given circle.

Pole $z - 2 = 0$

$$\Rightarrow z = 2$$

$$\Rightarrow x + jy = 2 + 0.j$$

Equating the coefficient of real and imaginary part, we get,

$$x = 2, \quad y = 0$$

Hence, we can write,

$$(x, y) = (2, 0)$$

$\therefore (x, y) = (2, 0)$ is on the circle c , then by Cauchy's Integral formula

$$\int_c \frac{f(z)}{z-a} dz = 2\pi i \times f(a)$$

$$\int_c \frac{f(z)}{z-2} dz = 2\pi i \times f(2) \quad [a = 2]$$

We have,

$$\int_c \frac{z}{z^2 - 3z + 2} dz$$

$$= \int_c \frac{z}{z^2 - 2z - z + 2} dz$$

$$= \int_c \frac{z}{z(z-2)-1(z-2)} dz$$

$$= \int_c \frac{z}{(z-1)(z-2)} dz$$

$$= \int_c \frac{\frac{z}{z-1}}{z-2} dz$$

$$\int_c \frac{f(z)}{z-a} dz = 2\pi i \times f(a)$$

$$\text{Here, } f(z) = \frac{z}{z-1}$$

$$\therefore f(2) = \frac{2}{2-1} = 2$$

So,

$$\int_c \frac{f(z)}{z-2} dz = 2\pi i \times f(2)$$

$$\int_c \frac{\frac{z}{z-1}}{z-2} dz = 2\pi i \times f(2)$$

$$\int_c \frac{\frac{z}{z-1}}{z-2} dz = 2\pi i \times 2$$

$$\int_c \frac{\frac{z}{z-1}}{z-2} dz = 4\pi i$$

Example 29:

$$\text{Evaluate } \int_c \frac{2z+1}{z^2+z} dz$$

Where c is the circle $|z| = \frac{1}{2}$

$$\sqrt{x^2+y^2} = \frac{1}{2}$$

$$\therefore x^2+y^2 = \frac{1}{4}$$

$$(x-0)^2 + (y-0)^2 = \left(\frac{1}{2}\right)^2 \quad \text{-----(i)}$$

[We have, $(x-a)^2 + (y-b)^2 = r^2$]

Which is the equation of a circle whose Center (0, 0), Radius = $\frac{1}{2}$

Poles: $z^2 + z = 0$

$z(z+1) = 0$

That is $z = 0, z = -1$

Now, Pole $z = 0$

$\Rightarrow x + jy = 0 + 0.j$

Equating the coefficient of real and imaginary part, we get,

$$x = 0, \quad y = 0$$

Hence, we can write,

$$(x, y) = (0, 0)$$

$\therefore (x, y) = (0, 0)$ is on the circle c, then by Cauchy's Integral formula

There is only one pole at $z = 0$ inside the given circle.

$$\int_c \frac{2z+1}{z^2+z} dz$$

$$\int_c \frac{2z+1}{z(z+1)} dz$$

$$= \int_c \frac{2z+1}{z} dz$$

Here, $f(z) = \frac{2z+1}{z+1}$

$$\therefore f(0) = \frac{1}{1} = 1$$

Hence, from Cauchy's Integral Formula:

$$\int_c \frac{f(z)}{z-a} dz = 2\pi i \times f(a)$$

$$\int_c \frac{f(z)}{z-0} dz = 2\pi i \times f(0) \quad [a = 0]$$

$$\int_c \frac{f(z)}{z} dz = 2\pi i \times f(0)$$

$$\int_c \frac{f(z)}{z} dz = \int_c \frac{2z+1}{z} dz = 2\pi i \times f(0)$$

$$\int_c \frac{f(z)}{z} dz = \int_c \frac{2z+1}{z} dz = 2\pi i \times 1$$

$$\int_c \frac{f(z)}{z} dz = \int_c \frac{\frac{2z+1}{z}}{z} dz = 2\pi i$$

Example 30. Show that $\oint_c \frac{e^{tz}}{z^2+1} dz = 2\pi i \sin t$, where c is the circle $|z| = 3$

Answer: Now, Poles: $z^2 + 1 = 0$

$$\Rightarrow z^2 = -1$$

$$\Rightarrow z^2 = i^2$$

$$\Rightarrow z = \pm i$$

$$\Rightarrow x + jy = 0 \pm 1.j$$

Equating the coefficient of real and imaginary part, we get,

$$x = 0, \quad y = 1 \quad \text{or} \quad x = 0, \quad y = -1$$

Hence, we can write,

$$(x, y) = (0, 1) \quad \text{or} \quad (x, y) = (0, -1)$$

Given

$$|z| = 3$$

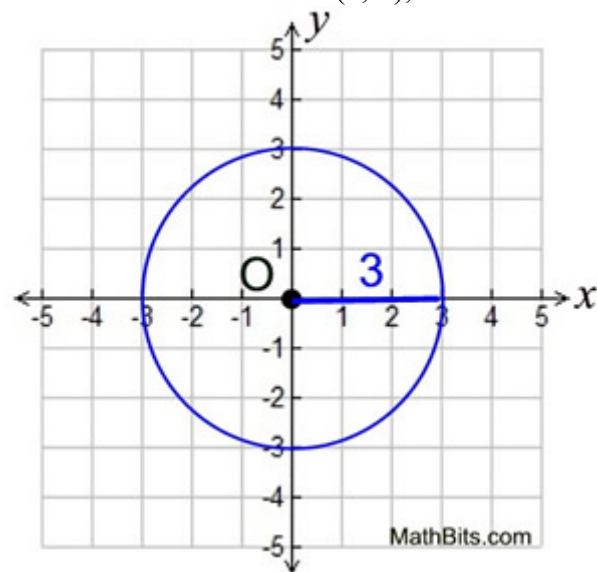
$$\Rightarrow \sqrt{x^2 + y^2} = 3$$

$$\Rightarrow x^2 + y^2 = 9$$

$$\Rightarrow x^2 + y^2 = 3^2$$

$$\Rightarrow x^2 + y^2 = 3^2$$

Which is the equation of a circle whose Center (0, 0), Radius = 3



$\therefore (x,y) = (0,1)$ and or $(x,y) = (0,-1)$ is on the circle c , then by Cauchy's Integral formula

$$\oint_c \frac{f(z)}{z-a} dz = 2\pi i \times f(a)$$

$$\oint_c \frac{f(z)}{z-i} dz = 2\pi i \times f(i) \text{------(i)}$$

And

$$\oint_c \frac{f(z)}{z+i} dz = 2\pi i \times f(-i) \text{------(ii)}$$

Given,

$$\oint_c \frac{e^{tz}}{z^2+1} dz = \oint_c \frac{e^{tz}}{(z+i)(z-i)} dz = \oint_c e^{tz} \frac{1}{(z+i)(z-i)} dz$$

$$\oint_c \frac{e^{tz}}{z^2+1} dz = \oint_c \frac{e^{tz}}{(z+i)(z-i)} dz = \frac{1}{2i} \oint_c e^{tz} \left[\frac{1}{(z-i)} - \frac{1}{(z+i)} \right] dz$$

$$\oint_c \frac{e^{tz}}{z^2+1} dz = \oint_c \frac{e^{tz}}{(z+i)(z-i)} dz = \frac{1}{2i} \oint_c \left[\frac{e^{tz}}{(z-i)} - \frac{e^{tz}}{(z+i)} \right] dz$$

$$\oint_c \frac{e^{tz}}{z^2+1} dz = \oint_c \frac{e^{tz}}{(z+i)(z-i)} dz = \frac{1}{2i} \left[\oint_c \frac{e^{tz}}{(z-i)} dz - \oint_c \frac{e^{tz}}{(z+i)} dz \right]$$

We have, Cauchy's Integral Formula:

$$\oint_c \frac{f(z)}{z-a} dz = 2\pi i \times f(a)$$

$$\text{For } \oint_c \frac{e^{tz}}{(z-i)} dz$$

$$f(z) = e^{tz}$$

$$\therefore f(i) = e^{it} \text{------(iii)}$$

And

$$\text{For } \oint_c \frac{e^{tz}}{(z+i)} dz$$

$$f(z) = e^{tz}$$

$$\therefore f(-i) = e^{-it} \text{------(iv)}$$

$$\begin{aligned} \oint_c \frac{e^{tz}}{z^2+1} dz &= \oint_c \frac{e^{tz}}{(z+i)(z-i)} dz = \frac{1}{2i} \left[\oint_c \frac{e^{tz}}{(z-i)} dz - \oint_c \frac{e^{tz}}{(z+i)} dz \right] \\ &= \frac{1}{2i} [2\pi i \times f(i) - 2\pi i \times f(-i)] \quad \left[\because \oint_c \frac{f(z)}{z-a} dz = 2\pi i \times f(a) \right] \\ &= \frac{1}{2i} [2\pi i \times e^{it} - 2\pi i \times e^{-it}] \end{aligned}$$

$$= 2\pi i \frac{1}{2i} [e^{it} - e^{-it}]$$

$$= 2\pi i \times \sin t \quad (\text{Proved}) \quad [\because \sin x = \frac{1}{2i}(e^{ix} - e^{-ix})]$$

We have,

$$e^{ix} = \cos x + i \sin x$$

$$e^{-ix} = \cos x - i \sin x$$

$$e^{ix} - e^{-ix} = 2i \sin x$$

$$\therefore \sin x = \frac{1}{2i}(e^{ix} - e^{-ix})$$

Example 31: Show that $\oint_c \frac{\sin 3z}{z + \frac{\pi}{2}} dz = 2\pi i$, where c is the circle $|z| = 5$

Poles: $z + \frac{\pi}{2} = 0$

$$\Rightarrow z = -\frac{\pi}{2}$$

Poles: $z + \frac{\pi}{2} = 0$

$$\Rightarrow z = -\frac{\pi}{2}$$

$$\Rightarrow x + jy = -\frac{\pi}{2} + j.0$$

So, $(x, y) = (-\frac{\pi}{2}, 0)$

Given

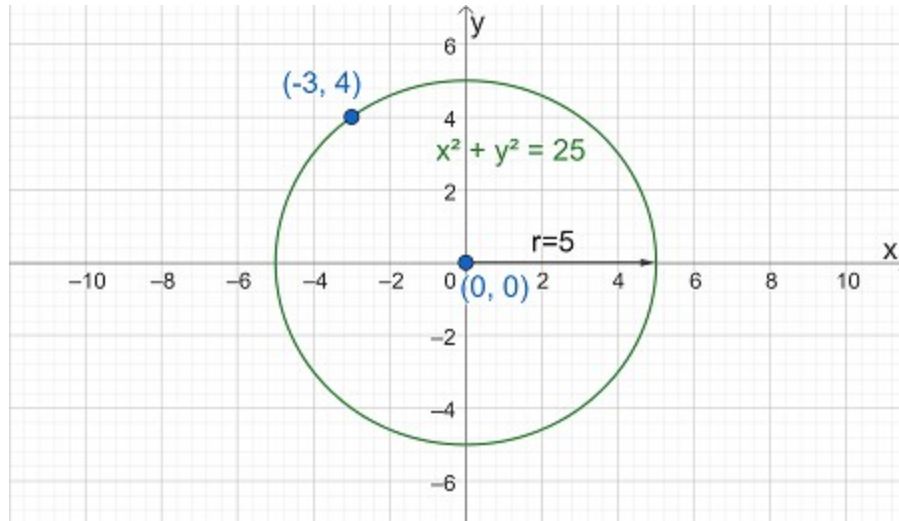
$$|z| = 5$$

$$\Rightarrow \sqrt{x^2 + y^2} = 5$$

$$\Rightarrow x^2 + y^2 = 25$$

$$\Rightarrow x^2 + y^2 = 5^2$$

Which is the equation of a circle whose Center (0, 0), Radius = 5



$\therefore (x,y) = (-\frac{\pi}{2}, 0)$ is on the circle c , then by Cauchy's Integral

$$\oint_c \frac{f(z)}{z-a} dz = 2\pi i \times f(a)$$

Here,

$$\oint_c \frac{\sin 3z}{z + \frac{\pi}{2}} dz$$

$$f(z) = \sin 3z$$

$$\therefore f(-\frac{\pi}{2}) = \sin(-3 \frac{\pi}{2})$$

$$\therefore f(-\frac{\pi}{2}) = -\sin(3 \frac{\pi}{2}) \quad [\sin(-\theta) = -\sin \theta]$$

$$\therefore f(-\frac{\pi}{2}) = -\sin(3 \times \frac{\pi}{2} + 0)$$

$$\therefore f(-\frac{\pi}{2}) = -\sin(3 \times 90^\circ + 0)$$

$$\therefore f(-\frac{\pi}{2}) = -(-\cos 0)$$

$$\therefore f(-\frac{\pi}{2}) = 1$$

$$\therefore \oint_c \frac{\sin 3z}{z + \frac{\pi}{2}} dz = 2\pi i \times f(-\frac{\pi}{2})$$

$$\therefore \oint_c \frac{\sin 3z}{z + \frac{\pi}{2}} dz = 2\pi i \times 1$$

$$\therefore \oint_{\gamma} \frac{\sin 3z}{z + \frac{\pi}{2}} dz = 2\pi i \text{ (Proved)}$$

Example 32:

Find the residue at pole of $\frac{1-2z}{z(z-1)(z-2)}$

Answer:

$$\text{Let, } f(z) = \frac{1-2z}{z(z-1)(z-2)}$$

$$\text{Poles } z(z-1)(z-2) = 0$$

$$z = 0; (z-1) = 0; (z-2) = 0$$

$$z = 0; z = 1; z = 2$$

$$\text{Residue of } f(z) \text{ at } (z=0) = \lim_{z \rightarrow 0} (z-0)f(z)$$

$$= \lim_{z \rightarrow 0} z \frac{1-2z}{z(z-1)(z-2)}$$

$$= \lim_{z \rightarrow 0} \frac{1-2z}{(z-1)(z-2)}$$

$$= \frac{1}{(-1)(-2)}$$

$$= \frac{1}{2}$$

$$\text{Residue of } f(z) \text{ at } (z=1) = \lim_{z \rightarrow 1} (z-1)f(z)$$

$$= \lim_{z \rightarrow 1} (z-1) \frac{1-2z}{z(z-1)(z-2)}$$

$$= \lim_{z \rightarrow 1} \frac{1-2z}{z(z-2)}$$

$$= \frac{1-2}{(1)(1-2)}$$

$$= 1$$

$$\text{Residue of } f(z) \text{ at } (z=2) = \lim_{z \rightarrow 2} (z-2)f(z)$$

$$= \lim_{z \rightarrow 2} (z-2) \frac{1-2z}{z(z-1)(z-2)}$$

$$= \lim_{z \rightarrow 2} \frac{1-2z}{z(z-1)}$$

$$= \frac{1-2 \times 2}{(2)(2-1)}$$

$$= -\frac{3}{2}$$

Or

Residue of $f(z)$ at $(z = 0)$

Given,

$$f(z) = \frac{1-2z}{z(z-1)(z-2)}$$

$$f(z) = \frac{1-2z}{z(z-1)(z-2)}$$

Residue of $f(z)$ at $(z = 0)$

$$\begin{aligned} &= \lim_{z \rightarrow 0} \frac{1-2z}{(z-1)(z-2)} \\ &= \frac{1}{(-1)(-2)} = \frac{1}{2} \end{aligned}$$

Residue of $f(z)$ at $(z = 1)$

Given,

$$f(z) = \frac{1-2z}{z(z-1)(z-2)}$$

$$f(z) = \frac{1-2z}{z(z-1)(z-2)}$$

Residue of $f(z)$ at $(z = 1)$

$$\begin{aligned} &= \lim_{z \rightarrow 1} \frac{1-2z}{z(z-2)} \\ &= \frac{1-2}{(1)(1-2)} = 1 \end{aligned}$$

Residue of $f(z)$ at $(z = 2)$

Given,

$$f(z) = \frac{1-2z}{z(z-1)(z-2)}$$

$$f(z) = \frac{1-2z}{z(z-1)(z-2)}$$

Residue of $f(z)$ at $(z = 2)$

$$= \lim_{z \rightarrow 2} \frac{1-2z}{z(z-1)}$$

$$\begin{aligned}
 &= \frac{1-2 \times 2}{(2)(2-1)} \\
 &= -\frac{3}{2}
 \end{aligned}$$

Example 33: Consider the function $f(z) = \frac{5}{i-3z}$; to find the poles and residue of the function

Answer:

Given $f(z) = \frac{5}{i-3z}$

To find the pole and residue, divide the numerator and denominator by -3

$$\therefore f(z) = \frac{-\frac{5}{3}}{z - \frac{i}{3}}$$

That is the pole of the function $z = \frac{i}{3}$

$$\begin{aligned}
 \text{And residue of } f(z) \text{ at } (z = \frac{i}{3}) &= \lim_{z \rightarrow \frac{i}{3}} (z - \frac{i}{3}) f(z) = \lim_{z \rightarrow \frac{i}{3}} (z - \frac{i}{3}) \frac{5}{i-3z} \\
 &= \lim_{z \rightarrow \frac{i}{3}} \left(\frac{3z-i}{3} \right) \frac{5}{i-3z} \\
 &= \lim_{z \rightarrow \frac{i}{3}} \left(\frac{3z-i}{3} \right) \frac{5}{-(3z-i)} \\
 &= -\frac{5}{3}
 \end{aligned}$$

Or

Given $f(z) = \frac{5}{i-3z}$

Pole : $i-3z = 0$

Pole : $-3z = -i$

Pole : $3z = i$

Pole : $z = i/3$

Given $f(z) = \frac{5}{i-3z}$

To find the residue, divide the numerator and denominator by -3

$$\therefore f(z) = \frac{-\frac{5}{3}}{z - \frac{i}{3}}$$

Hence, residue of $f(z) = -\frac{5}{3}$

Example 34: Consider the function $f(z) = \frac{z}{z^2 + 1}$; to find the poles and residue of the function

Given, $f(z) = \frac{z}{z^2 + 1}$

To find the pole and residue, we factorize the denominator $f(z) = \frac{z}{(z+i)(z-i)}$

Poles: $z^2 + 1 = 0$

Poles $z^2 = -1$

Poles: $z^2 = i^2$

Poles: $z = \pm i$

Hence there are two simple poles at $z = +i, -i$

And residue of $f(z)$ at $(z = i) = \lim_{z \rightarrow i} (z - i)f(z) = \lim_{z \rightarrow i} (z - i) \frac{z}{(z+i)(z-i)}$

$$= \lim_{z \rightarrow i} \frac{z}{(z+i)}$$

$$= \lim_{z \rightarrow i} \frac{i}{(i+i)} = \frac{i}{2i} = \frac{1}{2}$$

Or

To find the residue, at $z = i$, we separate the divergent part to obtain $f(z) = \frac{z}{(z+i)(z-i)}$

We can write, $f(z) = \frac{\frac{z}{z+i}}{(z-i)}$

Hence residue: $\text{Res}[f(z)]_{z=i} = \left[\frac{z}{z+i} \right]_{z=i} = \frac{i}{i+i} = \frac{1}{2}$

And

To find the residue, at $z = -i$, we separate the divergent part to obtain

$$f(z) = \frac{z}{(z+i)(z-i)}$$

$$f(z) = \frac{z}{z-i}$$

$$\text{Hence residue : Res}[f(z)]_{z=-i} = \left[\frac{z}{z-i} \right]_{z=-i} = \frac{-i}{-i-i} = \frac{1}{2}$$

https://www1.spms.ntu.edu.sg/~ydchong/teaching/09_contour_integration.pdf
file:///C:/Users/REZAUL/Downloads/basic_complex_int_HELM%20(1).pdf

*https://complex-analysis.com/content/complex_integration.html
https://www.academia.edu/39133549/Complex_Variables_with_Applications?email_work_card=title (vvvvvvvvvvvvvvv imp)*